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SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)
(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12 (B))
Pedatadepalli, Tadepalligudem-534101, W.G.Dist. (A.P)

B. Tech (Regular-Full time)

(Effective for the students admitted into I year from the Academic
Year **2023-24** onwards)

&

B.Tech.(Lateral Entry Scheme)

(Effective for the students admitted into II year through Lateral Entry
Scheme from the Academic Year **2024 - 25** onwards)

Academic Regulations, Course Structure and Detailed Syllabus

for

Computer Science and Engineering

&

Computer Science and Technology

V23

Academic Regulations



JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

(Established by Govt. of A.P., Act No.30 of 2008)

Kakinada – 533003, Andhra Pradesh, India, www.jntuk.edu.in

B. Tech (Regular-Full time)

(Effective for the students admitted into I year from the Academic Year **2023-24** onwards)

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B.Tech.(Lateral Entry Scheme)

(Effective for the students admitted into II year through Lateral Entry Scheme from the Academic Year **2024 - 25** onwards)

Academic Regulations (R23) for B. Tech (Regular-Full time)

(Effective for the students admitted into I year from
the Academic Year **2023-24** onwards)

1. Award of the Degree

- (a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:
 - (i) Pursues a course of study for not less than four academic years and not more than eight academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Eight years).
 - (ii) Registers for 160 credits and secures all 160 credits.
- (b) **Award of B.Tech. degree with Honors** if he/she fulfils the following:
 - (i) Student secures additional 15 credits fulfilling all the requisites of a B.Tech. program i.e., 160 credits.
 - (ii) Registering for Honors is optional.
 - (iii) Honors is to be completed simultaneously with B.Tech. programme.

2. Students, who fail to fulfil all the academic requirements for the award of the degree within eight academic years from the year of their admission, shall forfeit their seat in B.Tech. course and their admission stands cancelled. This clause shall be read along with clause 1 a) i).

3. Admissions

Admission to the B. Tech Program shall be made subject to the eligibility, qualifications and specialization prescribed by the A.P. State Government/University from time to time. Admissions shall be made either based on the merit rank obtained by the student in the common entrance examination conducted by the A.P. Government/University or any other order of merit approved by the A.P. Government/University, subject to reservations as prescribed by the Government/University from time to time.

4. Program related terms

Credit: A unit by which the course work is measured. It determines the number of hours of instruction required per week. One credit is equivalent to one hour of teaching (Lecture/Tutorial) or two hours of practical work/field work per week.

Credit Definition:

1 Hr. Lecture (L) per week	1 credit
1 Hr. Tutorial (T) per week	1 credit
1 Hr. Practical (P) per week	0.5 credit
2 Hrs. Practical (Lab) per week	1 credit

a) **Academic Year:** Two consecutive (one odd + one even) semesters constitute one academic year.

b) **Choice Based Credit System (CBCS):** The CBCS provides a choice for students to select from the prescribed courses.

5. Semester/Credits:

- i) A semester comprises 90 working days and an academic year is divided into two semesters.
- ii) The summer term is for eight weeks during summer vacation. Internship/ apprenticeship / work-based vocational education and training can be carried out during the summer term, especially by students who wish to exit after two semesters or four semesters of study.
- iii) Regular courses may also be completed well in advance through MOOCs satisfying prerequisites.

6. Structure of the Undergraduate Programme

All courses offered for the undergraduate program (B. Tech.) are broadly classified as follows:

S.No.	Category	Breakup of Credits (Total 160)	Percentage of total credits	AICTE Recommendation (%)
1.	Humanities and Social Science including Management (HM)	13	8 %	8 – 9%
2.	Basic Sciences (BS)	20	13 %	12 - 16%
3.	Engineering Sciences (ES)	23.5	14%	10 – 18%
4.	Professional Core (PC)	54.5	34 %	30 – 36%
5.	Electives – Professional (PE) & Open (OE); Domain Specific Skill Enhancement Courses (SEC)	33	21 %	19 - 23%
6.	Internships & Project work (PR)	16	10 %	8 – 11%
7.	Mandatory Courses (MC)	Non-credit	Non-credit	-

7. Course Classification:

All subjects/ courses offered for the undergraduate programme in Engineering & Technology (B.Tech. degree programmes) are broadly classified as follows:

S.No.	Broad Course Classification	Course Category	Description
1.	Foundation Courses	Foundation courses	Includes Mathematics, Physics and Chemistry; fundamental engineering courses; humanities, social sciences and management courses
2.	Core Courses	Professional Core Courses (PC)	Includes subjects related to the parent discipline/department/branch of Engineering
3.	Elective Courses	Professional Elective Courses (PE)	Includes elective subjects related to the parent discipline/department/ branch of Engineering
		Open Elective Courses (OE)	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering
		Domain specific skill enhancement courses (SEC)	interdisciplinary/job-oriented/domain courses which are relevant to the industry
4.	Project & Internships	Project	B.Tech. Project or Major Project
		Internships	Summer Internships – Community based and Industry Internships; Industry oriented Full Semester Internship
5.	Audit Courses	Mandatory non-credit courses	Covering subjects of developing desired attitude among the learners

8. Programme Pattern

- Total duration of the of B. Tech (Regular) Programme is four academic years.
- Each academic year of study is divided into two semesters.
- Minimum number of instruction days in each semester is 90 days.
- There shall be mandatory student induction program for freshers, with a three-week duration before the commencement of first semester. Physical activity, Creative Arts, Universal Human Values, Literary, Proficiency Modules, Lectures by Eminent People, Visits to local Areas, Familiarization to Dept./Branch & Innovations etc., are included as per the guidelines issued by AICTE.
- Health/wellness/yoga/sports and NSS /NCC /Scouts & Guides / Community service activities are made mandatory as credit courses for all the undergraduate students.
- Courses like Environmental Sciences, Indian Constitution, Technical Paper Writing & IPR are offered as non-credit mandatory courses for all the undergraduate students.
- Design Thinking for Innovation & Tinkering Labs are made mandatory as credit courses for all the undergraduate students.
- Increased flexibility for students through an increase in the elective component of the curriculum, with 05 Professional Elective courses and 04 Open Elective courses.

- ix. Professional Elective Courses, include the elective courses relevant to the chosen specialization/branch. Proper choice of professional elective courses can lead to students specializing in emerging areas within the chosen field of study.
- x. A total of 04 Open Electives are offered in the curriculum. A student can complete the requirement for B.Tech. Degree with a Minor within the 160 credits by opting for the courses offered through various verticals/tracks under Open Electives.
- xi. While choosing the electives, students shall ensure that they do not opt for the courses with syllabus contents similar to courses already pursued.
- xii. A pool of interdisciplinary/job-oriented/domain skill courses which are relevant to the industry are integrated into the curriculum of all disciplines. There shall be 05 skill-oriented courses offered during III to VII semesters. Among the five skill courses, four courses shall focus on the basic and advanced skills related to the domain/interdisciplinary courses and the other shall be a soft skills course.
- xiii. Students shall undergo mandatory summer internships, for a minimum of eight weeks duration at the end of second and third year of the programme. The internship at the end of second year shall be community oriented and industry internship at the end of third year.
- xiv. There shall also be mandatory full internship in the final semester of the programme along with the project work.
- xv. Undergraduate degree with Honors is introduced by the University for the students having good academic record.
- xvi. Each college shall take measures to implement Virtual Labs (<https://www.vlab.co.in>) which provide remote access to labs in various disciplines of Engineering and will help student in learning basic and advanced concept through remote experimentation. Student shall be made to work on virtual lab experiments during the regular labs.
- xvii. Each college shall assign a faculty advisor/mentor after admission to a group of students from same department to provide guidance in courses registration/career growth/placements/opportunities for higher studies/GATE/other competitive exams etc.
- xviii. Preferably 25% of course work for the theory courses in every semester shall be conducted in the blended mode of learning.

9. Evaluation Process

The performance of a student in each semester shall be evaluated subject wise with a maximum of 100 marks for theory and 100 marks for practical subject. Summer Internships shall be evaluated for 50 marks, Full Internship & Project work in final semester shall be evaluated for 200 marks, mandatory courses with no credits shall be evaluated for 30 mid semester marks.

A student has to secure not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester and end examination marks taken together for the theory, practical, design, drawing subject or project etc. In case of a mandatory course, he/she should secure 40% of the total marks.

Theory Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

- i) For theory subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End-Examination.
- ii) For practical subject, the distribution shall be 30 marks for Internal Evaluation and 70 marks for the End- Examination.
- iii) If any course contains two different branch subjects, the syllabus shall be written in two parts with 3 units each (Part-A and Part-B) and external examination question paper shall be set with two parts each for 35 marks.
- iv) If any subject is having both theory and practical components, they will be evaluated separately as theory subject and practical subject. However, they will be given same subject code with an extension of 'T' for theory subject and 'P' for practical subject.

a) Continuous Internal Evaluation

- i) For theory subjects, during the semester, there shall be two midterm examinations. Each midterm examination shall be evaluated for 30 marks of which 10 marks for objective paper (20 minutes duration), 15 marks for subjective paper (90 minutes duration) and 5 marks for assignment.
- ii) Objective paper shall contain for 05 short answer questions with 2 marks each or maximum of 20 bits for 10 marks. Subjective paper shall contain 3 either or type questions (totally six questions from 1 to 6) of which student has to answer one from each either-or type of questions. Each question carries 10 marks. The marks obtained in the subjective paper are condensed to 15 marks.

Note:

- The objective paper shall be prepared in line with the quality of competitive examinations questions.
 - The subjective paper shall contain 3 either or type questions of equal weightage of 10 marks. Any fraction shall be rounded off to the next higher mark.
 - The objective paper shall be conducted by the respective institution on the day of subjective paper test.
 - Assignments shall be in the form of problems, mini projects, design problems, slip tests, quizzes etc., depending on the course content. It should be continuous assessment throughout the semester and the average marks shall be considered.
- iii) If the student is absent for the mid semester examination, no re-exam shall be conducted and mid semester marks for that examination shall be considered as zero.
 - iv) First midterm examination shall be conducted for I, II units of syllabus with one either or type question from each unit and third either or type question from both the

units. The second midterm examination shall be conducted for III, IV and V units with one either or type question from each unit.

- v) Final mid semester marks shall be arrived at by considering the marks secured by the student in both the mid examinations with 80% weightage given to the better mid exam and 20% to the other.

For Example:

Marks obtained in first mid: 25

Marks obtained in second mid: 20

Final mid semester Marks: $(25 \times 0.8) + (20 \times 0.2) = 24$

If the student is absent for any one midterm examination, the final mid semester marks shall be arrived at by considering 80% weightage to the marks secured by the student in the appeared examination and zero to the other. For Example:

Marks obtained in first mid: Absent

Marks obtained in second mid: 25

Final mid semester Marks: $(25 \times 0.8) + (0 \times 0.2) = 20$

b) End Examination Evaluation:

End examination of theory subjects shall have the following pattern:

- i) There shall be 6 questions and all questions are compulsory.
- ii) Question I shall contain 10 compulsory short answer questions for a total of 20 marks such that each question carries 2 marks.
- iii) There shall be 2 short answer questions from each unit.
- a) In each of the questions from 2 to 6, there shall be either/or type questions of 10 marks each. Student shall answer any one of them.
- iv) The questions from 2 to 6 shall be set by covering one unit of the syllabus for each question.

End examination of theory subjects consisting of two parts of different subjects, for Example: Basic Electrical & Electronics Engineering shall have the following pattern:

- i) Question paper shall be in two parts viz., Part A and Part B with equal weightage of 35 marks each.
- ii) In each part, question 1 shall contain 5 compulsory short answer questions for a total of 5 marks such that each question carries 1 mark.
- iii) In each part, questions from 2 to 4, there shall be either/or type questions of 10 marks each. Student shall answer any one of them.
- iv) The questions from 2 to 4 shall be set by covering one unit of the syllabus for each question.

Practical Courses

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

- b) For practical courses, there shall be a continuous evaluation during the semester for 30 sessional marks and end examination shall be for 70 marks.
- c) Day-to-day work in the laboratory shall be evaluated for 15 marks by the concerned laboratory teacher based on the record/viva and 15 marks for the internal test.
- d) The end examination shall be evaluated for 70 marks, conducted by the concerned laboratory teacher and a senior expert in the subject from the same department.
 - Procedure: 20 marks
 - Experimental work & Results: 30 marks
 - Viva voce: 20 marks.

In a practical subject consisting of two parts (Eg: Basic Electrical & Electronics Engineering Lab), the end examination shall be conducted for 70 marks as a single laboratory in 3 hours. Mid semester examination shall be evaluated as above for 30 marks in each part and final mid semester marks shall be arrived by considering the average of marks obtained in two parts.

- e) For the subject having design and/or drawing, such as Engineering Drawing, the distribution of marks shall be 30 for mid semester evaluation and 70 for end examination.

Assessment Method	Marks
Continuous Internal Assessment	30
Semester End Examination	70
Total	100

Day-to-day work shall be evaluated for 15 marks by the concerned subject teacher based on the reports/submissions prepared in the class. And there shall be two midterm examinations in a semester for duration of 2 hours each for 15 marks with weightage of 80% to better mid marks and 20% for the other. The subjective paper shall contain 3 either or type questions of equal weightage of 5 marks. There shall be no objective paper in mid semester examination. The sum of day-to-day evaluation and the mid semester marks will be the final sessional marks for the subject.

The end examination pattern for Engineering Graphics, shall consists of 5 questions, either/or type, of 14 marks each. There shall be no objective type questions in the end examination. However, the end examination pattern for other subjects related to design/drawing , multiple branches, etc is mentioned along with the syllabus.

- f) There shall be no external examination for mandatory courses with zero credits. However, attendance shall be considered while calculating aggregate attendance and student shall be declared to have passed the mandatory course only when he/she secures 40% or more in the internal examinations. In case, the student fails, a re-

examination shall be conducted for failed candidates for 30 marks satisfying the conditions mentioned in item 1 & 2 of the regulations.

- g) The laboratory records and mid semester test papers shall be preserved for a minimum of 3 years in the respective institutions as per the University norms and shall be produced to the Committees of the University as and when the same are asked for.

10. Skill oriented Courses

- i) There shall be five skill-oriented courses offered during III to VII semesters.
- ii) Out of the five skill courses two shall be skill-oriented courses from the same domain. Of the remaining three skill courses, one shall be a soft skill course and the remaining two shall be skill-advanced courses from the same domain/Interdisciplinary/Job oriented.
- iii) The course shall carry 100 marks and shall be evaluated through continuous assessments during the semester for 30 sessional marks and end examination shall be for 70 marks. Day-to-day work in the class / laboratory shall be evaluated for 30 marks by the concerned teacher based on the regularity/assignments/viva/mid semester test. The end examination similar to practical examination pattern shall be conducted by the concerned teacher and an expert in the subject nominated by the principal.
- iv) The Head of the Department shall identify a faculty member as coordinator for the course. A committee consisting of the Head of the Department, coordinator and a senior Faculty member nominated by the Head of the Department shall monitor the evaluation process. The marks/grades shall be assigned to the students by the above committee based on their performance.
- v) The student shall be given an option to choose either the skill courses being offered by the college or to choose a certificate course being offered by industries/Professional bodies or any other accredited bodies. If a student chooses to take a Certificate Course offered by external agencies, the credits shall be awarded to the student upon producing the Course Completion Certificate from the agency. A committee shall be formed at the level of the college to evaluate the grades/marks given for a course by external agencies and convert to the equivalent marks/grades.
- vi) The recommended courses offered by external agencies, conversions and appropriate grades/marks are to be approved by the University at the beginning of the semester. The principal of the respective college shall forward such proposals to the University for approval.
- vii) If a student prefers to take a certificate course offered by external agency, the department shall mark attendance of the student for the remaining courses in that semester excluding the skill course in all the calculations of mandatory attendance requirements upon producing a valid certificate as approved by the University.

11. Massive Open Online Courses (MOOCs):

A Student has to pursue and complete one course compulsorily through MOOCs approved by the University. A student can pursue courses other than core through MOOCs and it is mandatory to complete one course successfully through MOOCs for awarding the degree. A student is not permitted to register and pursue core courses through MOOCs.

A student shall register for the course (Minimum of either 8 weeks or 12 weeks) offered through MOOCs with the approval of Head of the Department. The Head of the Department shall appoint one mentor to monitor the student's progression. The student needs to earn a certificate by passing the exam. The student shall be awarded the credits assigned in the curriculum only by submission of the certificate. Examination fee, if any, will be borne by the student.

Students who have qualified in the proctored examinations conducted through MOOCs platform can apply for credit transfer as specified and are exempted from appearing internal as well as external examination (for the specified equivalent credit course only) conducted by the university.

Necessary amendments in rules and regulations regarding adoption of MOOC courses would be proposed from time to time.

12. Credit Transfer Policy

Adoption of MOOCs is mandatory, to enable Blended model of teaching-learning as also envisaged in the NEP 2020. As per University Grants Commission (Credit Framework for Online Learning Courses through SWAYAM) Regulation, 2016, the University shall allow up to a maximum of 20% of the total courses being offered in a particular programme i.e., maximum of 32 credits through MOOCs platform.

- i) The University shall offer credit mobility for MOOCs and give the equivalent credit weightage to the students for the credits earned through online learning courses.
- ii) Student registration for the MOOCs shall be only through the respective department of the institution, it is mandatory for the student to share necessary information with the department.
- iii) Credit transfer policy will be applicable to the Professional & Open Elective courses only.
- iv) The concerned department shall identify the courses permitted for credit transfer.
- v) The University/institution shall notify at the beginning of semester the list of the online learning courses eligible for credit transfer.
- vi) The institution shall designate a faculty member as a Mentor for each course to guide the students from registration till completion of the credit course.
- vii) The university shall ensure no overlap of MOOC exams with that of the university examination schedule. In case of delay in results, the university will re-issue the marks sheet for such students.

- viii) Student pursuing courses under MOOCs shall acquire the required credits only after successful completion of the course and submitting a certificate issued by the competent authority along with the percentage of marks and grades.
- ix) The institution shall submit the following to the examination section of the university:
 - a) List of students who have passed MOOC courses in the current semester along with the certificate of completion.
 - b) Undertaking form filled by the students for credit transfer.
- x) The universities shall resolve any issues that may arise in the implementation of this policy from time to time and shall review its credit transfer policy in the light of periodic changes brought by UGC, SWAYAM, NPTEL and state government.

Note: Students shall be permitted to register for MOOCs offered through online platforms approved by the University from time to time.

13. Academic Bank of Credits (ABC)

The University has implemented Academic Bank of Credits (ABC) to promote flexibility in curriculum as per NEP 2020 to

- i. provide option of mobility for learners across the universities of their choice
- ii. provide option to gain the credits through MOOCs from approved digital platforms.
- iii. facilitate award of certificate/diploma/degree in line with the accumulated credits in ABC
- iv. execute Multiple Entry and Exit system with credit count, credit transfer and credit acceptance from students' account.

14. Mandatory Internships

Summer Internships : Two summer internships either onsite or virtual each with a minimum of 08 weeks duration, done at the end of second and third years, respectively are mandatory. It shall be completed in collaboration with local industries, Govt. Organizations, construction agencies, Power projects, software MNCs or any industries in the areas of concerned specialization of the Undergraduate program. One of the two summer internships at the end of second year (Community Service Project) shall be society oriented and shall be completed in collaboration with government organizations/NGOs & others. The other internship at the end of third year is Industry Internship and shall be completed in collaboration with Industries. The student shall register for the internship as per course structure after commencement of academic year. The guidelines issued by the APSCHE / University shall be followed for carrying out and evaluation of Community Service Project and Industry Internship.

Evaluation of the summer internships shall be through the departmental committee. A student will be required to submit a summer internship report to the concerned department and appear for an oral presentation before the departmental committee comprising of Head of the Department, supervisor of the internship and a senior faculty member of the department. A certificate of successful completion from industry shall

be included in the report. The report and the oral presentation shall carry 50% weightage each. It shall be evaluated for 50 external marks. There shall be no internal marks for Summer Internship. A student shall secure minimum 40% of marks for successful completion. In case, if a student fails, he/she shall reappear as and when semester supplementary examinations are conducted by the University.

Full Semester Internship and Project work: In the final semester, the student should mandatorily register and undergo internship (onsite/virtual) and in parallel he/she should work on a project with well-defined objectives. At the end of the semester the candidate shall submit an internship completion certificate and a project report. A student shall also be permitted to submit project report on the work carried out during the internship.

The project report shall be evaluated with an external examiner. The total marks for project work 200 marks and distribution shall be 60 marks for internal and 140 marks for external evaluation. The supervisor assesses the student for 30 marks (Report: 15 marks, Seminar: 15 marks). At the end of the semester, all projects shall be showcased at the department for the benefit of all students and staff and the same is to be evaluated by the departmental Project Review Committee consisting of supervisor, a senior faculty and HOD for 30 marks. The external evaluation of Project Work is a Viva-Voce Examination conducted in the presence of internal examiner and external examiner appointed by the University and is evaluated for 140 marks.

The college shall facilitate and monitor the student internship programs. Completion of internships is mandatory, if any student fails to complete internship, he/she will not be eligible for the award of degree. In such cases, the student shall repeat and complete the internship.

15. Guidelines for offering a Minor

To promote interdisciplinary knowledge among the students, the students admitted into B.Tech. in a major stream/branch are eligible to obtain degree in Minor in another stream.

- i) The Minor program requires the completion of 12 credits in Minor stream chosen.
- i) Two courses for 06 credits related to a Minor are to be pursued compulsorily for the minor degree, but maybe waived for students who have done similar/equivalent courses. If waived for a student, then the student must take an extra elective course in its place. It is recommended that students should complete the compulsory courses (or equivalents) before registering for the electives.
- ii) Electives (minimum of 2 courses) to complete a total of 12 credits.

Note: A total of 04 Open Electives are offered in the curriculum. A student can complete the requirement for Minor by opting for the courses offered through various verticals/tracks under Open Electives.

16. Guidelines for offering Honors

The objective of introducing B.Tech. (Hons.) is to facilitate the students to choose

additionally the specialized courses of their choice and build their competence in a specialized area in the UG level. The programme is a best choice for academically excellent students having good academic record and interest towards higher studies and research.

- i) Honors is introduced in the curriculum of all B. Tech. programs offering a major degree and is applicable to all B. Tech (Regular and Lateral Entry) students admitted in Engineering & Technology.
- ii) A student shall earn additional 15 credits for award of B.Tech.(Honors) degree from same branch/department/discipline registered for major degree. This is in addition to the credits essential for obtaining the Undergraduate degree in Major Discipline (i.e., 160 credits).
- iii) A student is permitted to register for Honors in IV semester after the results of III Semester are declared and students may be allowed to take maximum two subjects per semester pertaining to the Honors from V Semester onwards.
- iv) The concerned Principal of the college shall arrange separate class work and timetable of the courses offered under Honors program.
- v) Courses that are used to fulfil the student's primary major may not be double counted towards the Honors. Courses with content substantially equivalent to courses in the student's primary Major may not be counted towards the Honors.
- vi) Students can complete the courses offered under Honors either in the college or in online platforms like SWAYAM with a minimum duration of 12 weeks for a 3-credit course and 8 weeks duration for a 2-credit course satisfying the criteria for credit mobility. If the courses under Honors are offered in conventional mode, then the teaching and evaluation procedure shall be similar to regular B. Tech courses.
- vii) The attendance for the registered courses under Honors and regular courses offered for Major degree in a semester are to be considered separately.
- viii) A student shall maintain an attendance of 75% in all registered courses under Honors to be eligible for attending semester end examinations.
- ix) **A student registered for Honors shall pass in all subjects that constitute the requirement for the Honors degree program.** No class/division (i.e., second class, first class and distinction, etc.) shall be awarded for Honors degree programme.
- x) If a student drops or is terminated from the Honors program, the additional credits so far earned cannot be converted into open or core electives; they will remain extra. However, such students will receive a separate grade sheet mentioning the additional courses completed by them.
- xi) The Honors will be mentioned in the degree certificate as Bachelor of Technology (Honors) in XYZ. For example, B.Tech. (Honors) in Mechanical Engineering

Enrolment into Honors:

- i) Students of a Department/Discipline are eligible to opt for Honors program offered by the same Department/Discipline
- ii) The enrolment of student into Honors is based on the CGPA obtained in the major degree program. CGPA shall be taken up to III semester in case of regular entry students and only III semester in case of lateral entry students. Students having 7 CGPA without any backlog subjects will be permitted to register for

Honors.

- iii) If a student is detained due to lack of attendance either in Major or in Honors, registration shall be cancelled.
- iv) Transfer of credits from Honors to regular B. Tech degree and vice-versa shall not be permitted.
- v) Honors is to be completed simultaneously with a Major degree program.

Registration for Honors:

- i) The eligible and interested students shall apply through the HOD of his/her parent department. The whole process should be completed within one week before the start of every semester. Selected students shall be permitted to register the courses under Honors.
- ii) The selected students shall submit their willingness to the principal through his/her parent department offering Honors. The parent department shall maintain the record of student pursuing the Honors.
- iii) The students enrolled in the Honors courses will be monitored continuously. An advisor/mentor from parent department shall be assigned to a group of students to monitor the progress.
- iv) There is no fee for registration of subjects for Honors program offered in offline at the respective institutions.

17. Attendance Requirements:

- i) A student shall be eligible to appear for the University external examinations if he/she acquires a minimum of 40% attendance in each subject and 75% of attendance in aggregate of all the subjects. Condonation of shortage of attendance in aggregate up to 10% (65% and above and below 75%) in each semester may be granted by the College Academic Committee.
- ii) Shortage of Attendance below 65% in aggregate shall in NO CASE be condoned.
- iii) A stipulated fee shall be payable towards condonation of shortage of attendance to the University.
- iv) Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that class and their registration shall stand cancelled.
- v) A student will not be promoted to the next semester unless he satisfies the attendance requirements of the present semester. They may seek readmission for that semester from the date of commencement of class work.
- vi) If any candidate fulfils the attendance requirement in the present semester, he shall not be eligible for readmission into the same class.
- vii) If the learning is carried out in blended mode (both offline & online), then the total attendance of the student shall be calculated considering the offline and online attendance of the student.
- viii) For induction programme attendance shall be maintained as per AICTE norms.

18. Promotion Rules:

The following academic requirements must be satisfied in addition to the attendance requirements mentioned in section 16.

- i) A student shall be promoted from first year to second year if he/she fulfils the minimum attendance requirement as per university norms.
- ii) A student will be promoted from II to III year if he/she fulfils the academic requirement of securing 40% of the credits (any **decimal** fraction should be **rounded off** to **lower** digit) up to in the subjects that have been studied up to III semester.
- iii) A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any **decimal** fraction should be **rounded off** to **lower** digit) in the subjects that have been studied up to V semester.

And in case a student is detained for want of credits for a particular academic year by ii) & iii) above, the student may make up the credits through supplementary examinations and only after securing the required credits he/she shall be permitted to join in the V semester or VII semester respectively as the case may be.

- iv) When a student is detained due to lack of credits/shortage of attendance he/she may be re-admitted when the semester is offered after fulfilment of academic regulations. In such case, he/she shall be in the academic regulations into which he/she is readmitted.

19. Grading:

As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades and corresponding percentage of marks shall be followed:

After each course is evaluated for 100 marks, the marks obtained in each course will be converted to a corresponding letter grade as given below, depending on the range in which the marks obtained by the student fall.

Structure of Grading of Academic Performance

Range in which the marks in the subject fall	Grade	Grade points
		Assigned
90 & above	S (Superior)	10
80 - 89	A (Excellent)	9
70 - 79	B (Very Good)	8
60 - 69	C (Good)	7
50 - 59	D (Average)	6
40 - 49	E (Pass)	5
< 40	F (Fail)	0
Absent	Ab (Absent)	0

- i) A student obtaining Grade 'F' or Grade 'Ab' in a subject shall be considered failed and will be required to reappear for that subject when it is offered the next supplementary examination.
- ii) For non-credit audit courses, "Satisfactory" or "Unsatisfactory" shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA/Percentage.

Computation of Semester Grade Point Average (SGPA) and Cumulative GradePoint Average (CGPA):

The Semester Grade Point Average (SGPA) is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the courses taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.,

$$SGPA = \frac{\sum (C_i \times G_i)}{\sum C_i}$$

where, C_i is the number of credits of the i^{th} subject and G_i is the grade point scored by the student in the i^{th} course.

The Cumulative Grade Point Average (CGPA) will be computed in the same manner considering all the courses undergone by a student over all the semesters of a program, i.e.,

$$CGPA = \frac{\sum (C_i \times S_i)}{\sum C_i}$$

where " S_i " is the SGPA of the i^{th} semester and C_i is the total number of credits up to that semester.

Both SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

While computing the SGPA the subjects in which the student is awarded Zero grade points will also be included.

Grade Point: It is a numerical weight allotted to each letter grade on a 10-point scale.

Letter Grade: It is an index of the performance of students in a said course. Grades are denoted by the letters S, A, B, C, D and F.

Award of Class:

After a student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes:

Class Awarded	CGPA Secured
First Class with Distinction	≥ 7.5
First Class	$\geq 6.5 < 7.5$
Second Class	$\geq 5.5 < 6.5$
Pass Class	$\geq 5.0 < 5.5$

CGPA to Percentage conversion Formula – $(CGPA - 0.5) \times 10$

20. With-holding of Results

If the candidate has any dues not paid to the university or if any case of indiscipline or malpractice is pending against him/her, the result of the candidate shall be withheld in such cases.

21. Multiple Entry / Exit Option

(a) Exit Policy:

The students can choose to exit the four-year programme at the end of first/second/third year.

- i) **UG Certificate in (Field of study/discipline)** - Programme duration: First year (first two semesters) of the undergraduate programme, 40 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-credit job-specific internship/ apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.
- ii) **UG Diploma (in Field of study/discipline)** - Programme duration: First two years (first four semesters) of the undergraduate programme, 80 credits followed by an additional exit 10-credit bridge course(s) lasting two months, including at least 6-credit job-specific internship/ apprenticeship that would help the candidates acquire job-ready competencies required to enter the workforce.
- iii) **Bachelor of Science (in Field of study/discipline) i.e., B.Sc. Engineering in (Field of study/discipline)**- Programme duration: First three years (first six semesters) of the undergraduate programme, 120 credits.

(b) Entry Policy:

Modalities on multiple entry by the student into the B.Tech. programme will be provided in due course of time.

Note: The Universities shall resolve any issues that may arise in the implementation of Multiple Entry and Exit policies from time to time and shall review the policies in the light of periodic changes brought by UGC, AICTE and State government.

22. Gap Year Concept:

Gap year concept for Student Entrepreneur in Residence is introduced and outstanding students who wish to pursue entrepreneurship / become entrepreneur are allowed to take a break of one year at any time after II year to pursue full-time entrepreneurship programme/to establish startups. This period may be extended to two years at the most and these two years would not be counted for the time for the maximum time for graduation. The principal of the respective college shall forward such proposals submitted by the students to the University. An evaluation committee constituted by the University shall evaluate the proposal submitted by the student and the committee shall decide whether to permit the student(s) to avail the Gap Year or not

23. Transitory Regulations

Discontinued, detained, or failed candidates are eligible for readmission as and when the semester is offered after fulfilment of academic regulations. Candidates who have been detained for want of attendance or not fulfilled academic requirements or who have failed after having undergone the course in earlier regulations or have discontinued and wish to continue the course are eligible for admission into the unfinished semester from the date of commencement of class work with the same or equivalent subjects as and when subjects are offered, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

Candidates who are permitted to avail Gap Year shall be eligible for re-joining into the succeeding year of their B. Tech from the date of commencement of class work, subject to Section 2 and they will follow the academic regulations into which they are readmitted.

24. Minimum Instruction Days for a Semester:

The minimum instruction days including exams for each semester shall be 90 days.

25. Medium of Instruction:

The medium of instruction of the entire B. Tech undergraduate programme in Engineering & Technology (including examinations and project reports) will be in English only.

26. Student Transfers:

Student transfers shall be as per the guidelines issued by the Government of Andhra Pradesh and the Universities from time to time.

27. General Instructions:

- i. The academic regulations should be read as a whole for purpose of any interpretation.
- ii. Malpractices rules-nature and punishments are appended.
- iii. Where the words “he”, “him”, “his”, occur in the regulations, they also include “she”, “her”, “hers”, respectively.
- iv. In the case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Vice-Chancellor is final.
- v. The Universities may change or amend the academic regulations or syllabi at any time and the changes or amendments shall be made applicable to all the students on rolls with effect from the dates notified by the Universities.
- vi. In the case of any doubt or ambiguity in the interpretation of the guidelines given, the decision of the Vice-Chancellor / Head of the institution is final.

*** **

ACADEMIC REGULATIONS (R23)**FOR B.TECH. (LATERAL ENTRY SCHEME)**

(Effective for the students admitted into II year through Lateral Entry Scheme from the Academic Year 2024-25 onwards)

1. Award of the Degree

- (a) Award of the B.Tech. Degree / B.Tech. Degree with a Minor if he/she fulfils the following:
 - (i) Pursues a course of study for not less than three academic years and not more than six academic years. However, for the students availing Gap year facility this period shall be extended by two years at the most and these two years would in addition to the maximum period permitted for graduation (Six years).
 - (ii) Registers for 120 credits and secures all 120 credits.
- (b) **Award of B.Tech. degree with Honors** if he/she fulfils the following:
 - (i) Student secures additional 15 credits fulfilling all the requisites of a B.Tech. program i.e., 120 credits.
 - (ii) Registering for Honors is optional.
 - (iii) Honors is to be completed simultaneously with B.Tech. programme.

- 2. Students, who fail to fulfil the requirement for the award of the degree within six consecutive academic years from the year of admission, shall forfeit their seat.

3. Minimum Academic Requirements

The following academic requirements have to be satisfied in addition to the requirements mentioned in item no.2

- i. A student shall be deemed to have satisfied the minimum academic requirements and earned the credits allotted to each theory, practical, design, drawing subject or project if he secures not less than 35% of marks in the end examination and a minimum of 40% of marks in the sum total of the mid semester evaluation and end examination taken together.
- ii. A student shall be promoted from III year to IV year if he/she fulfils the academic requirements of securing 40% of the credits (any decimal fraction should be rounded off to lower digit) in the subjects that have been studied up to V semester.

And in case if student is already detained for want of credits for particular academic year, the student may make up the credits through supplementary exams of the above exams before the commencement of IV year I semester class work of next year.

4. Course Pattern

- i) The entire course of study is three academic years on semester pattern.
 - ii) A student eligible to appear for the end examination in a subject but absent at it or has failed in the end examination may appear for that subject at the next supplementary examination offered.
 - iii) When a student is detained due to lack of credits/shortage of attendance the student may be re-admitted when the semester is offered after fulfilment of academic regulations, the student shall be in the academic regulations into which he/she is readmitted.
5. All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

Course Structure & Syllabus

SEMESTER – I

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23CMT01	Basic Civil & Mechanical Engineering	ESC	3	0	0	3
2	V23CHT01	Applied Chemistry	BS&H	3	0	0	3
3	V23MAT01	Linear Algebra & Calculus	BS&H	3	0	0	3
4	V23CST01	Introduction to Programming	ESC	3	0	0	3
5	V23MET01	Engineering Graphics	ESC	1	0	4	3
6	V23CSL02	IT Workshop Lab	ESC	0	0	2	1
7	V23MEL01	Engineering Workshop Lab	ESC	0	0	3	1.5
8	V23CHL01	Applied Chemistry Lab	BS&H	0	0	2	1
9	V23CSL01	Computer Programming Lab	ESC	0	0	3	1.5
10	V23SPT01	NSS/NCC/Scouts & Guides/Community Service	BS&H	-	-	1	0.5
11	-	English Certificate Course	-	2	0	0	0
Total				15	0	15	20.5

SEMESTER – II

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23MAT02	Differential Equations & Vector Calculus	BS&H	3	0	0	3
2	V23PHT01	Engineering Physics	BS&H	3	0	0	3
3	V23ENT01	Communicative English	BS&H	2	0	0	2
4	V23EET01	Basic Electrical and Electronics Engineering	ESC	3	0	0	3
5	V23CST02	Data Structures	PCC	3	0	0	3
6	V23PHL01	Engineering Physics Lab	BS&H	0	0	2	1
7	V23ENL01	Communicative English Lab	BS&H	0	0	2	1
8	V23EEL01	Electrical and Electronics Engineering Workshop Lab	ESC	0	0	3	1.5
9	V23CSL03	Data Structures Lab	PCC	0	0	3	1.5
10	V23SPT02	Health and wellness, Yoga and sports	BS&H	-	-	1	0.5
Total				14	0	11	19.5

SEMESTER-III

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23MAT05	Discrete Mathematics & Graph Theory	BS&H	3	0	0	3
2	V23MBT51	Managerial Economics and Financial Analysis	Management Course- I	2	0	0	2
3	V23CST03	Digital Logic & Computer Organization	ESC	3	0	0	3
4	V23CST04	Advanced Data Structures & Algorithm Analysis	PCC	3	0	0	3
5	V23CST05	Object Oriented Programming Through Java	PCC	3	0	0	3
6	V23CSL04	Advanced Data Structures and Algorithm Analysis Lab	PCC	0	0	3	1.5
7	V23CSL05	Object Oriented Programming Through Java Lab	PCC	0	0	3	1.5
8	V23CSSE01	Python Programming Lab	SEC	0	1	2	2
9	V23MET09	Design Thinking & Innovation	BS&H	1	0	2	2
10	V23ENT02	Professional Communication Skills - I	Audit Course	2	0	0	0
Total				17	1	10	21

SEMESTER - IV

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23MBT53	Universal Human Values	BS&H	2	1	0	3
2	V23MAT07	Probability & Statistics	BS&H	3	0	0	3
3	V23CST06	Operating Systems	PCC	3	0	0	3
4	V23CST07	Database Management Systems	PCC	3	0	0	3
5	V23CST08	Software Engineering	PCC	2	1	0	3
6	V23CSL06	Operating Systems Lab	PCC	0	0	3	1.5
7	V23CSL07	Database Management Systems Lab	PCC	0	0	3	1.5
8	V23CSSE02	Full Stack Development –I	SEC	0	1	2	2
9	V23CEAC01	Environmental Science	Audit Course	2	0	0	-
10	V23ENT03	Professional Communication Skills - II	Audit Course	0	2	0	0
Total				15	5	8	20
Mandatory Community Service Project (V23COSP01) Internship of 08 weeks duration during summer vacation							
		Minor Course (Student may select from the specialized minors pool)	MC	3	0	3	
		Honors Course (Student may select from the honors pool)	HC	3	0	3	

SEMESTER-V

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23CST09	Data Warehousing and Data Mining	PCC	3	0	0	3
2	V23CST10	Computer Networks	PCC	3	0	0	3
3	V23CST11	Automata and Compiler Design	PCC	3	0	0	3
4	Professional Elective-I		PEC	3	0	0	3
	V23CSTPE01	1. Object Oriented Analysis and Design					
	V23CSTPE02	2. Artificial Intelligence					
	V23CSTPE03	3. Microprocessors & Microcontrollers					
	V23CSTPE04	4. Quantum Computing					
		5. 12 week MOOC Swayam/NPTEL course recommended by the BoS					
5	Open Elective -I		OEC	3	0	0	3
		Offline / 12 week MOOC Swayam/NPTEL course recommended by the BoS					
		Entrepreneurship Development & Venture Creation					
6	V23CSL08	Data Mining Lab	PCC	0	0	3	1.5
7	V23CSL09	Computer Networks Lab	PCC	0	0	3	1.5
8	V23CSSE03	Full Stack development-2	SEC	0	1	2	2
9	Engineering Science Course		ESC	0	0	2	1
	V23CSES01	Master Coding and Competitive Programming - Part-1					
	V23CSES02	User Interface Design using Flutter					
		SWAYAM Plus - Android Application Development (with Flutter)					
10	V23CSP01	Evaluation of Community Service Internship	CSI	-	-	-	2
11	V23ENT04	English for Employability(EE)		-	-	-	-
Total				15	01	10	23
		Minor Course (Student may select from the same specialized minors pool)	MC	3	0	0	3
		Minor Course (Student may select from the same specialized minors pool)	MC	3	0	0	3
		Honors Course (Student may select from the same honors pool)	HC	3	0	0	3
		Honors Course (Student may select from the same honors pool)	HC	3	0	0	3

SEMESTER-VI

S.No.	Course Code	Name of the Course		L	T	P	C
1	V23CST12	Machine Learning	PCC	3	0	0	3
2	V23CST13	Cloud Computing	PCC	3	0	0	3
3	V23CST14	Cryptography & Network Security	PCC	3	0	0	3
4	Professional Elective-II		PEC	3	0	0	3
	V23CSTPE05	1. Software Testing Methodologies					
	V23CSTPE06	2. Cyber Security					
	V23CSTPE07	3. DevOps					
	V23CSTPE08	4. Mobile Adhoc Networks					
		5. 12 week MOOC Swayam/NPTEL course recommended by the BoS					
5	Professional Elective-III		PEC	3	0	0	3
	V23CSTPE09	1. Software Project Management					
	V23CSTPE10	2. Natural Language Processing					
	V23CSTPE11	3. Big Data Analytics					
	V23CSTPE12	4. Distributed Operating System					
		5. 12 week MOOC Swayam/NPTEL course recommended by the BoS					
6	Open Elective -II		OEC	3	0	0	3
		Offline / 12 week MOOC Swayam/NPTEL course recommended by the BoS					
7	V23CSL10	Cloud Computing Lab	PCC	0	0	3	1.5
8	V23CSL11	Machine Learning Lab using Python	PCC	0	0	3	1.5
9	V23CSSE04	Master Coding and Competitive Programming - Part-2	SEC	0	1	2	2
		SWAYAM Plus - 21st Century Soft Skills					
10		Technical Paper Writing & IPR	Audit Course	2	0	0	-
Total				20	01	08	23
		Minor Course (Student may select from the same specialized minors pool)	MC	3	0	0	3
		Minor Course (Student may select from the same specialized minors pool)	MC	3	0	0	3
		Honors Course (Student may select from the honors pool)	HC	3	0	0	3
		Honors Course (Student may select from the honors pool)	HC	3	0	0	3

* Under Industry Internship interested students can pursue SWAYAM Plus courses viz., Hands-on Masterclass on Data Analytics OR Artificial Intelligence for Real-World Application

SYLLABUS

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CMT01
Name of the Course	Basic Civil and Mechanical Engineering					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

PART-A:

- CO1:** Apply knowledge of fundamental laws. to analyze KCL, KVL, series & parallel circuits, ac fundamentals, average, RMS values. (K3)
- CO2:** Understand the construction and operation of AC and DC Electrical machines, measuring instruments (PMMC, MI). (K2)
- CO3:** Understand the various sources of energy, types of power plants and layout representation of electrical power systems, electricity bill calculations and safety measures. (K2)

UNIT-I: Basics of Civil Engineering: Role of Civil Engineers in Society- Various Disciplines of Civil Engineering- Structural Engineering- Geo-technical Engineering- Transportation Engineering - Hydraulics and Water Resources Engineering - Environmental Engineering- Scope of each discipline - Building Construction and Planning- Construction Materials- Cement - Aggregate - Bricks- Cement concrete- Steel. Introduction to Prefabricated construction Techniques.

UNIT-II: Surveying: Objectives of Surveying- Horizontal Measurements- Angular Measurements- Introduction to Bearings Levelling instruments used for levelling - Simple problems on levelling and bearings- Contour mapping.

UNIT-III: Transportation Engineering Importance of Transportation in Nation's economic development- Types of Highway Pavements- Flexible Pavements and Rigid Pavements - Simple Differences. Basics of Harbour, Tunnel, Airport, and Railway Engineering.

Water Resources and Environmental Engineering: Introduction, Sources of water- Quality of water- Specifications- Introduction to Hydrology- Rainwater Harvesting- Water Storage and Conveyance Structures (Simple introduction to Dams and Reservoirs).

Textbooks:

1. Basic Civil Engineering, M.S. Palanisamy, , Tata McGraw Hill publications (India) Pvt. Ltd. Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers. 2022. First Edition.
3. Basic Civil Engineering, Satheesh Gopi, Pearson Publications, 2009, First Edition.

Reference Books:

1. Surveying, Vol- I and Vol-II, S.K. Duggal, Tata McGraw Hill Publishers 2019. Fifth Edition.
2. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi. 2016
3. Irrigation Engineering and Hydraulic Structures - Santosh Kumar Garg, Khanna Publishers, Delhi 2023. 38th Edition.
4. Highway Engineering, S.K. Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.
5. Indian Standard DRINKING WATER — SPECIFICATION IS 10500-2012.

PART B: BASIC MECHANICAL ENGINEERING

Course Outcomes: On completion of the course, the student should be able to

CO4: Understand the semiconductor devices –Characteristics of diodes, transistors. (K2)

CO5: Understand the basic electronic circuits - Rectifiers and power supplies, Amplifiers & Instrumentation. (K2)

CO6: Apply knowledge of Number Systems, Logic gates, Boolean algebra, Truth Tables to solve Simple combinational circuit. (K3)

UNIT-I: Introduction to Mechanical Engineering: Role of Mechanical Engineering in Industries and Society- Technologies in different sectors such as Energy, Manufacturing, Automotive, Aerospace, and Marine sectors. Engineering Materials - Metals-Ferrous and Non-ferrous, Ceramics, Composites, Smart materials.

UNIT-II: Manufacturing Processes: Principles of Casting, Forming, joining processes, Machining, Introduction to CNC machines, 3D printing, and Smart manufacturing. Thermal Engineering – working principle of Boilers, Otto cycle, Diesel cycle, Refrigeration and air-conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

UNIT-III Power plants: working principle of Steam, Diesel, Hydro, Nuclear power plants. Mechanical Power Transmission - Belt Drives, Chain, Rope drives, Gear Drives and their applications. Introduction to Robotics - Joints & links, configurations, and applications of robotics.

(Note: The subject covers only the basic principles of Civil and Mechanical Engineering systems. The evaluation shall be intended to test only the fundamentals of the subject)

Textbooks:

1. Internal Combustion Engines by V.Ganesan, By Tata McGraw Hill publications (India)Pvt. Ltd.
2. A Tear book of Theory of Machines by S.S. Rattan, Tata McGraw Hill Publications,(India) Pvt. Ltd.
3. An introduction to Mechanical Engg by Jonathan Wicker and Kemper Lewis, Cengage learning India Pvt. Ltd.

Reference Books:

1. Appuu Kuttan KK, Robotics, I.K. International Publishing House Pvt. Ltd. Volume-I
2. 3D printing & Additive Manufacturing Technology- L. Jyothish Kumar, Pulak MPandey, Springer publications
3. Thermal Engineering by Mahesh M Rathore Tata McGraw Hill publications (India) Pvt.Ltd.
4. G. Shanmugam and M.S.Palanisamy, Basic Civil and the Mechanical Engineering, Tata McGraw Hill publications (India) Pvt. Ltd.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CHT01
Name of the Course	Applied Chemistry					
Branch	(Common to EEE, ECE, CSE, IT) & allied branches)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Solution for the particle in the one dimensional box and applications of MO theory. (K3)

CO2: Classify the advanced materials and its applications in Technology. (K3)

CO3: Associate concepts of Electro chemical sensors and designing Electro chemical energy storages . (K3)

CO4: Summarize the importance of polymers and conducting polymers. (K3)

CO5: Compile the Advance spectroscopy and its applications. (K3)

UNIT-I :Structure and Bonding Models: Fundamentals of Quantum mechanics, Schrodinger Wave equation, significance of Ψ and Ψ^2 , particle in one dimensional box, molecular orbital theory – bonding in homo- and heteronuclear diatomic molecules – energy level diagrams of O₂ and CO, etc. π -molecular orbitals of butadiene and benzene, calculation of bond order.

UNIT-II: Modern Engineering materials Semiconductors : Introduction, basic concept, application Super conductors-Introduction basic concept, applications. **Super capacitors:** Introduction, Basic Concept-Classification – Applications.

Nano materials: Introduction, classification, properties and applications of Fullerenes, carbon nano tubes and Graphines nanoparticles.

UNIT-III: Electrochemistry and Applications: Electrochemical cell, Nernst equation, cell potential calculations and numerical problems, potentiometry- potentiometric titrations (redox titrations), concept of conductivity, conductivity cell, conductometric titrations (acid-base titrations). Electrochemical sensors – potentiometric sensors with examples, amperometric sensors with examples. Primary cells – Zinc-air battery, Secondary cells –lithium-ion batteries- working of the batteries including cell reactions; Fuel cells, hydrogen-oxygenfuel cell– working of the cells. Polymer Electrolyte Membrane Fuel cells (PEMFC).

UNIT-IV: Polymer Chemistry: Introduction to polymers, functionality of monomers, chain growth and step growth polymerization, coordination polymerization, with specific examples and mechanisms of polymer formation.

Plastics –Thermo and Thermosetting plastics, Preparation, properties and applications of – PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres. Elastomers–Buna-S, Buna-N–preparation, properties and applications.

Conducting polymers – polyacetylene, polyaniline, – mechanism of conduction and applications. Bio-Degradable polymers - Poly Glycolic Acid (PGA), Polyl Lactic Acid (PLA).

UNIT-V: Instrumental Methods and Applications: Electromagnetic spectrum. Absorption of radiation: Beer-Lambert's law. UV-Visible Spectroscopy, electronic transition, Instrumentation, IR spectroscopies, fundamental modes and selection rules, Instrumentation. Chromatography-Basic Principle, Classification-HPLC: Principle, Instrumentation and Applications.

Textbooks:

1. Jain and Jain, Engineering Chemistry, 16/e, DhanpatRai, 2013.
2. Peter Atkins, Julio de Paula and James Keeler, Atkins' Physical Chemistry, 10/e, Oxford University Press, 2010.

Reference Books:

1. Skoog and West, Principles of Instrumental Analysis, 6/e, Thomson, 2007.
2. J.D. Lee, Concise Inorganic Chemistry, 5th Edition, Wiley Publications, Feb.2008
3. Textbook of Polymer Science, Fred W. Billmayer Jr, 3rd Edition.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23MAT01
Name of the Course	Linear Algebra & Calculus					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Apply matrix algebra techniques to solve engineering problems. (K3)

CO2: Use Eigen values and Eigen vectors concept to find nature of quadratic form, inverse and powers of matrix. (K3)

CO3: Apply mean value theorems to real life problems. (K3)

CO4: Use the concepts of functions of several variables which is useful in optimization. (K3)

CO4: Apply double and triple integrals of functions of several variables in two dimensions using Cartesian and polar coordinates and in three dimensions using cylindrical and spherical coordinates. (K3)

UNIT-I: Matrices: Rank of a matrix by echelon form, normal form. Cauchy–Binet formulae (without proof). Inverse of Non- singular matrices by Gauss-Jordan method, System of linear equations: Solvingsystem of Homogeneous and Non-Homogeneous equations by Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods.

UNIT-II: Eigenvalues, Eigenvectors and Orthogonal Transformation: Eigenvalues, Eigenvectors and their properties, Diagonalization of a matrix, Cayley-Hamilton Theorem (without proof), finding inverse and power of a matrix by Cayley-Hamilton Theorem, Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-II: Calculus: Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem with their geometrical interpretation, Cauchy's mean value theorem, Taylor's and Maclaurin theorems with remainders (without proof), Problems and applications on the above theorems.

UNIT-IV: Partial differentiation and Applications (Multi variable calculus): Functions of several variables: Continuity and Differentiability, Partial derivatives, total derivatives, chain rule, Directional derivative, Taylor's and Maclaurin's series expansion of functions of two variables. Jacobians, Functional dependence, maxima and minima of functions of two variables, method of Lagrange multipliers.

UNIT-V: Multiple Integrals (Multi variable Calculus): Double integrals, triple integrals, change of order of integration, change of variables to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, Micheal Greenberg, , Pearson publishers, 9th edition
5. Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications, 2014, Third Edition (Reprint 2021).

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST01
Name of the Course	Introduction To Programming					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Understand basics of computers, the concept of algorithm and algorithmic thinking. **(K2)**

CO2: Analyse a problem and develop an algorithm to solve it. **(K3)**

CO3: Implement various algorithms using the C programming language. **(K3)**

CO4: Understand more advanced features of C language. **(K2)**

CO5: Develop problem-solving skills and the ability to debug and optimize the code. **(K3)**

UNIT- I: Introduction to Programming and Problem Solving: History of Computers, Basic organization of a computer: ALU, input-output units, memory, program counter, Introduction to Programming Languages, Basics of a Computer Program- Algorithms, flowcharts (Using Dia Tool), pseudo code. Introduction to Compilation and Execution, Primitive Data Types, Variables, and Constants, Basic Input and Output, Operations, Type Conversion, and Casting.

Problem solving techniques: Algorithmic approach, characteristics of algorithm, Problem solving strategies: Top-down approach, Bottom-up approach, Time and space complexities of algorithms.

UNIT-II: Control Structures: Simple sequential programs Conditional Statements (if, if-else, switch), Loops (for, while, do- while) Break and Continue.

UNIT-III: Arrays and Strings: Arrays indexing, memory model, programs with array of integers, two dimensional arrays, Introduction to Strings.

UNIT-IV: Pointers & User Defined Data types: Pointers, dereferencing and address operators, pointer and address arithmetic, array manipulation using pointers, User-defined data types-Structures and Unions.

UNIT-V: Functions & File Handling: Introduction to Functions, Function Declaration and Definition, Function call Return Types and Arguments, modifying parameters inside functions using pointers, arrays as parameters. Scope and Lifetime of Variables, Basics of File Handling

Note: The syllabus is designed with C Language as the fundamental language of implementation.

Textbooks:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice- Hall, 1988
2. Schaum's Outline of Programming with C, Byron S Gottfried, McGraw-Hill Education, 1996

Reference Books:

1. Computing fundamentals and C Programming, Balagurusamy, E., McGraw-Hill Education, 2008.
2. Programming in C, Rema Theraja, Oxford, 2016, 2nd edition
3. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE, 3rd edition.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	1	0	4	3	V23MET01
Name of the Course	Engineering Graphics					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the principles of engineering drawing, including engineering curves, scales, orthographic and isometric projections. (K3)
- CO2:** Draw and interpret orthographic projections of points, lines, planes and solids in front, top and side views. (K3)
- CO3:** Understand and draw projection of solids in various positions in first quadrant. (K3)
- CO4:** Explain principles behind development of surfaces. (K3)
- CO5:** Prepare isometric and perspective sections of simple solids. (K3)

UNIT- I: Introduction: Lines, Lettering and Dimensioning, Geometrical Constructions and Constructing regular polygons by general methods.

Curves: construction of ellipse, parabola and hyperbola by general, Cycloids, Involute, Normal and tangent to Curves. **Scales:** Plain scales, diagonal scales and vernier scales.

UNIT- II: Orthographic Projections: Reference plane, importance of reference lines or Plane, Projections of a point situated in any one of the four quadrants.

Projections of Straight Lines: Projections of straight lines parallel to both reference planes, perpendicular to one reference plane and parallel to other reference plane, inclined to one reference plane and parallel to the other reference plane. Projections of Straight Line Inclined to both the reference planes

projections of Planes: regular planes Perpendicular to both reference planes, parallel to one reference plane and inclined to the other reference plane; plane inclined to both the reference planes.

UNIT- III: Projections of Solids: Types of solids: Polyhedra and Solids of revolution. Projections of solids in simple positions: Axis perpendicular to horizontal plane, Axis perpendicular to vertical plane and Axis parallel to both the reference planes, Projection of Solids with axis inclined to one reference plane and parallel to another plane.

UNIT- IV: Sections of Solids: Perpendicular and inclined section planes, Sectional views and True shape of section, Sections of solids in simple position only.

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT- V: Conversion of Views: Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Computer graphics: Creating 2D&3D drawings of objects including PCB and Transformations using AutoCAD (*Not for end examination*).

Textbook:

1. N. D. Bhatt, Engineering Drawing, Charotar Publishing House, 2016.

Reference Books:

1. Engineering Drawing, K.L. Narayana and P. Kanniah, Tata McGraw Hill, 2013.
2. Engineering Drawing, M.B. Shah and B.C. Rana, Pearson Education Inc, 2009.
3. Engineering Drawing with an Introduction to AutoCAD, Dhananjay Jolhe, Tata McGraw Hill, 2017.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23CSL02
Name of the Course	IT Workshop					
Branch	(Common to all branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate Hardware troubleshooting (K3)
CO2: Understand Hardware components and inter dependencies. (K2)
CO3: Employ safeguard measures of computer systems from viruses/worms. (K3)
CO4: Report Document/ Presentation preparation. (K2)
CO5: Compute calculations using spreadsheets. (K3)

PC Hardware & Software Installation

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Also students need to go through the video which shows the process of assembling a PC. A video would be given as part of the course content.

Task 3: Every student should individually install MS windows on the personal computer. Lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux on the computer. This computer should have windows installed. The system should be configured as dual boot (VMWare) with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Task 5: Every student should install BOSS on the computer. The system should be configured as dual boot (VMWare) with both Windows and BOSS. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process they configure the TCP/IP setting. Finally students should demonstrate, to the instructor, how to access the websites and email. If there is no internet connectivity preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use the search engines. A few topics would be given to the students for which they need to search on Google. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computer to be safe on the internet. They need to customize their browsers to block pop ups, block active x downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of La TeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of La TeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using La TeXand word – Accessing, overview of toolbars, saving files, Usinghelp and resources, rulers, format painter in word.

Task 2: Using La TeX and Word to create a project certificate. Features to be covered:- Formatting Fonts in word, Drop Cap in word, Applying Text effects, Using Character Spacing, Borders and Colors, Inserting Header and Footer, Using Date and Time option in both La TeXand Word.

Task 3: Creating project abstract Features to be covered:-Formatting Styles, Inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered:- Table of Content, Newspaper columns, Images from files and clipart, Drawing toolbar and Word Art, Formatting Images, Textboxes, Paragraphs and Mail Merge in word.

EXCEL

Excel Orientation: The mentor needs to tell the importance of MS office or equivalent (FOSS)tool Excel as a Spreadsheet tool, give the details of the four tasks and features that would be covered in each. Using Excel – Accessing, overview of toolbars, saving excel files, Using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text

Task 2: Calculating GPA -. Features to be covered:- Cell Referencing, Formulae in excel – average, std. deviation, Charts, Renaming and Inserting worksheets, Hyper linking, Count function,

LOOKUP/VLOOKUP

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, Conditional formatting

POWER POINT

Task 1: Students will be working on basic power point utilities and tools which help them create basic power point presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive presentations - Hyperlinks, Inserting –Images, Clip Art, Audio, Video, Objects, Tables and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

AI TOOLS – ChatGPT

Task 1: Prompt Engineering: Experiment with different types of prompts to see how the model responds. Try asking questions, starting conversations, or even providing incomplete sentences to see how the model completes them.

- Ex: Prompt: "You are a knowledgeable AI. Please answer the following question: What is the capital of France?"

Task 2: Creative Writing: Use the model as a writing assistant. Provide the beginning of a story or a description of a scene, and let the model generate the rest of the content. This can be a fun way to brainstorm creative ideas

- Ex: Prompt: "In a world where gravity suddenly stopped working, people started floating upwards. Write a story about how society adapted to this new reality."

Task 3: Language Translation: Experiment with translation tasks by providing a sentence in one language and asking the model to translate it into another language. Compare the output to see how accurate and fluent the translations are.

- Ex: Prompt: "Translate the following English sentence to French: 'Hello, how are you doing today?'"

Reference Books:

1. Comdex Information Technology course tool kit, Vikas Gupta, WILEY Dream tech, 2003
2. The Complete Computer upgrade and repair book, Cheryl A Schmidt, WILEY Dream tech, 2013, 3rd edition
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education, 2012, 2nd edition
4. PC Hardware - A Handbook, Kate J. Chase, PHI (Microsoft)
5. LaTeX Companion, Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide, David Anfinson and Ken Quamme. – CISCO Press, Pearson Education, 3rd edition
7. IT Essentials PC Hardware and Software Labs and Study Guide, Patrick Regan – CISCO Press, Pearson Education, 3rd edition.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23MEL01
Name of the Course	Engineering Workshop					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Identify workshop tools and their operational capabilities. **(K2)**

CO2: Practice on manufacturing of components using workshop trades including fitting, carpentry, foundry and welding. **(K3)**

CO3: Apply fitting operations in various applications. **(K3)**

CO4: Apply basic electrical engineering knowledge for House Wiring Practice. **(K3)**

SYLLABUS

1. **Demonstration:** Safety practices and precautions to be observed in workshop.
2. **Wood Working:** Familiarity with different types of woods and tools used in wood working and make following joints.
 - a) Half – Lap joint b) Mortise and Tenon joint c) Corner Dovetail joint or Bridlejoint
3. **Sheet Metal Working:** Familiarity with different types of tools used in sheet metal working, Developments of following sheet metal job from GI sheets.
 - a) Tapered tray b) Conical funnel c) Elbow pipe d) Brazing
4. **Fitting:** Familiarity with different types of tools used in fitting and do the following fitting exercises.
 - a) V-fit b) Dovetail fit c) Semi-circular fit d) Bicycle tire puncture and change of two-wheeler tyre
5. **Electrical Wiring:** Familiarity with different types of basic electrical circuits and make the following connections.
 - a) Parallel and series b) Two-way switch c) Godown lighting
 - d) Tube light e) Three phase motor f) Soldering of wires
6. **Foundry Trade:** Demonstration and practice on Moulding tools and processes, Preparation of Green Sand Moulds for given Patterns.
7. **Welding Shop:** Demonstration and practice on Arc Welding and Gas welding. Preparation of Lap joint and Butt joint.
8. **Plumbing:** Demonstration and practice of Plumbing tools, Preparation of Pipe joints with coupling for same diameter and with reducer for different diameters.

Textbooks:

1. Basic Workshop Technology: Manufacturing Process, Felix W.; Independently Published, 2019. Workshop Processes, Practices and Materials; Bruce J. Black, Routledge publishers, 5th Edn. 2015.
2. A Course in Workshop Technology Vol I. & II, B.S. Raghuwanshi, Dhanpath Rai & Co., 2015 & 2017.

Reference Books:

1. Elements of Workshop Technology, Vol. I by S. K. Hajra Choudhury & Others, Media Promoters and Publishers, Mumbai. 2007, 14th edition
2. Workshop Practice by H. S. Bawa, Tata-McGraw Hill, 2004.
3. Wiring Estimating, Costing and Contracting; Soni P.M. & Upadhyay P.A.; Atul Prakashan, 2021-22.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23CHL01
Name of the Course	Applied Chemistry Lab					
Branch	(Common to EEE, ECE, CSE, IT & allied branches)					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

- | | |
|---|-------------|
| CO1: Determine the cell constant and conductance of solutions. | (K4) |
| CO2: Prepare advanced polymer Bakelite materials. | (K3) |
| CO3: Measure the strength of an acid present in secondary batteries. | (K5) |
| CO4: Analyse the IR spectra of some organic compounds. | (K4) |
| CO5: Calculate strength of acid in Pb-Acid battery. | (K4) |

List of Experiments:

1. Measurement of 10Dq by spectrophotometric method
2. Conductometric titration of strong acid vs. strong base
3. Conductometric titration of weak acid vs. strong base
4. Determination of cell constant and conductance of solutions
5. Potentiometry - determination of redox potentials and emfs
6. Determination of Strength of an acid in Pb-Acid battery
7. Preparation of a Bakelite
8. Verify Lambert-Beer's law
9. Wavelength measurement of sample through UV-Visible Spectroscopy
10. Identification of simple organic compounds by IR
11. Preparation of nanomaterials by precipitation method
12. Estimation of Ferrous Iron by Dichrometry

Reference:

- "Vogel's Quantitative Chemical Analysis 6th Edition 6th Edition" Pearson Publications by J. Mendham, R.C.Denney, J.D.Barnes and B. Sivasankar.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL01
Name of the Course	Computer Programming Lab					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Read, understand, and trace the execution of programs written in C language. **(K3)**

CO2: Select the right control structure for solving the problem. **(K3)**

CO3: Develop C programs which utilize memory efficiently using programming constructs like pointers. **(K3)**

CO4: Develop, Debug and Execute programs to demonstrate the applications of arrays, functions, basic concepts of pointers in C. **(K3)**

WEEK 1

Tutorial 1: Problem-solving using Computers.

Lab1: Familiarization with programming environment

- i) Basic Linux environment and its editors like Vi, Vim & Emacs etc.
- ii) Exposure to Turbo C, gcc
- iii) Writing simple programs using printf(), scanf()

WEEK 2

Tutorial 2: Problem-solving using Algorithms and Flow charts.

Lab 1: Converting algorithms/flow charts into C Source code.

Developing the algorithms/flowcharts for the following sample programs

- i) Sum and average of 3 numbers
- ii) Conversion of Fahrenheit to Celsius and vice versa
- iii) Simple interest calculation

WEEK 3

Tutorial 3: Variable types and type conversions:

Lab 3: Simple computational problems using arithmetic expressions.

- i) Finding the square root of a given number
- ii) Finding compound interest
- iii) Area of a triangle using heron's formulae
- iv) Distance travelled by an object

WEEK 4

Tutorial4: Operators and the precedence and as associativity:

Lab4: Simple computational problems using the operator' precedence and associativity

- i) Evaluate the following expressions.
 - a. $A+B*C+(D*E) + F*G$
 - b. $A/B*C-B+A*D/3$
 - c. $A+++B---A$
 - d. $J= (i++) + (++i)$
- ii) Find the maximum of three numbers using conditional operator
- iii) Take marks of 5 subjects in integers, and find the total, average in float

WEEK 5

Tutorial 5: Branching and logical expressions:

Lab 5: Problems involving if-then-else structures.

- i) Write a C program to find the max and min of four numbers using if-else.
- ii) Write a C program to generate electricity bill.
- iii) Find the roots of the quadratic equation.
- iv) Write a C program to simulate a calculator using switch case.
- v) Write a C program to find the given year is a leap year or not.

WEEK 6

Tutorial 6: Loops, while and for loops

Lab 6: Iterative problems e.g., the sum of series

- i) Find the factorial of given number using any loop.
- ii) Find the given number is a prime or not.
- iii) Compute sine and cos series
- iv) Checking a number palindrome
- v) Construct a pyramid of numbers.

WEEK 7:

Tutorial 7: 1 D Arrays: searching.

Lab 7: 1D Array manipulation, linear search

- i) Find the min and max of a 1-D integer array.
- ii) Perform linear search on 1D array.
- iii) The reverse of a 1D integer array
- iv) Find 2's complement of the given binary number.
- v) Eliminate duplicate elements in an array.

WEEK 8:

Tutorial 8: 2 D arrays, sorting and Strings.

Lab 8: Matrix problems, String operations, Bubble sort

- i) Addition of two matrices
- ii) Multiplication two matrices
- iii) Sort array elements using bubble sort
- iv) Concatenate two strings without built-in functions
- v) Reverse a string using built-in and without built-in string functions

WEEK 9:

Tutorial 9: Pointers, structures and dynamic memory allocation

Lab 9: Pointers and structures, memory dereference.

- i) Write a C program to find the sum of a 1D array using malloc()
- ii) Write a C program to find the total, average of n students using structures
- iii) Enter n students data using calloc() and display failed students list
- iv) Read student name and marks from the command line and display the student details along with the total.
- v) Write a C program to implement realloc()

WEEK 10:

Tutorial 10: Bitfields, Self-Referential Structures, Linked lists

Lab 10 : Bitfields, linked lists

Read and print a date using dd/mm/yyyy format using bit-fields and differentiate the same without using bit- fields

- i) Create and display a singly linked list using self-referential structure.
- ii) Demonstrate the differences between structures and unions using a C program.

- iii) Write a C program to shift/rotate using bitfields.
- iv) Write a C program to copy one structure variable to another structure of the same type.

WEEK 11:

Tutorial 11: Functions, call by value, scope and extent,

Lab 11: Simple functions using call by value, solving differential equations using Euler's theorem.

- i) Write a C function to calculate NCR value.
- ii) Write a C function to find the length of a string.
- iii) Write a C function to transpose of a matrix.
- iv) Write a C function to demonstrate numerical integration of differential equations using Euler's method

WEEK 12:

Tutorial 12: Recursion, the structure of recursive calls

Lab 12: Recursive functions

- i) Write a recursive function to generate Fibonacci series.
- ii) Write a recursive function to find the lcm of two numbers.
- iii) Write a recursive function to find the factorial of a number.
- iv) Write a C Program to implement Ackermann function using recursion.
- v) Write a recursive function to find the sum of series.

WEEK 13:

Tutorial 13: Call by reference, dangling pointers

Lab 13: Simple functions using Call by reference, Dangling pointers.

- i) Write a C program to swap two numbers using call by reference.
- ii) Demonstrate Dangling pointer problem using a C program.
- iii) Write a C program to copy one string into another using pointer.
- iv) Write a C program to find no. of lowercase, uppercase, digits and other characters using pointers.

WEEK 14:

Tutorial 14: File handling

Lab 14: File operations

- i) Write a C program to write and read text into a file.
- ii) Write a C program to write and read text into a binary file using fread() and fwrite()
- iii) Copy the contents of one file to another file.
- iv) Write a C program to merge two files into the third file using command-line arguments.
- v) Find no. of lines, words and characters in a file
- vi) Write a C program to print last n characters of a given file.

Textbooks:

1. Ajay Mittal, Programming in C: A practical approach, Pearson.
2. Byron Gottfried, Schaum's Outline of Programming with C, McGraw Hill

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice-Hall of India
2. C Programming, A Problem-Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE.

Semester	I	L	T	P	C	COURSE CODE
Regulation	V23	0	0	1	0.5	V23SPT01
Name of the Course	NSS/NCC/SCOUTS & GUIDES/COMMUNITY SERVICE					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the importance of discipline, character and service motto. (K2)
- CO2:** Solve some societal issues by applying acquired knowledge, facts, and techniques. (K3)
- CO3:** Examine human relationships by analyzing social problems. (K3)
- CO4:** Determine to extend their help for the fellow beings and downtrodden people. (K4)
- CO5:** Develop leadership skills and civic responsibilities. (K3)

UNIT-I: Orientation: General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance.

Activities:

- i) Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills
- ii) Conducting orientations programs for the students –future plans-activities-releasing road map etc.
- iii) Displaying success stories-motivational biopics- award winning movies on societal issues etc.
- iv) Conducting talent show in singing patriotic songs-paintings- any other contribution.

UNIT-II: Nature & Care Activities:

- i) Best out of waste competition.
- ii) Poster and signs making competition to spread environmental awareness.
- iii) Recycling and environmental pollution article writing competition.
- iv) Organising Zero-waste day.
- v) Digital Environmental awareness activity via various social media platforms.
- vi) Virtual demonstration of different eco-friendly approaches for sustainable living.
- vii) Write a summary on any book related to environmental issues.

UNIT-III: Community Service Activities:

- i) Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media- authorities- experts-etc.
- ii) Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS,
- iii) Conducting consumer Awareness. Explaining various legal provisions etc.
- iv) Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education.
- v) Any other programmes in collaboration with local charities, NGOs etc.

Reference Books:

1. Nirmalya Kumar Sinha & Surajit Majumder, *A Text Book of National Service Scheme* Vol;I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2. *Red Book - National Cadet Corps* – Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3. Davis M. L. and Cornwell D. A., “Introduction to Environmental Engineering”, McGraw Hill, New York 4/e 2008
4. Masters G. M., Joseph K. and Nagendran R. “Introduction to Environmental Engineering and Science”, Pearson Education, New Delhi. 2/e 2007
5. Ram Ahuja. *Social Problems in India*, Rawat Publications, New Delhi.

General Guidelines:

1. Institutes must assign slots in the Timetable for the activities.
2. Institutes are required to provide instructor to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totaling to 90 marks.

A student shall be evaluated by the concerned teacher for 10 marks by conducting vivavoce on the sub

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23MAT02
Name of the Course	Differential Equations and Vector Calculus					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Apply the techniques of ordinary differential equations of first order and first degree related to various engineering fields. (K3)
- CO2:** Apply the techniques of linear differential equations of higher order related to various engineering fields. (K3)
- CO3:** Solve the first order linear PDE and homogeneous PDE by appropriate methods. (K3)
- CO4:** Interpret the physical meaning of gradient, curl and divergence. (K3)
- CO5:** Find the line, surface and volume integrals and their relations using vector calculus techniques. (K3)

UNIT-I: Differential equations of first order and first degree: Linear differential equations – Bernoulli's equations- Exact equations and equations reducible to exact form. Applications: Newton's Law of cooling – Law of natural growth and decay- Electrical circuits.

UNIT-II: Linear differential equations of higher order (Constant Coefficients): Definitions, homogenous and non-homogenous, complimentary function, general solution, particular integral, Wronskian, Method of variation of parameters. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.

UNIT-III: Partial Differential Equations: Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous Linear Partial differential equations with constant coefficients.

UNIT-IV: Vector differentiation: Scalar and vector point functions, vector operator Del, Del applies to scalar point functions- Gradient, Directional derivative, del applied to vector point functions-Divergence and Curl, vector identities.

UNIT-V: Vector integration: Line integral-circulation-work done, surface integral-flux, Green's theorem in the plane(without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and related problems.

Textbooks:

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, 2017, 44th Edition
2. Advanced Engineering Mathematics, Erwin Kreyszig, John Wiley & Sons, 2018, 10th Edition.

Reference Books:

1. Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2. Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3. Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5. Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017.

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23PHT01
Name of the Course	Engineering Physics					
Branch	(Common for all branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Examine the intensity variation of light due to polarization, interference and diffraction. (K3)
CO2: Describe the basics of crystals and their structures. (K2)
CO3: Interpret the various types of polarization of dielectrics and classify the magnetic materials. (K3)
CO4: Explain the basic concepts of Quantum Mechanics and apply it to one dimensional motion of particles. (K2)
CO5: Examine the type of semiconductor using Hall effect. (K3)

UNIT-I : Wave Optics: Interference: Introduction - Principle of superposition –Interference of light - Interference in thin films (Reflection Geometry) & applications - Colours in thin films- Newton's Rings, Determination of wavelength and refractive index.

Diffraction: Introduction - Fresnel and Fraunhofer diffractions - Fraunhofer diffraction due to single slit, double slit & N-slits (Qualitative) – Diffraction Grating - Dispersive power and resolving power of Grating (Qualitative). **Polarization:** Introduction -Types of polarization - Polarization by reflection, refraction and Double refraction - Nicol's Prism -Half wave and Quarter wave plates.

UNIT-II: Crystallography and X-ray diffraction: Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Bravais Lattices – crystal systems (3D) – coordination number - packing fraction of SC, BCC & FCC - Miller indices – separation between successive (hkl) planes.

X-ray diffraction: Bragg's law - X-ray Diffractometer – crystal structure determination by Laue's and powder methods

UNIT-III: Dielectric and Magnetic Materials: Dielectric Materials: Introduction - Dielectric polarization - Dielectric polarizability, Susceptibility, Dielectric constant and Displacement Vector – Relation between the electric vectors - Types of polarizations- Electronic (Quantitative), Ionic (Quantitative) and Orientation polarizations (Qualitative) - Lorentz internal field - Clausius- Mossotti equation - complex dielectric constant – Frequency dependence of polarization – dielectric loss

Magnetic Materials: Introduction - Magnetic dipole moment - Magnetization-Magnetic susceptibility and permeability – Atomic origin of magnetism - Classification of magnetic materials: Dia, para, Ferro, anti-ferro & Ferri magnetic materials - Domain concept for Ferromagnetism & Domain walls (Qualitative) - Hysteresis - soft and hard magnetic materials.

UNIT-IV: Quantum Mechanics and Free electron Theory: Quantum Mechanics: Dual nature of matter – Heisenberg's Uncertainty Principle – Significance and properties of wave function – Schrodinger's time independent and dependent wave equations– Particle in a one-dimensional infinite potential well.

Free Electron Theory: Classical free electron theory (Qualitative with discussion of merits and demerits) – Quantum free electron theory – electrical conductivity based on quantum free electron theory - Fermi-Dirac distribution - Density of states - Fermi energy.

UNIT-V: Semiconductors: Semiconductors: Formation of energy bands – classification of crystalline solids - Intrinsic semiconductors: Density of charge carriers – Electrical conductivity – Fermi level – Extrinsic semiconductors: density of charge carriers – dependence of Fermi energy on carrier concentration and temperature - Drift and diffusion currents – Einstein's equation – Hall effect and its applications.

Textbooks:

1. A Text book of Engineering Physics, M. N. Avadhanulu, P.G.Kshirsagar & TVS ArunMurthy, S. Chand Publications, 11th Edition 2019.
2. Engineering Physics - D.K.Bhattacharya and Poonam Tandon, Oxford press (2015)

Reference Books:

1. Engineering Physics - B.K. Pandey and S. Chaturvedi, Cengage Learning 2021.
2. Engineering Physics - Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018.
3. Engineering Physics” - Sanjay D. Jain, D. Sahasrabudhe and Girish, University Press.2010
4. Engineering Physics - M.R. Srinivasan, New Age international publishers (2009).

Web Resources: <https://www.loc.gov/rr/scitech/selected-internet/physics.html>

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	2	V23ENT01
Name of the Course	Communicative English					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Identify the main idea of the text using skimming and scanning methods of reading and Parts of Speech. Frame Sentences and questions using appropriate Punctuation and Vocabulary. **(K2)**
- CO2:** Interpret the poem and write summary, using appropriate vocabulary. Gain mastery over articles, prepositions and improve writing skills. **(K3)**
- CO3:** Find the success formula after reading the text in detail to answer questions. Use appropriate tense and concord, and get acquainted with summarizing, note-making and paraphrasing. **(K3)**
- CO4:** Employ reading skills to comprehend the given inspirational story. Find suitable vocabulary and format to draft Letters and Resume. **(K3)**
- CO5:** Appraise the delivered lecture and text, recognize the contextual vocabulary, write Essays and prepare Presentations. **(K4)**

UNIT-I: Lesson: HUMAN VALUES: Gift of Magi (Short Story)

- Listening:** Identifying the topic, the context and specific pieces of information by listening to short audio texts and answering a series of questions.
- Speaking:** Asking and answering general questions on familiar topics such as home, family, work, studies and interests; introducing oneself and others.
- Reading:** Skimming to get the main idea of a text; scanning to look for specific pieces of information.
- Writing:** Mechanics of Writing- questions
- Vocabulary:** Synonyms, Antonyms, Affixes (Prefixes/Suffixes), Root words.

UNIT-II: Lesson: NATURE: The Brook by Alfred Tennyson (Poem)

- Listening:** Answering a series of questions about main ideas and supporting ideas after listening to audio texts.
- Speaking:** Discussion in pairs/small groups on specific topics followed by short structure talks.
- Reading:** Identifying sequence of ideas; recognizing verbal techniques that help to link the ideas in a paragraph together.
- Writing:** Structure of a paragraph - Paragraph writing (specific topics) Grammar: Cohesive devices - linkers, use of articles and zero article; prepositions.
- Vocabulary:** Homonyms, Homophones, Homographs.

UNIT-III: Lesson: BIOGRAPHY: Elon Musk

- Listening:** Listening for global comprehension and summarizing what is listened to.
- Speaking:** Discussing specific topics in pairs or small groups and reporting what is discussed
- Reading:** Reading a text in detail by making basic inferences -recognizing and interpreting specific context clues; strategies to use text clues for comprehension.
- Writing:** Summarizing, Note-making, paraphrasing
- Grammar:** Verbs - tenses; subject-verb agreement; Compound words, Collocations
- Vocabulary:** Compound words, Collocations

UNIT-IV: Lesson: INSPIRATION: The Toys of Peace by Saki

- Listening:** Making predictions while listening to conversations/ transactional dialogues without video; listening with video.
- Speaking:** Role plays for practice of conversational English in academic contexts (formal and informal) - asking for and giving information/directions.
- Reading:** Studying the use of graphic elements in texts to convey information, reveal trends/patterns/relationships, communicate processes or display complicated data.
- Writing:** Letter Writing: Official Letters, Resumes
- Grammar:** Reporting verbs, Direct & Indirect speech, Active & Passive Voice
- Vocabulary:** Words often confused, Jargons

UNIT-V: Lesson: MOTIVATION: The Power of Intrapersonal Communication (An Essay)

- Listening:** Identifying key terms, understanding concepts and answering a series of relevant questions that test comprehension.
- Speaking:** Formal oral presentations on topics from academic contexts
- Reading:** Reading comprehension.
- Writing:** Writing structured essays on specific topics.
- Grammar:** Editing short texts –identifying and correcting common errors in grammar and usage (articles, prepositions, tenses, subject verb agreement)
- Vocabulary:** Technical Jargons.

Text Books

1. Pathfinder: Communicative English for Undergraduate Students, 1st Edition, Orient Black Swan, 2023 (Units 1,2 & 3)
2. Empowering with Language by Cengage Publications, 2023 (Units 4 & 5)

Reference Books:

1. Dubey, Sham Ji & Co. English for Engineers, Vikas Publishers, 20202.
2. Bailey, Stephen. Academic writing: A Handbook for International Students. Routledge, 2014.
3. Murphy, Raymond. English Grammar in Use, Fourth Edition, Cambridge University Press, 2019.
4. Lewis, Norman. Word Power Made Easy- The Complete Handbook for Building a Superior Vocabulary. Anchor, 2014.

Web Resources:

GRAMMAR:

1. www.bbc.co.uk/learningenglish
2. <https://dictionary.cambridge.org/grammar/british-grammar/>
3. www.eslpod.com/index.html
4. <https://www.learngrammar.net/>
5. <https://english4today.com/english-grammar-online-with-quizzes/>
6. <https://www.talkenglish.com/grammar/grammar.aspx>

VOCABULARY

1. <https://www.youtube.com/c/DailyVideoVocabulary/videos>
2. https://www.youtube.com/channel/UC4cmBAit8i_NJZE8qK8sfpA

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23EET01
Name of the Course	Basic Electrical & Electronics Engineering					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Apply Knowledge of fundamental laws to analyze AC and DC Electrical circuits. (K3)
CO2: Understand the construction and operation of electrical machines and measuring instruments. (K2)
CO3: Understand the layout of various power plants, electricity bill and safety measures. (K2)
CO4: Understand the working of various semiconductor devices. (K2)
CO5: Understand the basic electronic circuits. (K2)
CO6: Apply knowledge of digital electronics to solve various combinational circuits. (K3)

PART A: BASIC ELECTRICAL ENGINEERING

UNIT-I: DC & AC Circuits

DC Circuits: Electrical circuit elements (R, L and C), Ohm's Law and its limitations, KCL & KVL, series, parallel, series-parallel circuits, Super Position theorem, Simple numerical problems.

AC Circuits: A.C. Fundamentals: Equation of AC Voltage and current, waveform, time period, frequency, amplitude, phase, phase difference, average value, RMS value, form factor, peak factor, Voltage and current relationship with phasor diagrams in R, L, and C circuits, Concept of Impedance, Active power, reactive power and apparent power, Concept of power factor (Simple Numerical problems).

UNIT-II: Machines and Measuring Instruments

Machines: Construction, principle and operation of (i) DC Motor, (ii) DC Generator, (iii) Single Phase Transformer, (iv) Three Phase Induction Motor and (v) Alternator, Applications of electrical machines.

Measuring Instruments: Construction and working principle of Permanent Magnet Moving Coil (PMMC), Moving Iron (MI) Instruments and Wheat Stone bridge.

UNIT-III: Energy Resources, Electricity Bill & Safety Measures

Energy Resources: Conventional and non-conventional energy resources; Layout and operation of various Power Generation systems: Hydel, Nuclear, Solar & Wind power generation.

Electricity bill: Power rating of household appliances including air conditioners, PCs, Laptops, Printers, etc. Definition of "unit" used for consumption of electrical energy, two-part electricity tariff, calculation of electricity bill for domestic consumers.

Equipment Safety Measures: Working principle of Fuse and Miniature circuit breaker(MCB), merits and demerits. Personal safety measures: Electric Shock, Earthing and its types, Safety Precautions to avoid shock.

Textbooks:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Reference Books:

1. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, Mc Graw Hill, 2019, Fourth Edition
2. Principles of Power Systems, V.K. Mehtha, S.Chand Technical Publishers, 2020
3. Basic Electrical Engineering, T. K. Nagsarkar and M. S. Sukhija, Oxford University Press, 2017
4. Basic Electrical and Electronics Engineering, S. K. Bhattacharya, Person Publications, 2018, Second Edition.

Web Resources:

1. <https://nptel.ac.in/courses/108105053>
2. <https://nptel.ac.in/courses/108108076>

PART B: BASIC ELECTRONICS ENGINEERING

UNIT-I: SEMICONDUCTOR DEVICES

Introduction - Evolution of electronics – Vacuum tubes to nano electronics - Characteristics of PN Junction Diode — Zener Effect — Zener Diode and its Characteristics. Bipolar Junction Transistor — CB, CE, CC Configurations and Characteristics — Elementary Treatment of Small Signal CE Amplifier.

UNIT-II: BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION

Rectifiers and power supplies: Block diagram description of a dc power supply, working of a full wave bridge rectifier, capacitor filter (no analysis), working of simple zener voltage regulator. Amplifiers: Block diagram of Public Address system, Circuit diagram and working of common emitter (RC coupled) amplifier with its frequency response. Electronic Instrumentation: Block diagram of an electronic instrumentation system.

UNIT-III: DIGITAL ELECTRONICS

Overview of Number Systems, Logic gates including Universal Gates, BCD codes, Excess-3 code, Gray code, Hamming code. Boolean Algebra, Basic Theorems and properties of Boolean Algebra, Truth Tables and Functionality of Logic Gates – NOT, OR, AND, NOR, NAND, XOR and XNOR. Simple combinational circuits–Half and Full Adders. Introduction to sequential circuits, Flip flops, Registers and counters (Elementary Treatment only)

Textbooks:

1. R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.
2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009

Reference Books:

1. R. S. Sedha, A Textbook of Electronic Devices and Circuits, S. Chand & Co, 2010.
2. Santiram Kal, Basic Electronics- Devices, Circuits and IT Fundamentals, Prentice Hall, India, 2002.
3. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009.

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST02
Name of the Course	Data Structures					
Branch	(Common to CSE, IT & allied branches)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. **(K2)**
- CO2:** Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation. **(K3)**
- CO3:** Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. **(K3)**
- CO4:** Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues, and apply them appropriately to solve data management challenges. **(K3)**
- CO5:** Develop novel solutions to small scale programming challenges involving data structures such as Trees, hashing. **(K3)**

UNIT-I: Introduction to Linear Data Structures: Definition and importance of linear data structures, Abstract data types(ADTs) and their implementation, Overview of time and space complexity analysis for linear data structures. Searching Techniques: Linear & Binary Search, Sorting Techniques: Bubble sort, Selection sort, Insertion Sort

UNIT-II: Linked Lists: Singly linked lists: representation and operations, doubly linked lists and circular linked lists, Comparing arrays and linked lists, Applications of linked lists.

UNIT-III: Stacks: Introduction to stacks: properties and operations, implementing stacks using arrays and linked lists, Applications of stacks in expression evaluation, backtracking, reversing list etc.

UNIT-IV: Queues: Introduction to queues: properties and operations, implementing queues using arrays and linked lists, Applications of queues in breadth-first search, scheduling, etc.

Dequeues: Introduction to dequeues (double-ended queues), Operations on dequeues and their applications.

UNIT-V: Trees: Introduction to Trees, Binary Search Tree Insertion, Deletion & Traversal

Hashing: Brief introduction to hashing and hash functions, Collision resolution techniques: chaining and open addressing, Hash tables: basic implementation and operations, Applications of hashing in unique identifier generation, caching, etc.

Textbooks:

1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson- Freed, Silicon Press, 2008

Reference Books:

1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein
5. Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms" by Robert Sedgewick.

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23PHL01
Name of the Course	Engineering Physics Lab					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | |
|---|-------------|
| CO1: Operate optical instruments like travelling microscope and spectrometer. | (K3) |
| CO2: Estimate the wavelengths of different colours using diffraction grating. | (K2) |
| CO3: Plot the intensity of the magnetic field of circular coil carrying current with distance. | (K3) |
| CO4: Demonstrate the various experiments in the areas of optics, mechanics and Electronics. | (K3) |
| CO5: Calculate the band gap of a given semiconductor. | (K3) |
| CO6: Estimate Hall voltage and Hall coefficient of a given semiconductor using Hall effect. | (K2) |

List of Experiments:

1. Determination of radius of curvature of a given Plano-convex lens by Newton's rings.
2. Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
3. Verification of Brewster's law
4. Determination of dielectric constant using charging and discharging method.
5. Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
6. Determination of wavelength of Laser light using diffraction grating.
7. Estimation of Planck's constant using photoelectric effect.
8. Determination of the resistivity of semiconductors by four probe methods.
9. Determination of energy gap of a semiconductor using p-n junction diode.
10. Magnetic field along the axis of a current carrying circular coil by Stewart Gee's Method.
11. Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall effect.
12. Determination of temperature coefficients of a thermistor.
13. Determination of acceleration due to gravity and radius of Gyration by using a compound pendulum.
14. Determination of magnetic susceptibility by Kundt's tube method.
15. Determination of rigidity modulus of the material of the given wire using Torsional pendulum.
16. Sonometer: Verification of laws of stretched string.
17. Determination of young's modulus for the given material of wooden scale by non-uniform bending (or double cantilever) method.
18. Determination of Frequency of electrically maintained tuning fork by Melde's experiment.

Note: Any TEN of the listed experiments are to be conducted. Out of which any TWO experiments may be conducted in virtual mode.

References:

- A Textbook of Practical Physics - S. Balasubramanian, M.N. Srinivasan, S. Chand Publishers, 2017.

Web Resources

- www.vlab.co.in
- <https://phet.colorado.edu/en/simulations/filter?subjects=physics&type=html.prototype>

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23ENL01
Name of the Course	Communicative English Lab					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Identify vowels, consonant sounds, neutralization and accent rules for better listening and speaking comprehension. (K1)
- CO2:** Collect suitable expressions and vocabulary to participate in JAM and Role plays. (K1)
- CO3:** Arrange your ideas to draft an email, Resume and SOP. (K1)
- CO4:** Organize your ideas and prepare to give presentations in a professional manner and Debate rationally and cogently. (K3)
- CO5:** Show team spirit and communicative skills in group discussion and preparation to attend interviews. (K3)

List of Topics:

1. Vowels & Consonants
2. Neutralization/Accent Rules
3. Communication Skills & JAM
4. Role Play or Conversational Practice
5. E-mail Writing
6. Resume Writing, Cover letter, SOP
7. Group Discussions-methods & practice
8. Debates - Methods & Practice
9. PPT Presentations/ Poster Presentation
10. Interviews Skills

Suggested Software:

- Walden Infotech
- Young India Films

Reference Books:

1. Raman Meenakshi, Sangeeta-Sharma. *Technical Communication*. Oxford Press.2018.
2. Taylor Grant: *English Conversation Practice*, Tata McGraw-Hill Education India,2016
3. Hewing's, Martin. *Cambridge Academic English (B2)*. CUP, 2012.
4. J. Sethi & P.V. Dhamija. *A Course in Phonetics and Spoken English*, (2nd Ed), Kindle, 2013

Web Resources:

Spoken English:

1. www.esl-lab.com
2. www.englishmedialab.com
3. www.englishinteractive.net
4. <https://www.britishcouncil.in/english/online>
5. <http://www.letstalkpodcast.com/>
6. https://www.youtube.com/c/mmmEnglish_Emma/featured
7. <https://www.youtube.com/c/ArnelsEverydayEnglish/featured>
8. <https://www.youtube.com/c/engvidAdam/featured>
9. <https://www.youtube.com/c/EnglishClass101/featured>
10. <https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists>
11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw

Voice & Accent:

1. <https://www.youtube.com/user/letstalkaccent/videos>
2. <https://www.youtube.com/c/EngLanguageClub/featured>
3. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc
4. https://www.youtube.com/channel/UCNfm92h83W2i2ije5Xwp_IA

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23EEI01
Name of the Course	Electrical & Electronics Engineering Workshop					
Branch	(Common to All branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Analyse the Electrical circuit design concept, power and power factor. (K4)
CO2: Understand the operation and characteristics of Electrical Machines. (K2)
CO3: Analyse various domestic loads of energy bill calculation. (K4)
CO4: Understand the various characteristics of electronic devices. (K2)
CO5: Development of the digital circuits for electronic device applications. (K3)

Activities:

1. Familiarization of commonly used Electrical & Electronic Workshop Tools: Bread board, Solder, cables, relays, switches, connectors, fuses, Cutter, plier, screwdriver set, wire stripper, flux, knife/blade, soldering iron, de-soldering pump etc.
 - Provide some exercises so that hardware tools and instruments are learned to be used by the students.
2. Familiarization of Measuring Instruments like Voltmeters, Ammeters, multimeter, LCR-Q meter,
 - Power Supplies, CRO, DSO, Function Generator, Frequency counter.
 Provide some exercises so that measuring instruments are learned to be used by the students.
3. Components:
 - Familiarization/Identification of components (Resistors, Capacitors, Inductors, Diodes, transistors, IC's etc.) – Functionality, type, size, colour coding package, symbol, cost etc.
 - Testing of components like Resistor, Capacitor, Diode, Transistor, ICs etc. - Compare values of components like resistors, inductors, capacitors etc with the measured values by using instruments

PART A: ELECTRICAL ENGINEERING LAB

List of experiments:

1. Verification of KCL and KVL
2. Verification of Superposition theorem
3. Measurement of Resistance using Wheat stone bridge
4. Magnetization Characteristics of DC shunt Generator
5. Measurement of Power and Power factor using Single-phase wattmeter
6. Measurement of Earth Resistance using Megger
7. Calculation of Electrical Energy for Domestic Premises

Reference Books:

1. Basic Electrical Engineering, D. C. Kulshreshtha, Tata McGraw Hill, 2019, First Edition
2. Power System Engineering, P.V. Gupta, M.L. Soni, U.S. Bhatnagar and A. Chakrabarti, Dhanpat Rai & Co, 2013
3. Fundamentals of Electrical Engineering, Rajendra Prasad, PHI publishers, 2014, Third Edition

Note: Minimum Six Experiments to be performed.

PART B: ELECTRONICS ENGINEERING LAB

Course Outcomes: At the end of the course, the student will be able to

CO1: Identify & testing of various electronic components.

CO2: Understand the usage of electronic measuring instruments.

CO3: Plot and discuss the characteristics of various electron devices.

CO4: Explain the operation of a digital circuit.

List of Experiments:

1. Plot V-I characteristics of PN Junction diode A) Forward bias B) Reverse bias.

2. Plot V – I characteristics of Zener Diode and its application as voltage Regulator.

3. Implementation of half wave and full wave rectifiers

Plot Input & Output characteristics of BJT in CE and CB configurations

5. Frequency response of CE amplifier.

6. Simulation of RC coupled amplifier with the design supplied

7. Verification of Truth Table of AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR gates using ICs.

8. Verification of Truth Tables of S-R, J-K& D flip flops using respective ICs.

Tools / Equipment Required: DC Power supplies, Multi meters, DC Ammeters, DC Voltmeters,

AC Voltmeters, CROs, all the required active devices.

References:

1. R. L. Boylestad & Louis Nashlesky, Electronic Devices & Circuit Theory, Pearson Education, 2021.

2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009

3. R. T. Paynter, Introductory Electronic Devices & Circuits – Conventional Flow Version, Pearson Education, 2009.

Note: Minimum Six Experiments to be performed. All the experiments shall be implemented using both Hardware and Software.

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23CSL03
Name of the Course	Data Structures Lab					
Branch	(Common to EEE, ECE, CSE, IT & allied branches)					

Syllabus Details

Course Outcomes: At the end of the Course student will be able to:

CO1: Explain the role of linear data structures in organizing and accessing data efficiently in algorithms. **(K3)**

CO2: Design, implement, and apply linked lists for dynamic data storage, demonstrating understanding of memory allocation. **(K3)**

CO3: Develop programs using stacks to handle recursive algorithms, manage program states, and solve related problems. **(K3)**

CO4: Apply queue-based algorithms for efficient task scheduling and breadth-first traversal in graphs and distinguish between dequeues and priority queues and apply them appropriately to solve data management challenges. **(K3)**

CO5: Recognize scenarios where hashing is advantageous, and design hash-based solutions for specific problems. **(K3)**

List of Experiments:

Exercise 1: Array Manipulation

- i) Write a program to reverse an array.
- ii) C Programs to implement the Searching Techniques Linear & Binary Search
- iii) C Programs to implement Sorting Techniques Bubble, Selection and Insertion Sort

Exercise 2: Linked List Implementation

- i) Implement a singly linked list and perform insertion and deletion operations.
- ii) Develop a program to reverse a linked list iteratively and recursively.
- iii) Solve problems involving linked list traversal and manipulation.

Exercise 3: Linked List Applications

- i) Create a program to detect and remove duplicates from a linked list.
- ii) Implement a linked list to represent polynomials and perform addition.
- iii) Implement a double-ended queue (deque) with essential operations.

Exercise 4: Double Linked List Implementation

- i) Implement a doubly linked list and perform various operations to understand its properties and applications.
- ii) Implement a circular linked list and perform insertion, deletion, and traversal.

Exercise 5: Stack Operations

- i) Implement a stack using arrays and linked lists.
- ii) Write a program to evaluate a postfix expression using a stack.
- iii) Implement a program to check for balanced parentheses using a stack.

Exercise 6: Queue Operations

- i) Implement a queue using arrays and linked lists.
- ii) Develop a program to simulate a simple printer queue system.
- iii) Solve problems involving circular queues.

Exercise 7: Stack and Queue Applications

- i) Use a stack to evaluate an infix expression and convert it to postfix.
- ii) Create a program to determine whether a given string is a palindrome or not.
- iii) Implement a stack or queue to perform comparison and check for symmetry.

Exercise 8: Binary Search Tree

- i) Implementing a BST using Linked List.
- ii) Traversing of BST.

Exercise 9: Hashing

- i) Implement a hash table with collision resolution techniques.
- ii) Write a program to implement a simple cache using hashing.

Textbooks:

- 1. Data Structures and algorithm analysis in C, Mark Allen Weiss, Pearson, 2nd Edition.
- 2. Fundamentals of data structures in C, Ellis Horowitz, Sartaj Sahni, Susan Anderson- Freed, Silicon Press, 2008

Reference Books:

- 1. Algorithms and Data Structures: The Basic Toolbox by Kurt Mehlhorn and Peter Sanders
- 2. C Data Structures and Algorithms by Alfred V. Aho, Jeffrey D. Ullman, and John E. Hopcroft
- 3. Problem Solving with Algorithms and Data Structures" by Brad Miller and David Ranum
- 4. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein

Algorithms in C, Parts 1-5 (Bundle): Fundamentals, Data Structures, Sorting, Searching, and Graph Algorithms by Robert Sedgewick.

Semester	II	L	T	P	C	COURSE CODE
Regulation	V23	0	0	1	0.5	V23SPT02
Name of the Course	Health And Wellness, Yoga And Sports					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Discuss the importance of yoga and sports for Physical fitness and sound health. (K2)
CO2: Demonstrate an understanding of health-related fitness components. (K3)
CO3: Compare and contrast various activities that help enhance their health. (K4)
CO4: Assess current personal fitness levels. (K3)
CO5: Develop Positive Personality. (K3)

UNIT-I: Concept of health and fitness, Nutrition and Balanced diet, basic concept of immunity Relationship between diet and fitness, Globalization and its impact on health, Body Mass Index(BMI) of all age groups.

Activities:

- Organizing health awareness programmes in community
- Preparation of health profile
- Preparation of chart for balance diet for all age groups

UNIT-II: Concept of yoga, need for and importance of yoga, origin and history of yoga in Indian context, classification of yoga, Physiological effects of Asanas- Pranayama and meditation, stress management and yoga, Mental health and yoga practice.

Activities:

Yoga practices – Asana, Kriya, Mudra, Bandha, Dhyana, Surya Namaskar

UNIT-III: Concept of Sports and fitness, importance, fitness components, history of sports, Ancient and Modern Olympics, Asian games and Commonwealth games.

Activities:

- Participation in one major game and one individual sport viz., Athletics, Volleyball, Basketball, Handball, Football, Badminton, Kabaddi, Kho-kho, Table tennis, Cricket etc.
Practicing general and specific warm up, aerobics
- Practicing cardiorespiratory fitness, treadmill, run test, 9 min walk, skipping and running.

Reference Books:

- Gordon Edlin, Eric Golanty. Health and Wellness, 14th Edn. Jones & Bartlett Learning, 2022
- T.K.V.Desikachar. The Heart of Yoga: Developing a Personal Practice
- Archie J.Bahm. Yoga Sutras of Patanjali, Jain Publishing Company, 1993
- Wiseman, John Lofty, SAS Survival Handbook: The Ultimate Guide to Surviving Anywhere Third Edition, William Morrow Paperbacks, 2014
- The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc. 2014

General Guidelines:

- Institutes must assign slots in the Timetable for the activities of Health/Sports/Yoga.
- Institutes must provide field/facility and offer the minimum of five choices of as many as Games/Sports.
- Institutes are required to provide sports instructor / yoga teacher to mentor the students.

Evaluation Guidelines:

- Evaluated for a total of 100 marks.
- A student can select 6 activities of his/her choice with a minimum of 01 activity per unit. Each activity shall be evaluated by the concerned teacher for 15 marks, totaling to 90 marks.
- A student shall be evaluated by the concerned teacher for 10 marks by conducting viva voce on the subject.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23MAT05
Name of the Course	Discrete Mathematics & Graph Theory					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Develop skills required for solving mathematical problems using mathematical logic. **(K3)**
CO2: Demonstrate the set theory principles, relations and functions in real-time situations. **(K3)**
CO3: Apply the knowledge of combinatorics and recurrence relations in formulating and solving complex problems. **(K3)**
CO4: Apply the graph theory principles and techniques in computer science-related problems. **(K3)**
CO5: Find shortest paths and minimal spanning trees using prim's and Kruskal's algorithms, BFS and DFS algorithms. **(K3)**

UNIT-I: Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalence of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises, Indirect Method of Proof, Predicate Calculus: Predicates, Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT-II: Set Theory: Sets: Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Operations, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Functions: Bijective, Composition, Inverse, Permutation, and Recursive Functions, Lattice and its Properties.

UNIT-III: Combinatorics and Recurrence Relations: Basis of Counting, Permutations, Permutations with Repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and Theorems.

Recurrence Relations: Generating Functions, Function of Sequences, Partial Fractions, Calculating Coefficient of Generating Functions, Recurrence Relations, Formulation as Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Inhomogeneous Recurrence Relations

UNIT-IV: Graph Theory: Basic Concepts, Graph Theory and its Applications, Subgraphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs,

UNIT-V: Multi Graphs: Multigraphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Colouring and Covering, Chromatic Number, Spanning Trees, Prim's and Kruskal's Algorithms, BFS and DFS Spanning Trees.

Text Books:

1. Discrete Mathematical Structures with Applications to Computer Science, J. P.Tremblay and P. Manohar, Tata McGraw Hill.
2. Elements of Discrete Mathematics-A Computer Oriented Approach, C. L.Liu and D. P. Mohapatra, 3rd Edition, Tata McGraw Hill.
3. Theory and Problems of Discrete Mathematics, Schaum's Outline Series, Seymour Lipschutz and Marc Lars Lipson, 3rd Edition, McGraw Hill.

Reference Books:

1. Discrete Mathematics for Computer Scientists and Mathematicians, J. L.Mott, A.Kandel and T. P. Baker, 2nd Edition, Prentice Hall of India.
2. Discrete Mathematical Structures, Bernand Kolman, Robert C. Busby and SharonCutler Ross, PHI.
3. Discrete Mathematics, S. K. Chakraborty and B.K. Sarkar, Oxford, 2011.
4. Discrete Mathematics and its Applications with Combinatorics and GraphTheory, K. H. Rosen, 7th Edition, Tata McGraw Hill.

Syllabus Details

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23MBT51
Name of the Course	Managerial Economics and Financial Analysis					
Branch	Common to CSE and CST					

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1 :** Understand the basic concepts of managerial economics, demand, and elasticity of demand and methods of demand forecasting. (K2)
- CO2 :** Interpret production concept, least cost combinations and various costs concepts indecision making. (K3)
- CO3 :** Differentiate various Markets and Pricing methods a long with Business Cycles. (K2)
- CO4 :** Prepare financial statements and its analysis. (K3)
- CO5 :** Assess various investment project proposals with the help of Capital Budgeting techniques for decision making. (K3)

UNIT-I : Introduction to Managerial Economics and demand Analysis : Definition, Nature and Scope of Managerial Economics – Managerial Economics and its relation with other subjects- Concept of Demand : Meaning, Types, Determinants- Law of Demand its Exceptions – Elasticity of Demand and its Types and Measurement – Demand fore casting and its measuring methods.

UNIT-II: Production and Cost Analysis : Production function – Iso – cost- Law of Variable proportions- Economies of Scale-Cost Concepts : Opportunity Cost, Fixed vs Variable Cost, Explicit Costs vs Implicit Costs- Cost Volume Profit analysis- Determination of Break Even Point – BEP Chart (Simple Problems).

UNIT-III: Introduction to Markets, Pricing Policies & forms of Organizations and Business Cycles : Market Structures : Perfect Competition, Monopoly, Monopolistic and Oligopoly – Features – Price-Output Determination – Methods of Pricing – Evolution of Business Forms – Features of Sole Trader – Partnership – Joint Stock Company – State / Public Enterprises. Business Cycle : Meaning, Features, Phases of Business Cycle.

UNIT-IV : Introduction to Financial Accounting & Analysis: Introduction to Double Entry System – Accounting Cycle : Journal, Ledger, Trail balance, Preparation of Financial Statements – Trading Account, Profit & Loss Account and Balance Sheet- Ratio Analysis – (Simple Problems).

UNIT-V : Capital and Capital Budgeting : Meaning of Capital and Capital Budgeting – Need for Capital Budgeting – Capital budgeting process – Techniques of Capital Budgeting ;Traditional and Modern Methods.

Text Books:

1. Dr. N. Apparao, Dr. P. Vijaykumar : _ Managerial Economics and Financial Analysis, Cengage Publications, NewDelhi – 2011.
2. Dr. A.R. Aryasri – Managerial Economics and Financial Analysis, TMH 2011.
3. Prof. J.V. Prabhakararao, Prof.P. Venkatarao._ Managerial Economics and Financial Analysis, ‘Ravindra Publication.

References :

1. Dr. B. Kuberudu and Dr. T.V. Ramana : Managerial Economics & Financial Analysis, Himalaya Publishing House, 2014.
2. V. Maheswari ; Managerial Economics, Sultan Chand. 2014
3. Suma Damodaran : Managerial Economics, Oxford 2011.
4. Vanitha Agarwal : Managerial Economics, Pearson Publications 2011.
5. Sanjay Dhameje : Financial Accounting for Managers, Pearson
6. Maheswari : Managerial Accounting, Vikas Publications.
7. S.A. Siddiqui & A.S. Siddiqui : Managerial Economics and Financial Analysis, New Age International Publishers, 2012.
8. Ramesh Singh, Indian Economy, 7th Edn., TMH 2015.
9. Pankaj Tandon A Text Book of Micro economic Theory, Sage Publishers, 2015
10. Shailaja Gajjala and Usha Munipalle, Universities Press, 2012.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST03
Name of the Course	Digital Logic & Computer Organization					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain different Data Representation and various Combinational Digital Logic Circuits. (K2)
CO2: Explain various Sequential Digital Logic Circuits and basic Structure of Computers. (K2)
CO3: Describe the basic concepts of Computer Arithmetic and Processor Organization. (K2)
CO4: Illustrate different types of Memory and organization of Memory. (K2)
CO5: Demonstrate the ways of accessing various Interfacing devices with Processor. (K3)

UNIT-I:Data Representation: Binary Numbers, Fixed Point Representation. Floating Point Representation. Signed binary numbers.

Digital Logic Circuits I: Basic Logic Functions, Minimization of Logic expressions. K-Map Simplification, Combinational Circuits, Decoders, Multiplexers.

UNIT-II: Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters.

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi-computers, Computer Generations, Von-Neumann Architecture

UNIT-III: Computer Arithmetic: Addition and Subtraction of Signed Numbers, Design of Fast Adders, Multiplication of Positive Numbers, Signed-operand Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations.

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Multi programmed Control

UNIT-IV: The Memory Organization: Basic Concepts, Semiconductor RAM Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage

UNIT-V: Input / Output Organization: Accessing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 6th edition, McGraw-Hill
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education.

Reference Books:

1. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson.
2. Computer Systems Architecture, M. Moris Mano, 3rd Edition, Pearson
3. Computer Organization and Design, David A. Paterson, John L. Hennessy, Elsevier
4. Fundamentals of Logic Design, Roth, 5th Edition, Thomson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/103/106103068/>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST04
Name of the Course	Advanced Data Structures & Algorithm Analysis					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate asymptotic notation and non-linear data structures like AVL and B-Trees. (K3)
CO2: Demonstrate graphs and divide and conquer technique. (K3)
CO3: Use greedy approach to determine various problems. (K3)
CO4: Apply Dynamic Programming and Backtracking techniques. (K3)
CO5: Solve different problems using Branch and Bound ,NP Hard and NP Complete. (K3)

UNIT–I: Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations.
 AVL Trees – Creation, Insertion, Deletion operations and Applications.
 B-Trees– Creation, Insertion, Deletion operations and Applications.

UNIT–II: Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications.
Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications
Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen’s matrix multiplication, Convex Hull.

UNIT–III: Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths– Dijkstra's .

UNIT–IV: Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths–General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem.
Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem.

UNIT–V: Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem.

NP Hard and NP Complete Problems: Basic Concepts, Cook’s theorem (without Proof).

NP Hard Graph Problems: Clique Decision Problem (CDP), Traveling Salesperson Decision Problem (TSP).

NP Hard Scheduling Problems: Job Shop Scheduling.

Textbooks:

1. Fundamentals of Data Structures Horowitz, Ellis; Sahni, Sartaj; Mehta, Dinesh, 2ndEdition Universities Press

Reference Books:

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Trembley & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1997.
4. Data Structures using C & C++: Langsam, Augenstein & Tanenbaum, Pearson, 1995

5. Algorithms + Data Structures & Programs:, N. Wirth, PHI
6. Fundamentals of Data Structures in C++: Horowitz Sahni & Mehta, Galgotia Pub.
7. Data structures in Java:, Thomas Standish, Pearson Education Asia

Online Learning Resources:

1. https://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://peterindia.net/Algorithms.html>
3. Abdul Bari, Introduction to Algorithms (youtube.com)

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST05
Name of the Course	Object Oriented Programming Through Java					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Demonstrate the object-oriented programming principles with Java programming environment. (K3)

CO2:Demonstrate the concepts like classes, objects, argument passing mechanism, overloading and overriding. (K3)

CO3:Illustrate the concepts of arrays, inheritance and interfaces. (K3)

CO4: Demonstrate packages, java libraries, exception handling, java I/O and File concepts. (K3)

CO5:Illustrate the concepts of string handling, multithreading and Java FX GUI. (K3)

UNIT-I: Object Oriented Programming: Basic concepts, Principles, Program Structure in Java: Introduction, Writing Simple Java Programs, Elements or Tokens in Java Programs, Java Statements, Command Line Arguments, User Input to Programs, Escape Sequences Comments, Programming Style.

Data Types, Variables, and Operators: Introduction, Data Types in Java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attribute Final, Introduction to Operators, Precedence and Associativity of Operators, Assignment Operator (=), Basic Arithmetic Operators, Increment (++) and Decrement (- -) Operators, Ternary Operator, Relational Operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if-else Expressions, Ternary Operator?, Switch Statement, Iteration Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

UNIT-II: Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, Overloaded Methods, Overloaded Constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Nesting of Methods, Overriding Methods, Attributes Final and Static.

UNIT-III: Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on Array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three-dimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritances, Universal Super Class-Object Class, Inhibiting Inheritance of Class Using Final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple Interfaces, Nested Interfaces, Inheritance of Interfaces, Default Methods in Interfaces, Static Methods in Interface, Functional Interfaces, Annotations.

UNIT–IV: Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Java.lang Package and its Classes, Class Object, Enumeration, class Math, Wrapper Classes, Auto- boxing and Auto-unboxing, Java util Classes and Interfaces, Formatter Class, Random Class, Time Package, Class Instant (java.time.Instant), Formatting for Date/Time in Java, Temporal Adjusters Class, Temporal Adjusters Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords throws and throw, try, catch, and finally Blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Scanner class, Files in Java(Text Book 2)

UNIT–V: String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching; Class String Buffer.

Multithreaded Programming: Introduction, Need for Multiple Threads Multithreaded Programming for Multi-core Processor, Thread Class, Main Thread-Creation of New Threads, Thread States, Thread Priority-Synchronization, Deadlock and Race Situations, Inter-thread Communication - Suspending, Resuming, and Stopping of Threads.

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Textbooks:

1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105191/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL04
Name of the Course	Advanced Data Structures & Algorithm Analysis Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate programs on AVL Trees and Heap trees. **(K3)**
CO2: Develop programs on Sorting algorithms and Graph traversal algorithms. **(K3)**
CO3: Develop programs using Greedy and Dynamic programming technique. **(K3)**
CO4: Develop programs using Backtracking and branch & Bound technique. **(K3)**

Experiments covering the Topics:

- Operations on AVL trees, Heap Trees
- Graph Traversals
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Travelling Salesperson problem
- Optimal Binary Search Trees
- N-Queens Problem
- Job Sequencing

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file. And implement insert and delete operation on the constructed tree. Write contents of tree into a new file using in-order.
2. Construct B-Tree an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct Min and Max Heap using arrays, delete any element and display the content of the Heap.
4. Demonstrate BFT and DFT for given graph, when graph is represented by
 - a) Adjacency Matrix
 - b) Adjacency Lists
5. Develop a program for finding the biconnected components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes(Average, Worst and Best cases).
7. Examine the performance of Single Source Shortest Paths using Greedy method when the graph is represented by adjacency matrix and adjacency lists.
8. Demonstrate Job Sequencing with deadlines using Greedy strategy.
9. Demonstrate a program to solve 0/1Knapsack problem Using Dynamic Programming.
10. Demonstrate N-Queens Problem Using Backtracking.
11. Use Backtracking strategy to solve 0/1 Knapsack problem.
12. Demonstrate Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures, Horowitz Ellis, SahniSartaj, Mehta, Dinesh, 2ndEdition, Universities Press
2. Computer Algorithms/C++ Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2ndEdition, University Press
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
4. An introduction to Data Structures with applications, Trembley& Sorenson, McGraw Hill

Online Learning Resources:

1. <http://cse01-iiith.vlabs.ac.in/>
2. <http://peterindia.net/Algorithms.html>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL05
Name of the Course	Object Oriented Programming Through Java Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1:Construct programs to handle classes and objects. **(K3)**

CO2: Develop programs that incorporate inheritance and interfaces. **(K3)**

CO3:Construct programs on exception handling and File I/O. **(K3)**

CO4:Develop programs using multithreading and Java FX. **(K3)**

Experiments covering the Topics:

- Object Oriented Programming fundamentals- data types, control structures
- Classes, methods, objects, Inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI

Exercise – 1:

- a) Develop a JAVA program to display default value of all primitive data type of JAVA
- b) Develop a java program that display the roots of a quadratic equation $ax^2+bx=0$. Find the discriminate D and basing on value of D, describe the nature of root.

Exercise – 2:

- a) Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
- b) Develop a JAVA program to sort for an element in a given list of elements using bubble sort
- c) Develop a JAVA program using String Buffer to delete, remove character.

Exercise – 3:

- a) Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.
- b) Develop a JAVA program to implement method overloading.
- c) Develop a JAVA program to implement constructor.
- d) Develop a JAVA program to implement constructor overloading.

Exercise – 4:

- a) Develop a JAVA program to implement Single Inheritance.
- b) Develop a JAVA program to implement multi level Inheritance.
- c) Develop a JAVA program for abstract class to find areas of different shapes.

Exercise – 5:

- a) Develop a JAVA program on “super” keyword.
- b) Develop a JAVA program to implement Interface. What kind of Inheritance can be achieved?
- c) Develop a JAVA program that implements Runtime polymorphism.

Exercise – 6:

- a) Develop a JAVA program that describes exception handling mechanism.
- b) Develop a JAVA program on Multiple catch clauses.

- c) Develop a JAVA program to generate Built-in Exceptions.
- d) Develop a JAVA program to generate User Defined Exception.

Exercise – 7:

- a) Develop a JAVA program that creates threads by extending Thread class. First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds,(Repeat the same by implementing Runnable)
- b) Develop a program on isAlive() and join ().
- c) Demonstrate Daemon Threads.
- d) Demonstrate Producer Consumer Problem.

Exercise – 8:

- a) Develop a JAVA program that import and use the user defined packages.
- b) Demonstrate a GUI that display text in label and image in an ImageView (use JavaFX) Without writing any code.
- c) Develop a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI

Textbooks:

- 1. JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.
- 2. Joy with JAVA, Fundamentals of Object Oriented Programming, DebasisSamanta, MonalisaSarma, Cambridge, 2023.
- 3. JAVA 9 for Programmers, Paul Deitel, Harvey Deitel, 4th Edition, Pearson.

Reference Books:

- 1. The complete Reference Java, 11th edition, Herbert Schildt, TMH
- 2. Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson

Online Learning Resources:

- 1. <https://nptel.ac.in/courses/106/105/106105191/>
- 2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012880464547618816347_shared/overview.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE01
Name of the Course	Python Programming Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate basic concepts and control structures in python Programming. **(K3)**

CO2: Demonstrate functions and packages. **(K3)**

CO3: Construct python programs using structured data types. **(K3)**

CO4: Develop programs on Files, Exception handling and OOPs Concepts. **(K3)**

CO5: Construct programs for Data Analysis using Num Py and Pandas. **(K3)**

UNIT–I: History of Python Programming Language, Thrust Areas of Python, Installing Anaconda Python Distribution, Installing and Using Jupyter Notebook.

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and Is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if...elif...else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample Experiments:

1. Develop a program to find the largest element among three Numbers.
2. Develop a Program to display all prime numbers within an interval
3. Develop a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.
 - i) Arithmetic Operators ii) Relational Operators iii) Assignment Operators iv) Logical Operators
 - v) Bit wise Operators vi) Ternary Operator vii) Membership Operators viii) Identity Operators
5. Develop a program to add and multiply complex numbers
6. Develop a program to print multiplication table of a given number.

UNIT–II: Functions: Built-In Functions, Commonly Used Modules, Function Definition and Calling the function, return Statement and void Function, Scope and Lifetime of Variables, Default Parameters, Keyword Arguments, *args and **kwargs, Command Line Arguments.

Strings: Creating and Storing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and Joining, String Methods, Formatting Strings.

Lists: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-In Functions Used on Lists, List Methods, del Statement.

Sample Experiments:

1. Develop a program to define a function with multiple return values.
2. Develop a program to define a function using default arguments.
3. Develop a program to find the length of the string without using any library functions.
4. Develop a program to check if the substring is present in a given string or not.
5. Develop a program to perform the given operations on a list:
 - i. addition ii. Insertion iii. slicing
6. Develop a program to perform any 5 built-in functions by taking any list.

UNIT–III: Dictionaries: Creating Dictionary, Accessing and Modifying key: value Pairs in Dictionaries, Built-In Functions Used on Dictionaries, Dictionary Methods, del Statement.

Tuples and Sets: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-In Functions Used on Tuples, Relation between Tuples and Lists, Relation between Tuples and Dictionaries, Using zip() Function, Sets, Set Methods, Frozenset.

Sample Experiments:

1. Develop a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
2. Develop a program to count the number of vowels in a string (No control flow allowed).
3. Develop a program to check if a given key exists in a dictionary or not.
4. Develop a program to add a new key-value pair to an existing dictionary.
5. Develop a program to sum all the items in a given dictionary.

UNIT–IV: Files: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, Pickle Module, Reading and Writing CSV Files, Python os and os.path Modules.

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes Vs Data Attributes, Encapsulation, Inheritance, Polymorphism.

Sample Experiments:

1. Develop a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
2. Develop a Python program to print each line of a file in reverse order.
3. Develop a Python program to compute the number of characters, words and lines in a file.
4. Develop a program to create, display, append, insert and reverse the order of the items in the array.
5. Develop a program to add, transpose and multiply two matrices.
6. Develop a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT–V: Introduction to Data Science: Functional Programming, JSON and XML in Python, NumPy with Python, Pandas.

Sample Experiments:

1. Develop a Python program to check whether a JSON string contains complex object or not.
2. Demonstrate NumPy arrays creation using array () function.
3. Demonstrate use of ndim, shape, size, dtype.
4. Demonstrate basic slicing, integer and Boolean indexing.
5. Develop a Python program to find min, max, sum, cumulative sum of array
6. Prepare a dictionary with at least five keys and each key represent value as a list where this list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows:
 - a) Apply head () function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
7. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in matplotlib.

Reference Books:

1. Gowrishankar S, Veena A., Introduction to Python Programming, CRC Press.
2. Python Programming, S Sridhar, J Indumathi, V M Hariharan, 2nd Edition, Pearson, 2024
3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources:

1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
<https://www.coursera.org/learn/python?specialization=python#syllabus>.

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	1	0	2	2	V23MET09
Name of the Course	Design Thinking & Innovation					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Define the concepts related to design thinking. **(K1)**
CO 2: Explain the fundamentals of Design Thinking and innovation. **(K2)**
CO 3: Apply the design thinking techniques for solving problems in various sectors. **(K3)**
CO 4: Analyse to work in a multidisciplinary environment. **(K4)**
CO 5: Evaluate the value of creativity. **(K5)**

UNIT-I: Introduction to Design Thinking: Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.

UNIT-II: Design Thinking Process: Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development.

Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.

UNIT-III: Innovation: Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity.

Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.

UNIT-IV: Product Design: Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.

Activity: Importance of modeling, how to set specifications, Explaining their own product design.

UNIT-V: Design Thinking in Business Processes: Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes.

Activity: How to market our own product, about maintenance, Reliability and plan for startup.

Textbooks:

1. Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2. Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.

Reference Books:

1. David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2. Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3. William lidwell, Kritinaholden, & Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4. Chesbrough, H, The era of open innovation, 2003.

Online Learning Resources:

- <https://nptel.ac.in/courses/110/106/110106124/>
- <https://nptel.ac.in/courses/109/104/109104109/>
- https://swayam.gov.in/nd1_noc19_mg60/preview
- https://onlinecourses.nptel.ac.in/noc22_de16/preview

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	0	V23ENT02
Name of the Course	Professional Communication Skills – I					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Use vocabulary in regular chores of life with accuracy, make meaningful sentences, and describe people and their traits vividly. **(K3)**

CO2: Distinguish between places of pilgrimage and holiday spots; describe incidents, things and process; and frame questions, statements and expressions. **(K4)**

CO3: Demonstrate their knowledge of idioms which are similar to those of native speakers while speaking and writing and use phrases clearly and precisely to articulate their views that compare and contrast indianisms with native expressions and avoid common errors. **(K3)**

CO4: Employ the vocabulary of netizens with ease and walk through the letters and emails for effective official correspondence and infer the accurate meaning of the homophones that are often confusing. **(K3)**

CO5: Summarize their profile; introduce themselves as well as others by incorporating their accomplishments and Sketch stories and anecdotes in an interesting and engaging manner that arouses curiosity of the audience. **(K5)**

UNIT-I: BUILDING VOCABULARY FOR DAILY ACTIVITIES NAMES: Things- Kitchen Utensils – Occupation- tools – spices- vegetables –flowers - sciences of study – Professions. Framing Questions – statements – expressions related to the Vocabulary taught.

PEOPLE: Describing people -Physical characteristics,-Mental attributes – various professions Framing Questions – statements – expressions related to the Vocabulary taught.

ACTIVITY: Related to the topics learnt in Unit – 1

UNIT-II: BUILDING VOCABULARY FOR PLACES, THINGS & PROCESS PLACES:

Describing favorite place – famous place- Places of Pilgrimage.

THINGS: Describing a thing- Describe an incident or an event.

PROCESS: Describe a process –Recipe – experiment –Entrance test application.

Framing Questions – statements – expressions related to the Vocabulary taught.

ACTIVITY: Related to the topics learnt in Unit – II.

UNIT-III: NATIVE EXPRESSIONS – Idioms and Phrases – in day to day activities for different occasions - Usage written & spoken –

PHRASES with as—as expressions – used to compare & contrast

COMMON MISTAKES- in spoken & written

INDIANISMS- Most often used expressions – accepted in India – found place in Dictionary

ACTIVITY: Related to the topics learnt in Unit – III

UNIT-IV: NET VOCABULARY: Acronyms and abbreviations that are most often used

HOMOPHONES :Words often confused – Spelling & Pronunciation

Letter Writing :Formal& Informal- Letters for all occasions

Email Writing :Business mails – project status mails – informative mails

ACTIVITY :Related to the topics learnt in Unit – IV

UNIT-V SELF-INTRODUCTION: Basic information - Academic and personal - interests– strengths and weaknesses – goal. **PROFILE BUILDING:** Resume writing – CV Building – Types

STORYTELLING WITH CREATIVITY: Reading and Narrating a story – narrating anecdotes

ACTIVITY :Related to the topics learnt in Unit – V

Reference Books:

1. Lewis Norman, Word Power Made Easy (2008). Goyal Publishers & Distributors Pvt. Ltd.
2. Sunita Mishra & C. Muralikrishna, Communication Skills for Engineers (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
3. Chaturvedi PD & Chaturvedi Mukesh, Business Communication (2006). Dorling Kindersley (India) Pvt. Ltd., licensees of Pearson Education in South Asia.
4. **Joshi Manik, Popular English Idioms and Phrases: English Idiomatic Expressions (2013).**
5. **Joshi Manik, Homonyms, Homophones and Homographs: Vocabulary Building (2014).**
6. Gupta S.C. A Handbook for Letter Writing (2018). Arihant Publishers
7. Lisa McGrimmon, The Resume Writing Guide: A Step-by-Step Workbook for Creating a Winning Resume (2013). CareerChoiceGuide; 2nd edition.
8. Sawhney, Clifford. Improve your Word Power (2013). V&S Publishers

Web References: (NET Vocabulary)

1. <https://www.grammarly.com/blog/texting-abbreviations/>
2. <https://www.slicktext.com/blog/2019/02/text-abbreviations-guide/>
3. <https://www.webopedia.com/reference/text-abbreviations/>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	2	1	0	3	V23MBT53
Name of the Course	Universal Human Values					
Branch	(Common to All Branches of Engineering)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the fundamentals of Natural Acceptance, Happiness and Prosperity. (K2)
CO2: Identify one's self, and one's surroundings(family, society, nature). (K2)
CO3: Apply what they have learnt to their own self in different day-to-day settings in real life. (K3)
CO4: Explain human values with human relationship and human society. (K2)
CO5: Identify the need for universal human values and harmonious existence. (K2)
CO6: Develop as socially and ecologically responsible engineers. (K2)

Course Topics

The course has 28 lectures and 14 tutorials in 5 modules. The lectures and tutorials are of 1- hour duration. Tutorial sessions are to be used to explore and practice what has been proposed during the lecture sessions.

The Teacher's Manual provides the outline for lectures as well as practice sessions. The teacher is expected to present the issues to be discussed as propositions and encourage the students to have a dialogue.

UNIT-I: Introduction to Value Education (6 lectures and 3 tutorials for practice session)

- Lecture 1: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education)
- Lecture 2: Understanding Value Education
- Tutorial 1: Practice Session PS1 Sharing about Oneself
- Lecture 3: self-exploration as the Process for Value Education
- Lecture 4: Continuous Happiness and Prosperity – the Basic Human Aspirations
- Tutorial 2: Practice Session PS2 Exploring Human Consciousness
- Lecture 5: Happiness and Prosperity – Current Scenario
- Lecture 6: Method to Fulfill the Basic Human Aspirations
- Tutorial 3: Practice Session PS3 Exploring Natural Acceptance

UNIT-II: Harmony in the Human Being (6 lectures and 3 tutorials for practice session)

- Lecture 7: Understanding Human being as the Co-existence of the self and the body.
- Lecture 8: Distinguishing between the Needs of the self and the body
- Tutorial 4: Practice Session PS4 Exploring the difference of Needs of self and body.
- Lecture 9: The body as an Instrument of the self
- Lecture 10: Understanding Harmony in the self
- Tutorial 5: Practice Session PS5 Exploring Sources of Imagination in the self
- Lecture 11: Harmony of the self with the body
- Lecture 12: Programme to ensure self-regulation and Health
- Tutorial 6: Practice Session PS6 Exploring Harmony of self with the body

UNIT-III: Harmony in the Family and Society (6 lectures and 3 tutorials for practice session)

- Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction
- Lecture 14: 'Trust' – the Foundational Value in Relationship
- Tutorial 7: Practice Session PS7 Exploring the Feeling of Trust
- Lecture 15: 'Respect' – as the Right Evaluation
- Tutorial 8: Practice Session PS8 Exploring the Feeling of Respect
- Lecture 16: Other Feelings, Justice in Human-to-Human Relationship
- Lecture 17: Understanding Harmony in the Society
- Lecture 18: Vision for the Universal Human Order
- Tutorial 9: Practice Session PS9 Exploring Systems to fulfil Human Goal

UNIT-IV: Harmony in the Nature/Existence (4 lectures and 2 tutorials for practice session)

Lecture 19: Understanding Harmony in the Nature

Lecture 20: Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature

Tutorial 10: Practice Session PS10 Exploring the Four Orders of Nature

Lecture 21: Realizing Existence as Co-existence at All Levels

Lecture 22: The Holistic Perception of Harmony in Existence

Tutorial 11: Practice Session PS11 Exploring Co-existence in Existence.

UNIT-V: Implications of the Holistic Understanding – a Look at Professional Ethics (6 lectures and 3 tutorials for practice session)

Lecture 23: Natural Acceptance of Human Values Lecture 24: Definitiveness of (Ethical) Human Conduct

Tutorial 12: Practice Session PS12 Exploring Ethical Human Conduct

Lecture 25: A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order

Lecture 26: Competence in Professional Ethics

Tutorial 13: Practice Session PS13 Exploring Humanistic Models in Education

Lecture 27: Holistic Technologies, Production Systems and Management Models-Typical Case Studies

Lecture 28: Strategies for Transition towards Value-based Life and Profession

Tutorial 14: Practice Session PS14 Exploring Steps of Transition towards Universal Human Order

Practice Sessions for UNIT I – Introduction to Value Education PS1

Sharing about Oneself

PS2 Exploring Human Consciousness PS3

Exploring Natural Acceptance

Practice Sessions for UNIT II – Harmony in the Human Being PS4

Exploring the difference of Needs of self and body

PS5 Exploring Sources of Imagination in the self PS6

Exploring Harmony of self with the body

Practice Sessions for UNIT III – Harmony in the Family and Society PS7

Exploring the Feeling of Trust

PS8 Exploring the Feeling of Respect

PS9 Exploring Systems to fulfil Human Goal

Practice Sessions for UNIT IV – Harmony in the Nature (Existence) PS10

Exploring the Four Orders of Nature

PS11 Exploring Co-existence in Existence

Practice Sessions for UNIT V – Implications of the Holistic Understanding – a Look at Professional Ethics

PS12 Exploring Ethical Human Conduct

PS13 Exploring Humanistic Models in Education

PS14 Exploring Steps of Transition towards Universal Human Order

READINGS:

Textbook and Teachers Manual

a. The Textbook

R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1

b. The Teacher's Manual

R R Gaur, R Asthana, G P Bagaria, *Teachers' Manual for A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2

Reference Books

1. *JeevanVidya: EkParichaya*, A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
2. *Human Values*, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. *The Story of Stuff* (Book).
4. *The Story of My Experiments with Truth* - by Mohandas Karamchand Gandhi
5. *Small is Beautiful* - E. F Schumacher.
6. *Slow is Beautiful* - Cecile Andrews
7. *Economy of Permanence* - J C Kumarappa
8. *Bharat Mein Angreji Raj* – PanditSunderlal
9. *Rediscovering India* - by Dharampal
10. *Hind Swaraj or Indian Home Rule* - by Mohandas K. Gandhi
11. *India Wins Freedom* - Maulana Abdul Kalam Azad
12. *Vivekananda* - Romain Rolland (English)
13. *Gandhi* - Romain Rolland (English)

Mode of Conduct:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, explore and verify them.

Tutorial hours are to be used for practice sessions.

While analyzing and discussing the topic, the faculty mentor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements.

In the discussions, particularly during practice sessions (tutorials), the mentor encourages the student to connect with one's own self and do self-observation, self-reflection and self- exploration.

Scenarios may be used to initiate discussion. The student is encouraged to take up "ordinary" situations rather than "extra-ordinary" situations. Such observations and their analyses are shared and discussed with other students and faculty mentor, in a group sitting.

Tutorials (experiments or practical) are important for the course. The difference is that the laboratory is everyday life, and practical are how you behave and work in real life. Depending on the nature of topics, worksheets, home assignment and/or activity are included. The practice sessions (tutorials) would also provide support to a student in performing actions commensurate to his/her beliefs. It is intended that this would lead to development of commitment, namely behaving and working based on basic human values.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23MAT07
Name of the Course	Probability & Statistics					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Classify the concepts of data science and its importance. (K2)
CO2: Interpret the association of characteristics and through correlation and regression tools. (K3)
CO3: Apply discrete and continuous probability distributions. (K3)
CO4: Construct the components of a classical hypothesis test. (K3)
CO5: Apply the statistical inferential methods to test the hypothesis based on small and large samplings. (K3)

Unit-I: Descriptive statistics and methods for data science: Data science – Statistics Introduction – Population vs Sample –Collection of data – primary and Secondary data–Type of variable: dependent and independent Categorical and Continuous Variables –Data visualization – Measures of Central tendency – Measures of Variability – Skewness – Kurtosis.

UNIT-II: Correlation and Regression: Correlation–Correlation coefficient–Rank correlation. Linear Regression: Straight line – Multiple Linear Regression - Regression coefficients and properties – Curvilinear Regression: Parabola – Exponential – Power curves.

UNIT-III: Probability and Distributions: Probability– Conditional probability and Baye's theorem – Random variables – Discrete and Continuous random variables–Distribution functions–Probability mass function, Probability density function and Cumulative distribution functions–Mathematical Expectation and Variance – Binomial, Poisson, Uniform and Normal distributions. Expectation and Variance – Binomial, Poisson, Uniform and Normal distributions.

UNIT-IV: Sampling Theory: Introduction – Population and Samples – Sampling distribution of Means and Variance (Definition only)–Point and Interval estimations–Maximum error of estimate–Central Limit theorem (without proof)–Estimation using t, χ^2 , and F-distributions.

UNIT-V: Tests of Hypothesis: Introduction – Hypothesis – Null and Alternative Hypothesis – Type I and Type II errors – Level of significance – One tail and two-tail tests – Test of significance for large samples and Small Samples: Single and difference means – Single and two proportions –Student's t-test, F-test, chi-square test.

Text Books:

1. Miller and Freund's, Probability and Statistics for Engineers, 7/e, Pearson, 2008.
2. S. C. Gupta and V.K. Kapoor, Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.

Reference Books:

1. Shron L. Myers, Keying Ye, Ronald E Walpole, Probability and Statistics Engineers and the Scientists, 8th Edition, Pearson 2007.
2. Jay I. Devore, Probability and Statistics for Engineering and the Sciences, 8th Edition, Cengage.
3. Sheldon M. Ross, Introduction to probability and statistics Engineers and the Scientists, 4th Edition, Academic Foundation, 2011.
4. Johannes Ledolter and Robert V. Hogg, Applied statistics for Engineers and Physical Scientists, 3rd Edition, Pearson, 2010.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST06
Name of the Course	Operating Systems					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Describe Operating System Services, System Calls, Building, Boosting and Debugging. **(K2)**
CO2: Illustrate Process Management Concepts and CPU Scheduling Algorithms. **(K3)**
CO3: Demonstrate Process Synchronization primitives and Deadlock handling techniques. **(K3)**
CO4: Illustrate Memory Management Techniques and Page Replacement Algorithms. **(K3)**
CO5: Describe File System Concepts and Mass Storage Structures. **(K2)**

UNIT-I: Operating Systems Overview: Introduction, Operating system functions, Operating systems operations, Computing environments, Free and Open-Source Operating Systems

System Structures: Operating System Services, User and Operating-System Interface, system calls, Types of System Calls, system programs, Operating system Design and Implementation, Operating system structure, Building and Booting an Operating System, Operating system debugging

UNIT-II: Processes: Process Concept, Process scheduling, Operations on processes, Inter-process communication. **Threads and Concurrency:** Multithreading models, Thread libraries, Threading issues.

CPU Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling.

UNIT-III: Synchronization Tools: The Critical Section Problem, Peterson's Solution, Mutex Locks, Semaphores, Monitors, Classic problems of Synchronization.

Deadlocks: system Model, Deadlock characterization, Methods for handling Deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from Deadlock.

UNIT-IV: Memory-Management Strategies: Introduction, Contiguous memory allocation, Paging, Structure of the Page Table, Swapping.

Virtual Memory Management: Introduction, Demand paging, Copy-on-write, Page replacement, Allocation of frames, Thrashing

Storage Management: Overview of Mass Storage Structure, HDD Scheduling.

UNIT-V: File System: File System Interface: File concept, Access methods, Directory Structure; File system **Implementation:** File-system structure, File-system Operations, Directory implementation, Allocation method, Free space management; File-System Internals: File-System Mounting, Partitions and Mounting, File Sharing.

Protection: Goals of protection, Principles of protection, Protection Rings, Domain of protection, Access matrix.

Textbooks:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson , 2016

Reference Books:

1. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018
2. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/106/106106144/>
2. <http://peterindia.net/OperatingSystems.html>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST07
Name of the Course	Database Management Systems					
Branch	Common to CSE and CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Describe Database systems, various Data models and Database architecture. **(K2)**
CO2: Develop various real time applications using Relational algebra and Relational calculus. **(K3)**
CO3: Illustrate Nested queries and different types of joins. **(K3)**
CO4: Apply various Normalization techniques to refine schema. **(K3)**
CO5: Explain Transaction management, Concurrency control and Database indexing techniques. **(K2)**

UNIT-I: Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications. Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database.

Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-II: Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational Calculus.

BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update).

UNIT-III: SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and non-updatable), relational set operations.

UNIT-IV: Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form(BCNF), MVD, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT-V: Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for Serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm.

Introduction to Indexing Techniques: B+ Trees, operations on B+Trees, Hash Based Indexing.

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Introduction to Database Systems, 8th edition, C J Date, Pearson.
2. Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Carlos Coronel, Steven Morris, Peter Robb, Cengage Learning.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105175/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST08
Name of the Course	Software Engineering					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate the evolution, development practices, and various software life cycle models. **(K3)**
CO2: Illustrate project management techniques and explain requirements analysis and specification. **(K3)**
CO3: Develop effective software designs, using agile methodologies and user interface principles. **(K3)**
CO4: Apply coding practices, conduct thorough testing, and manage software quality. **(K3)**
CO5: Distinguish the use of CASE tools, manage software maintenance, and implement software reuse strategies. **(K4)**

UNIT-I: Introduction: Evolution, Software development projects, Exploratory style of software developments, Emergence of software engineering, Notable changes in software development practices, Computer system engineering.

Software Life Cycle Models: Basic concepts, Waterfall model and its extensions, Rapid application development, Agile development model, Spiral model.

UNIT-II: Software Project Management: Software project management complexities, Responsibilities of a software project manager, Metrics for project size estimation, Project estimation techniques, Empirical Estimation techniques, COCOMO, Halstead's software science, risk management.

Requirements Analysis And Specification: Requirements gathering and analysis, Software Requirements Specification (SRS), Formal system specification, Axiomatic specification, Algebraic specification, Executable specification and 4GL.

UNIT-III: Software Design: Overview of the design process, How to characterize a good software design? Layered arrangement of modules, Cohesion and Coupling. approaches to software design.

Agility: Agility and the Cost of Change, Agile Process, Extreme Programming (XP), Other Agile Process Models, Tool Set for the Agile Process (Text Book 2)

Function-Oriented Software Design: Overview of SA/SD methodology, Structured analysis, Developing the DFD model of a system, Structured design, Detailed design, and Design Review.

User Interface Design: Characteristics of a good user interface, Basic concepts, Types of user interfaces, Fundamentals of component-based GUI development, and user interface design methodology.

UNIT-IV: Coding And Testing: Coding, Code review, Software documentation, Testing, Black-box testing, White-Box testing, Debugging, Program analysis tools, Integration testing, Testing object-oriented programs, Smoke testing, and Some general issues associated with testing.

Software Reliability and Quality Management: Software reliability. Statistical testing, Software quality, Software quality management system, ISO 9000.SEI Capability maturity model. Few other important quality standards, and Six Sigma.

UNIT-V: Computer-Aided Software Engineering (Case): CASE and its scope, CASE environment, CASE support in the software life cycle, other characteristics of CASE tools, Towards second generation CASE Tool, and Architecture of a CASE Environment.

Software Maintenance: Characteristics of software maintenance, Software reverse engineering, Software maintenance process models and Estimation of maintenance cost.

Software Reuse: reuse- definition, introduction, reason behind no reuse so far, Basic issues in any reuse program, A reuse approach, and Reuse at organization level.

Textbooks:

1. Fundamentals of Software Engineering, Rajib Mall, 5th Edition, PHI.
2. Software Engineering A practitioner's Approach, Roger S. Pressman, 9th Edition, Mc- Graw Hill International Edition.

Reference Books:

1. Software Engineering, Ian Sommerville, 10th Edition, Pearson.
2. Software Engineering, Principles and Practices, Deepak Jain, Oxford University Press.

Online Learning Resources:

1. <https://nptel.ac.in/courses/106/105/106105182/>
2. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_012605895063871_48827_shared/overview
3. https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_013382690411003_904735_shared/overview

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL06
Name of the Course	Operating Systems Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Practice of UNIX fundamentals, commands & system calls. **(K3)**

CO2: Construct programs using CPU scheduling algorithms. **(K3)**

CO3: Demonstrate IPC, semaphores, monitors, deadlocks. **(K3)**

CO4: Develop programs to solve Page replacement algorithms, file allocation strategies and memory allocation strategies. **(K3)**

Experiments covering the Topics:

- UNIX fundamentals, commands & system calls
- CPU Scheduling algorithms, thread processing
- IPC, semaphores, monitors, deadlocks
- Page replacement algorithms, file allocation strategies
- Memory allocation strategies

Sample Programs:

1. Practice Basic UNIX Commands.
2. Develop programs using the following UNIX operating system calls fork, exec, getpid, exit, wait, close, stat, opendir and readdir
3. Demonstrate UNIX commands like cp, ls, grep, etc.,
4. Demonstrate the following CPU scheduling algorithms
a) FCFS b) SJF c) Priority d) Round Robin
5. Develop a program to control the number of ports opened by the operating system with
a) Semaphore b) Monitors.
6. Illustrate concurrent execution of threads using pthreads library.
7. Solve producer-consumer problem using Semaphores.
8. Demonstrate the following memory allocation methods for fixed partition
a) First fit b) Worst fit c) Best fit
9. Develop programs to simulate the following page replacement algorithms
a) FIFO b) LRU c) LFU
10. Develop program to simulate Paging Technique of memory management.
11. Demonstrate Bankers Algorithm for Dead Lock avoidance and prevention.
12. Develop program to simulate the following file allocation strategies
a) Sequential b) Indexed c) Linked
13. Download and install nachos operating system and experiment with it.

Reference Books:

1. Operating System Concepts, Silberschatz A, Galvin P B, Gagne G, 10th Edition, Wiley, 2018.
2. Modern Operating Systems, Tanenbaum A S, 4th Edition, Pearson, 2016
3. Operating Systems -Internals and Design Principles, Stallings W, 9th edition, Pearson, 2018.
4. Operating Systems: A Concept Based Approach, D.M Dhamdhare, 3rd Edition, McGraw- Hill, 2013

Online Learning Resources:

1. <https://www.cse.iitb.ac.in/~mythili/os/>
2. <http://peterindia.net/OperatingSystems.html>
3. www.cs.washington.edu/~tom/nachos

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL07
Name of the Course	Database Management Systems Lab					
Branch	Common to CSE, CST, CSE(AI) , AI&ML and CSE(DS)					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Construct SQL queries using DDL,DML and DCL commands. (K3)
CO2: Experiment with nested queries, built-in functions and various constraints. (K3)
CO3: Construct PL/SQL programs on Control structures, Cursors and Triggers . (K3)
CO4: Develop a Java program to connect with a database using JDBC. (K3)

Experiments covering the Topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions,
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers,
- Database connectivity- ODBC/JDBC

Sample Programs:

1. Demonstrate Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Construct queries (along with sub queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.
3. Construct queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Construct queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)
5.
 - i. Develop a simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)
 - ii. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Develop programs using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE- APPLICATION ERROR.
8. Develop programs using procedures, passing parameters IN and OUT of PROCEDURES.
9. Develop programs on stored functions, invoke functions in SQL Statements and complex functions.
10. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.
11. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers
12. Construct a table and perform the search operation on table using indexing and non- indexing techniques.
13. Develop a Java program that connects to a database using JDBC
14. Develop a Java program to connect to a database using JDBC and insert values into it
15. Develop a Java program to connect to a database using JDBC and delete values from it

Textbooks:

1. Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH (For Chapters 2, 3, 4)
2. Database System Concepts, 5th edition, Silberschatz, Korth, Sudarsan, TMH (For Chapter 1 and Chapter 5)

Reference Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilesh Shah, "Database Systems Using Oracle", PHI, 2007
 Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE02
Name of the Course	FULL STACK DEVELOPMENT – I					
Branch	Common to CSE, CST, CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate HTML elements and their attributes for designing static web pages. **(K3)**

CO2: Apply appropriate CSS styles to HTML elements. **(K3)**

CO3: Demonstrate JavaScript Pre-defined and User-defined Objects. **(K3)**

CO4: Develop dynamic web pages and validate forms using JavaScript. **(K3)**

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets, Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript - internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Sample Programs:

1. Lists, Links and Images

- a) Develop a HTML program, to explain the working of lists.
Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.
- b) Develop a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.
- c) Develop a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
- d) Develop a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

2. HTML Tables, Forms and Frames

- a) Develop a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>,<td> and attributes: border, rowspan, colspan)
- b) Develop a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).
- c) Develop a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).
- d) Develop a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame □ image, second frame □ paragraph, third frame □ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

- a) Develop a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags.
- b) Develop a HTML program, to embed audio and video into HTML web page.
- c) Develop a program to apply different types (or levels of styles or style specification formats)- inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

- a) Develop a program to apply different types of selector forms
 - i. Simple selector (element, id, class, group, universal)
 - ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
 - iii. Pseudo-class selector
 - iv. Pseudo-element selector
 - v. Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a) Develop a program to demonstrate the various ways you can reference a color in CSS.
- b) Develop a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
- c) Develop a program using the following terms related to CSS font and text:
 - i. font-size ii. font-weight iii. font-style
 - iv. text-decoration v. text-transformation vi. text-alignment
- d) Develop a program, to explain the importance of CSS Box model using
 - i. Content ii. Border iii. Margin iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

- a) Develop a program to embed internal and external JavaScript in a web page.
- b) Develop a program to explain the different ways for displaying output.
- c) Develop a program to explain the different ways for taking input.
- d) Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

7. Java Script Pre-defined and User-defined Objects

- a) Develop a program using document object properties and methods.
- b) Develop a program using window object properties and methods.
- c) Develop a program using array object properties and methods.
- d) Develop a program using math object properties and methods.
- e) Develop a program using string object properties and methods.
- f) Develop a program using regex object properties and methods.
- g) Develop a program using date object properties and methods.
- h) Develop a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. Java Script Conditional Statements and Loops

- a) Develop a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words "LARGER NUMBER" in an information message dialog. If the numbers are equal, output HTML text as "EQUAL NUMBERS".
- b) Develop a program to display week days using switch case.
- c) Develop a program to print 1 to 10 numbers using for, while and do-while loops.
- d) Develop a program to print data in object using for-in, for-each and for-of loops
- e) Develop a program to determine whether a given number is an 'ARMSTRONG NUMBER' or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $13 + 53 + 33 = 153$]
- f) Develop a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1- 10's, 1-2's & 1-1's)

9. Java Script Functions and Events

- a) Design an appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- b) Prepare a HTML file having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
 - i. Factorial of that number
 - ii. Fibonacci series up to that number
 - iii. Prime numbers up to that number
 - iv. Is it palindrome or not
- c) Develop a program to validate the following fields in a registration page
 - i. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
 - ii. Mobile (only numbers and length 10 digits)
 - iii. E-mail (should contain format like xxxxxxxx@xxxxxx.xxx)

Reference Books:

1. Programming the World Wide Web, 7th Edition, Robert W. Sebesta, Pearson, 2013.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jones & Bartlett Learning, 2019 (Chapters 1-11).
3. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasanth Subramanian, 2nd edition, APress, O'Reilly.

Online Learning Resources:

1. <https://www.w3schools.com/html>
2. <https://www.w3schools.com/css>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	-	V23CEAC01
Name of the Course	Environmental Science					
Branch	Common to CSE, CST, CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate multi disciplinary nature of environmental studies and various renewable and non-renewable resources. **(K3)**

CO2: Outline flow and bio-geo- chemical cycles and ecological Pyramids. **(K4)**

CO3: Ascertain various causes of pollution and solid waste management and related preventive measures. **(K5)**

CO4: Discriminate Understand the rainwater harvesting, watershed management, ozone layer depletion and waste land reclamation. **(K4)**

CO5: Illustrate the causes of population explosion, value education and welfare programs. **(K3)**

UNIT-I: Multidisciplinary Nature of Environmental Studies: – Definition, Scope and Importance – Need for Public Awareness. Natural Resources : Renewable and non-renewable resources – Natural resources and associated problems – Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people – Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. – Energy resources:

UNIT-II: Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem:

- a. Forest ecosystem.
- b. Grassland ecosystem
- c. Desert ecosystem
- d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity and Its Conservation : Introduction and Definition: genetic, species and ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

UNIT-III: Environmental Pollution: Definition, Cause, effects and control measures of:

- a. Air Pollution.
- b. Water pollution
- c. Soil pollution
- d. Marine pollution
- e. Noise pollution
- f. Thermal pollution
- g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides.

UNIT-IV Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people; its problems and concerns. Case studies – Environmental ethics: Issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies – Wasteland reclamation. – Consumerism and waste products. –

Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act – Issues involved in enforcement of environmental legislation – Public awareness.

UNIT-V: Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.

Textbooks:

1. Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2. Palaniswamy, Environmental Studies, 2nd Edn., Pearson Education, 2014.
3. Azeem Unnisa S, Environmental Studies, Academic Publishing Company, 2021.
4. Raghavan Nambiar K, Text book of Environmental Studies for Undergraduate Courses as per UGC model Syllabus, Sci Tech Publications (India) Pvt. Ltd, 2010.

Reference Books:

1. Deeksha Dave and E.Sai Baba Reddy, Textbook of Environmental Science, 2nd Edn., Cengage Publications, 2012.
2. Anji Reddy M, Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3. Sharma J P, Comprehensive Environmental Studies, Laxmi publications, 2006.
4. Glynn Henry J and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	0	2	0	0	V23ENT03
Name of the Course	Professional Communication Skills – II					
Branch	Common to All Branches					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Define the meaning and significance of Soft Skills in contrast to Hard Skills and English, and learn how to enhance them. **(K1)**
- CO2:** Identify SWOT and demonstrate soft skills competently in the workplace through intrapersonal skills. **(K2)**
- CO3:** Demonstrate interpersonal skills by displaying leadership and teamwork in the working environment for professional excellence. **(K3)**
- CO4:** Employ oral communication skills effectively and have an approachable demeanour in the workplace. **(K3)**
- CO5:** Produce literacy skills effortlessly, meeting the requirements of the corporate world. **(K3)**

UNIT – I:- Introduction : Definition & Meaning, Importance, Soft Skills vs Hard Skills, Soft Skills vs English- Improving Techniques.

UNIT – II: Intra-Personal : Definition & Meaning—Importance—SWOT analysis—Goal Setting—Emotional Intelligence—Attitudinal Skills—Problem Solving—Time Management—Stress Management.

UNIT – III: Inter-Personal: Definition & Meaning—Importance—Communication skills—Teamwork, Managerial skills —Negotiation skills—Leadership skills—Corporate etiquette.

UNIT – IV: Oral Communication Skills (L&S) & Kinesics:

Listening: Importance-Improving Tips for Listening & Practice.

Kinesics: Eye contact, Facial expressions, Gestures, Appearance, and Dress code.

Speaking: Importance and Suprasegmental Features, Celebrity and Behavioral Interviews.

UNIT – V: Literacy Skills (R&W):

Reading: Strategies, Reading Comprehension (Literal and Inferential).

Writing: Technical Report, Resume with Cover Letter.

Text Books:

- 1) Sherfield, M. Robert et al, Cornerstone Developing Soft Skills, 4/e, Pearson Publication, New Delhi, 2014.
- 2) Alka Wadkar, Life Skills for Success, 1/e, Sage Publications India Private Limited, 2016.

Reference Books:

1. Sambaiah.M. Technical English, Wiley publishers India. New Delhi. 2014.
2. Gangadhar Joshi, From Campus to Corporate, SAGE TEXT.
3. Alex.K, Soft Skills, 3rd ed. S. Chand Publication, New Delhi, 2014.
4. Meenakshi Raman and Sangita Sharma, Technical Communication: Principle and Practice, Oxford University Press, 2009.
5. Shalini Varma, Body Language for Your Success Mantra, 4/e, S. Chand Publication, New Delhi, 2014.
6. Stephen Covey, Seven Habits of Highly Effective People, JMD Book, 2013.

Online Learning Resources:

- https://onlinecourses.nptel.ac.in/noc20_hs60/preview
- <http://www.youtube.com/@softskillsdevelopment6210>
- https://youtube.com/playlist?list=PLLy_2iUCG87CQhELCytvXh0E_y-bOO1_q&si=Fs05Xh8ZrOPsR8F4
- <https://www.coursera.org/learn/people-soft-skills-assessment?language=English>
- <https://www.edx.org/learn/soft-skills>

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST09
Name of the Course	Data Warehousing & Data Mining					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Discuss the basic concepts and techniques of data warehousing and data mining. **(K2)**

CO2: Demonstrate the types of the data to be mined and apply pre-processing methods on raw data. **(K3)**

CO3: Illustrate various Classification Techniques. **(K3)**

CO4: Discover interesting patterns, analyze supervised and unsupervised models and estimate the accuracy of the algorithms. **(K3)**

CO5: Use different clustering techniques to cluster data. **(K3)**

UNIT I: Data Warehousing and Online Analytical Processing: Basic concepts, Data Warehouse Modeling: Data Cube and OLAP, Data Warehouse Design and Usage, Data Warehouse Implementation, Cloud Data Warehouse, Data Mining and Pattern Mining, Technologies, Applications, Major issues, Data Objects & Attribute Types, Basic Statistical Descriptions of Data, Data Visualization, Measuring Data Similarity and Dissimilarity.

UNIT II: Data Preprocessing: An Overview, Data Cleaning, Data Integration, Data Reduction, Data Transformation and Data Discretization.

UNIT III: Classification: Basic Concepts, General Approach to solving a classification problem, Decision Tree Induction: Attribute Selection Measures, Tree Pruning, Scalability and Decision Tree Induction, Visual Mining for Decision Tree Induction, Bayesian Classification Methods: Bayes Theorem, Naïve Bayes Classification, Rule-Based Classification, Model Evaluation and Selection.

UNIT IV: Association Analysis: Problem Definition, Frequent Item set Generation, Rule Generation: Confident Based Pruning, Rule Generation in Apriori Algorithm, **Improving the efficiency of Apriori**, Compact Representation of frequent item sets, FP-Growth Algorithm.

UNIT V: Cluster Analysis: Overview, Basics and Importance of Cluster Analysis, Clustering techniques, Different Types of Clusters; K-means: The Basic K-means Algorithm, K-means Additional Issues, Bi-secting K Means, Agglomerative Hierarchical Clustering: Basic Agglomerative Hierarchical Clustering Algorithm DBSCAN: Traditional Density Center-Based Approach, DBSCAN Algorithm, Strengths and Weaknesses.

Text Books:

1. Data Mining concepts and Techniques, 3rd edition, Jiawei Han, Michel Kamber, Elsevier, 2011.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson, 2012.

Reference Books:

1. Data Mining: VikramPudi and P. Radha Krishna, Oxford Publisher.
2. Data Mining Techniques, Arun K Pujari, 3rd edition, Universities Press, 2013.
3. (NPTEL course by Prof. Pabitra Mitra) http://onlinecourses.nptel.ac.in/noc17_mg24/preview
4. http://www.saedsayad.com/data_mining_map.htm

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST10
Name of the Course	Computer Networks					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Illustrate fundamentals of networks, Reference Models and Transmission Media. (K2)

CO2: Demonstrate Error Detection & Correction and Data Link Layer protocols. (K3)

CO3: Illustrate the protocols of MAC sub layer and Channelization. (K2)

CO4: Utilize connection and connectionless services and Apply Routing and Congestion Control Algorithms. (K3)

CO5: Explain Transport Layer and Application Layer protocols. (K2)

UNIT I: Introduction: Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP.

Physical Layer –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and introduction about unguided media.

UNIT II: Data link layer: Design issues, **Framing:** fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer, **Elementary Data Link Layer protocols:** Simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel. **Sliding window protocol:** One bit, Go back N, Selective Repeat-Stop and wait protocol, Data link layer in HDLC, point to point protocol (PPP)

UNIT III: Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, **Controlled Access:** Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA).

Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet(100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.

UNIT IV: The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service-Comparison of Virtual Circuit and Datagram Networks,

Routing Algorithms-The Optimality principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion control, Congestion prevention policies, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm-Leaky bucket & Token bucket.

Internet Working: How networks differ- How networks can be connected- Tunnelling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6.

UNIT V: The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol-User datagram-UDP services-UDP applications- Transmission control protocol: TCP services- TCP features- Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP.

Application Layer — World Wide Web: HTTP, Electronic Mail-Architecture- web based mail- email security-TELENET-local versus remote Logging-Domain Name System.

Text Books:

1. Computer Networks, Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks, Behrouz A. Forouzan, Fifth Edition TMH.

Reference Books:

1. Data Communications and Networks- Achut S Godbole, AtulKahate, 2nd Edition, McGrawHill
2. Computer Networks, Mayank Dave, 1st Edition, CENGAGE.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST11
Name of the Course	Automata and Compiler Design					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- | | |
|--|-------------|
| CO1: Construct Finite Automata and Regular Expressions. | (K3) |
| CO2: Describe the Compilation Process and Lexical Analysis. | (K2) |
| CO3: Construct Top down and Bottom up Parsing Techniques. | (K3) |
| CO4: Produce Intermediate Code Generation and Runtime Environments. | (K3) |
| CO5: Explain Code Optimization and Code Generation. | (K2) |

UNIT I: Formal Language and Regular Expressions: Alphabet, Strings, Language, Finite Automaton- Design of DFA, Design of NFA, Equivalence between NFA and DFA, Finite Automata with ϵ -Transition, Equivalence between NFA and ϵ -NFA. **Regular Expression:** Regular expressions Equivalence between Regular Expressions and Finite Automata, Chomsky Hierarchy.

UNIT II: Compiler: Definition, Structure of a compiler. **Lexical Analysis:** The Role of the Lexical Analyzer, Specification of Tokens, Recognition of Tokens and the Lexical-Analyzer Generator-Lex. **Context Free grammars:** Context free grammars, derivation, parse trees, Ambiguous Grammar, Writing a Grammar-Elimination of Left Recursion, Left Factoring.

UNIT III: Top Down Parsing: First and Follow, LL(1) Grammars, **Bottom-Up Parsing:** Bottom Up Parser Classification, Reductions, Handle Pruning, Shift-Reducing, Constructing SLR Parsing Tables, construction of CLR (1), LALR Parsing tables, Comparison of all Bottom Up approaches.

UNIT IV: Semantic Analysis: Syntax Directed Definitions, Evaluation Orders for SDD's
Intermediate Code Generation: Variants of Syntax Trees, Three-Address Code, Basic blocks and Flow graphs, Control Flow. **Run-Time Environments:** Storage Organization, Stack Allocation of Space, Heap Management

UNIT V: Code optimization: Machine Independent Optimization. The principle sources of Optimization, optimization of Basic blocks, peep hole Optimization, Introduction to Data flow Analysis.

Code generation: Issues in design of code generation, The target Language, Address in the target code, A Simple Code generation.

Text Books:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcroft, R.Motwani and J.D.Ullman, 3rd Edition, Pearson, 2008.
2. Compilers Principles, Techniques and Tools Aho, Ullman, Ravisethi, Pearson Education.

Reference Books:

1. Louden: "Compiler Construction, Principles & Practice", 1st Edition, Thomson Press, 2006.
 2. Tremblay J P, Sorenson G P: "The Theory & Practice of Compiler writing", 1st Edition, BSP Publication, 2010.
- Theory of Computation, V. Kulkarni, Oxford University Press, 2013.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE01
Name of the Course	Object Oriented Analysis and Design (Professional Elective-I)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Illustrate the inherent complexity and attributes of software systems and Design structured solutions for complex problems by applying principles of system architecture. **(K3)**
- CO2:** Use the significance and principles of object-oriented modeling and apply UML concepts to represent structural aspects of software systems using appropriate diagrams. **(K3)**
- CO3:** Develop detailed class and object models and differentiate between basic and advanced structural constructs such as interfaces, packages, and roles for effective software design. **(K3)**
- CO4:** Construct behavioral models such as use case, interaction, and activity diagrams to illustrate user interactions and system functionalities in real-world applications. **(K3)**
- CO5:** Design comprehensive behavioral and architectural models using state charts, component, and deployment diagrams to demonstrate system dynamics and implementation strategies. **(K3)**

UNIT I: Introduction: The Structure of Complex systems, The Inherent Complexity of Software, Attributes of Complex System, Organized and Disorganized Complexity, Bringing Order to Chaos, Designing Complex Systems. **Case Study:** System Architecture: Satellite-Based Navigation

UNIT II: Introduction to UML: Importance of modeling, principles of modeling, object-oriented modeling, conceptual model of the UML, Architecture, and Software Development Life Cycle. **Basic Structural Modeling:** Classes, Relationships, common Mechanisms, and diagrams. **Case Study:** Control System: Traffic Management.

UNIT III: Class & Object Diagrams: Terms, concepts, modeling techniques for Class & Object Diagrams. **Advanced Structural Modeling:** Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. **Case Study:** AI: Cryptanalysis.

UNIT IV: Basic Behavioral Modeling-I: Interactions, Use cases, Use case Diagrams, Interaction diagrams, Activity Diagrams. **Case Study:** Web Application: Vacation Tracking System

UNIT V: Advanced Behavioral Modeling: Events and signals, state machines, processes and Threads, time and space, state chart diagrams. **Architectural Modeling:** Component, Deployment, Component diagrams and Deployment diagrams. **Case Study:** Weather Forecasting

Text Books:

1. Grady BOOCH, Robert A. Maksimchuk, Michael W. ENGLE, Bobbi J. Young, Jim Conallen, Kellia Houston, "Object- Oriented Analysis and Design with Applications", 3rd Edition, 2013, PEARSON.
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, 1st Edition, Pearson Education.

Reference Books:

1. Meilir Page-Jones: Fundamentals of Object-Oriented Design in UML, 1st Edition, Pearson Education.
2. Pascal Roques: Modeling Software Systems Using UML2, 2004, WILEY- Dreamtech India Pvt. Ltd.
3. Atul Kahate: Object Oriented Analysis & Design, 2004, The McGraw-Hill Companies.
4. Applying UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, 2nd Edition, Craig Larman, Pearson Education.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE02
Name of the Course	Artificial Intelligence (Professional Elective-I)					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Understand the fundamental concepts of Artificial Intelligence. **(K2)**

CO2: Employ various search strategies and game-playing algorithms to solve AI problems. **(K3)**

CO3: Apply knowledge representation techniques and probabilistic reasoning methods to model AI systems. **(K3)**

CO4: Use logic based inference techniques and learning methods for intelligent decision- making. **(K3)**

CO5: Demonstrate the principles of expert system architecture and knowledge acquisition in AI applications. **(K3)**

UNIT I: Introduction: AI problems, foundation of AI and history of AI intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT II : Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversarial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.

UNIT III: Representation of Knowledge: Knowledge representation issues, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems. Reasoning under uncertainty, review of probability, Bayes' probabilistic interferences and Dempstershafer theory.

UNIT IV: Logic concepts: First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts forward chaining, Backward chaining, Resolution, Learning from observation Inductive learning, Decision trees, Explanation based learning, Statistical Learning methods, Reinforcement Learning.

UNIT V: Expert Systems: Architecture of expert systems, Roles of expert systems – Knowledge Acquisition Meta knowledge Heuristics. Typical expert systems – MYCIN, DART, XCON: Expert systems shells. **AI Agent Communication Protocols:** Model Context Protocol (MCP), Agent-to-Agent Protocol (A2A) / Inter-agent Protocol and Agent Communication Protocol (ACP).

Textbooks:

1. S. Russel and P. Norvig, "Artificial Intelligence – A Modern Approach", Second Edition, Pearson Education.
2. Kevin Night and Elaine Rich, Nair B., "Artificial Intelligence (SIE)", McGraw Hill

Reference Books:

1. David Poole, Alan Mackworth, Randy Goebel, "Computational Intelligence: a logical approach", Oxford University Press.
2. G. Luger, "Artificial Intelligence: Structures and Strategies for complex problemsolving", Fourth Edition, Pearson Education.
3. J. Nilsson, "Artificial Intelligence: A new Synthesis", Elsevier Publishers.
4. Artificial Intelligence, Saroj Kaushik, CENGAGE Learning.

Online Learning Resources:

1. <https://ai.google/>
2. https://swayam.gov.in/nd1_noc19_me71/preview

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE03
Name of the Course	Microprocessors & Microcontrollers(Professional Elective-I)					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Understand the fundamental concepts of Artificial Intelligence. **(K2)**

CO2: Employ various search strategies and game-playing algorithms to solve AI problems. **(K3)**

CO3: Apply knowledge representation techniques and probabilistic reasoning methods to model AI systems. **(K3)**

CO4: Use logic based inference techniques and learning methods for intelligent decision- making. **(K3)**

CO5: Demonstrate the principles of expert system architecture and knowledge acquisition in AI applications. **(K3)**

UNIT I: 8086 Architecture: Main features, pin diagram/description, 8086 microprocessor family, internal architecture, bus interfacing unit, execution unit, interrupts and interrupt response, 8086 system timing, minimum mode and maximum mode configuration.

UNIT II: 8086 Programming: Program development steps, instructions, addressing modes, assembler directives, writing simple programs with an assembler, assembly language program development tools.

UNIT III: 8086 Interfacing: Semiconductor memories interfacing (RAM, ROM), Intel 8255 programmable peripheral interface, Interfacing switches and LEDs, Interfacing seven segment displays, software and hardware interrupt applications, Intel 8251 USART architecture and interfacing, Intel 8237a DMA controller, stepper motor, A/D and D/A converters, Need for 8259 programmable interrupt controllers.

UNIT IV: Microcontroller, Architecture of 8051, Special Function Registers(SFRs), I/O Pins Ports and Circuits, Instruction set, Addressing modes, Assembly language programming.

UNIT V: Interfacing Microcontroller, Programming 8051 Timers, Serial Port Programming, Interrupts Programming, LCD & Keyboard Interfacing, ADC, DAC & Sensor Interfacing, External Memory Interface, Stepper Motor and Waveform generation, Comparison of Microprocessor, Microcontroller, PIC and ARM processors

Textbooks:

1. Microprocessors and Interfacing – Programming and Hardware by Douglas V Hall, SSSP Rao, Tata McGraw Hill Education Private Limited, 3rd Edition, 1994.
2. K M Bhurchandi, A K Ray, Advanced Microprocessors and Peripherals, 3rd edition, McGraw Hill Education, 2017.
3. Raj Kamal, Microcontrollers: Architecture, Programming, Interfacing and System Design, 2nd edition, Pearson, 2012.

Reference Books:

1. Ramesh S Gaonkar, Microprocessor Architecture Programming and Applications with the 8085, 6th edition, Penram International Publishing, 2013.
2. Kenneth J. Ayala, The 8051 Microcontroller, 3rd edition, Cengage Learning, 2004.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE04
Name of the Course	Quantum Computing(Professional Elective-I)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Understand the foundational principles of quantum mechanics relevant to computing. **(K2)**

CO2: Explain quantum computing or related interdisciplinary fields (mathematics, physics). **(K2)**

CO3: Construct various physical quantum units of information. **(K3)**

CO4: Demonstrate the various quantum Algorithms. **(K3)**

CO5: Illustrate the role of noise in quantum systems and the necessity for error correction. **(K3)**

UNIT I: History of Quantum Computing: Importance of Mathematics, Physics and Biology. Introduction to Quantum Computing: Bits Vs Qubits, Classical Vs Quantum logical operations

UNIT II: Background Mathematics: Basics of Linear Algebra, Hilbert space, Probabilities and measurements. Background Physics: Paul's exclusion Principle, Superposition, Entanglement and super-symmetry, density operators and correlation, basics of quantum mechanics, Measurements in bases other than computational basis. Background Biology: Basic concepts of Genomics and Proteomics (Central Dogma)

UNIT III: Qubit: Physical implementations of Qubit. Qubit as a quantum unit of information. The Bloch sphere Quantum Circuits: single qubit gates, multiple qubit gates, designing the quantum circuits. Bell states.

UNIT IV: Quantum Algorithms: Classical computation on quantum computers. Relationship between quantum and classical complexity classes. Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor's factorization algorithm, Grover's search algorithm.

UNIT V: Noise and error correction: Graph states and codes, Quantum error correction, fault-tolerant computation. Quantum Information and Cryptography: Comparison between classical and quantum information theory. Quantum Cryptography, Quantum teleportation

Textbooks:

1. Quantum Computation and Quantum Information, Nielsen M. A., Cambridge
2. Programming Quantum Computers, Essential Algorithms and Code Samples, Eric R Johnson, Nic Harrigan, Mercedes Ginemo, Segovia, Oreilly

Reference Books:

1. Quantum Computing for Computer Scientists, Noson S. Yanofsk, Mirco A. Mannucci
2. Principles of Quantum Computation and Information, Benenti G., Casati G. and Strini G., Vol.I: Basic Concepts, Vol II
3. Basic Tools and Special Topics, World Scientific. Pittenger A. O., An Introduction to Quantum Computing Algorithms.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL08
Name of the Course	Data Mining Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Describe Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment. **(K2)**

CO2: Develop a data warehouse or data mart to present information needed by management in a form that is usable. **(K3)**

CO3: Practice hands-on experience working with all real data sets. **(K3)**

CO4: Compute real data sets using popular data mining tools such as WEKA, Python Libraries. **(K3)**

CO5: Develop ability to design various algorithms based on data mining tools. **(K3)**

List of Experiments

(Software Requirements: The tasks should be implementing by using WEKA Tool/Python/R-Tool/Rapid Tool/Oracle Data mining)

1. Illustrate the Creation of a Data Warehouse.

- Build Data Warehouse/Data Mart (using open source tools like Pentaho Data Integration Tool, Pentaho Business Analytics; or other data warehouse tools like Microsoft-SSIS, Informatica, Business Objects, etc.,)
- Design multi-dimensional data models namely Star, Snowflake and Fact Constellation schemas for any one enterprise (ex. Banking, Insurance, Finance, Healthcare, manufacturing, Automobiles, sales etc).
- Develop ETL scripts and implement using data warehouse tools.
- Perform Various OLAP operations such slice, dice, roll up, drill up and pivot

2. Describe the machine learning tool “WEKA”

- Explore WEKA Data Mining/Machine Learning Toolkit.
 - Downloading and/or installation of WEKA data mining toolkit.
 - Understand the features of WEKA toolkit such as Explorer, Knowledge Flow interface, Experimenter, command-line interface.
 - Navigate the options available in the WEKA (ex. Select attributes panel, Preprocess panel, Classify panel, Cluster panel, Associate panel and Visualize panel)
- Study the arff file format Explore the available data sets in WEKA. Load a data set (ex. Weather dataset, Iris dataset, etc.)
- Load each dataset and observe the following:
 1. List the attribute names and they types
 2. Number of records in each dataset
 3. Identify the class attribute (if any)
 4. Plot Histogram
 5. Determine the number of records for each class.
 6. Visualize the data in various dimensions.

3. Demonstrate data preprocessing tasks and also performing association rule mining on data sets using WEKA/Python.

- Explore various options available in Weka for preprocessing data and apply Unsupervised filters like Discretization, Resample filter, etc. on each dataset
- Load weather. nominal, Iris, Glass datasets into Weka and run Apriori Algorithm with different support and confidence values.
- Study the rules generated. Apply different discretization filters on numerical attributes and run the Apriori association rule algorithm. Study the rules generated.
- Derive interesting insights and observe the effect of discretization in the rule generation process.

4. Demonstrate performing classification on data sets

- Load each dataset and run 1d3, J48 classification algorithm. Study the classifier output. Compute entropy values, Kappa statistic.
- Extract if-then rules from the decision tree generated by the classifier, Observe the confusion matrix.
- Load each dataset into Weka/Python and perform k-Nearest Neighbour classification. Interpret the results obtained.
- Plot RoC Curves
- Compare classification results of ID3, J48, Naïve-Bayes and k-NN classifiers for each dataset, and deduce which classifier is performing best and poor for each dataset and justify.

5. Demonstrate performing clustering of data sets

- Load each dataset into Weka/R and run simple k-means clustering algorithm with different values of k (number of desired clusters).
- Study the clusters formed. Observe the sum of squared errors and centroids, and derive insights.
- Explore other clustering techniques available in Weka/R.
- Explore visualization features of Weka/R to visualize the clusters. Derive interesting insights and explain.

6. Demonstrate knowledge flow application on data sets into Weka/R

- Develop a knowledge flow layout for finding strong association rules by using Apriori, FP Growth algorithms
 - Set up the knowledge flow to load an ARFF (batch mode) and perform a cross validation using J48 algorithm
 - Demonstrate plotting multiple ROC curves in the same plot window by using j48 and Random forest tree.
7. Demonstrate ZeroR technique on Iris dataset (by using necessary preprocessing technique(s)) and share your observations
 8. Develop a java program to prepare a simulated data set with unique instances.
 9. Develop a Python program to generate frequent item sets / association rules using Apriori algorithm
 10. Develop a program to calculate chi-square value using Python/R. Report your observation.
 11. Develop a program of Naive Bayesian classification using Python/R programming language.
 12. Implement a Java/R program to perform Apriori algorithm
 13. Develop a R program to cluster your choice of data using simple k-means algorithm using JDK
 14. Develop a program of cluster analysis using simple k-means algorithm Python/R programming language.
 15. Develop a program to compute/display dissimilarity matrix (for your own dataset containing at least four instances with two attributes) using Python
 16. Visualize the datasets using matplotlib in python/R.(Histogram, Box plot, Bar chart, Pie chart etc.,)

Textbooks:

1. Data Mining concepts and Techniques, 3rd edition, Jiawei Han, Michel Kamber, Elsevier, 2011.
2. Introduction to Data Mining: Pang-Ning Tan & Michael Steinbach, Vipin Kumar, Pearson, 2012.

Reference Books:

1. Data Mining: Practical Machine Learning Tools and Techniques, IanH.Witten, Eibe Frank, Mark A.Hall,3rdEdition,MorganKaufmannPublishers.
2. Data Mining Concepts and Techniques,JiaweiHan,MichelineKamber,JianPei,3rd Edition, Morgan Kaufmann Publishers.
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach,VipinKumar,1stEdition,Pearson Education Inc.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL09
Name of the Course	Computer Networks Lab					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Objectives: After Successful completion of the Course, the student will be able to:

CO 1:Develop programs on data link layer framing methods and checksum. (K3)

CO 2:Develop programs to implement Error detection technique and Sliding window protocol. (K3)

CO 3:Develop programs to implement Routing and Congestion Control Algorithms. (K3)

CO 4:Demonstrate various networking tools like Wireshark, NS2 and Nmap. (K3)

List of Experiments

1. Study of Network devices in detail and connect the computers in Local Area Network.
2. Develop a Program to implement the data link layer framing methods such as
 - i) Character stuffing ii) bit stuffing.
3. Develop a Program to implement data link layer framing method checksum.
4. Develop a program for Hamming Code generation for error detection and correction.
5. Develop a Program to implement on a data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP.
6. Develop a Program to implement Sliding window protocol for Goback N.
7. Develop a Program to implement Sliding window protocol for Selective repeat.
8. Develop a Program to implement Stop and Wait Protocol.
9. Develop a program for congestion control using leaky bucket algorithm
10. Develop a Program to implement Dijkstra's algorithm to compute the Shortest path through a graph.
11. Develop a Program to implement Distance vector routing algorithm by obtaining routing table at each node (Take an example subnet graph with weights indicating delay between nodes).
12. Develop a Program to implement Broadcast tree by taking subnet of hosts.
13. Wireshark
 - i. Packet Capture Using Wire shark
 - ii. Starting Wire shark
 - iii. Viewing Captured Traffic
 - iv. Analysis and Statistics & Filters.
14. Discover active hosts, open ports and running services in a network using Nmap
15. Find Operating System in a host using Nmap
16. Do the following using NS2 Simulator
 - i. NS2 Simulator-Introduction
 - ii. Simulate to Find the Number of Packets Dropped
 - iii. Simulate to Find the Number of Packets Dropped by TCP/UDP
 - iv. Simulate to Find the Number of Packets Dropped due to Congestion
 - v. Simulate to Compare Data Rate& Throughput.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE03
Name of the Course	Full Stack Development-2 (Skill Enhancement Course)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the key concepts of ExpressJS such as routing, HTTP methods, middleware, templating, and handling form data. **(K2)**
- CO2:** Apply ExpressJS to implement cookies, sessions, authentication, database operations, and RESTful APIs in server-side applications. **(K3)**
- CO3:** Develop ReactJS components using JSX, props, state, and styling to handle events and user interactions. **(K3)**
- CO4:** Analyse ReactJS techniques such as conditional rendering, forms, routing, hooks, and screen updates to manage dynamic content. **(K4)**
- CO5:** Apply MongoDB CRUD operations to create and manage databases, collections, and documents for full-stack applications. **(K3)**

Experiments covering the Topics:

- ExpressJS – Routing, HTTP Methods, Middleware, Templating, Form Data
- ExpressJS – Cookies, Sessions, Authentication, Database, RESTful APIs
- ReactJS – Render HTML, JSX, Components – function & Class, Props and States, Styles, Respond to Events
- ReactJS – Conditional Rendering, Rendering Lists, React Forms, React Router, Updating the Screen
- ReactJS – Hooks, Sharing data between Components, Applications – To-do list and Quiz
- MongoDB – Installation, Configuration, CRUD operations, Databases, Collections and Records

Sample Experiments:

1. ExpressJS – Routing, HTTP Methods, Middleware.

- a. Develop a program to define a route, Handling Routes, Route Parameters, Query Parameters and URL building.
- b. Develop a program to accept data, retrieve data and delete a specified resource using http methods.
- c. Develop a program to show the working of middleware.

2. ExpressJS – Templating, Form Data

- a. Develop a program using templating engine.
- b. Develop a program to work with form data.

3. ExpressJS – Cookies, Sessions, Authentication

- a. Develop a program for session management using cookies and sessions.
- b. Develop a program for user authentication.

4. ExpressJS – Database, RESTful APIs

- a. Develop a program to connect MongoDB database using Mongoose and perform CRUD operations.
- b. Develop a program for single page application using RESTful APIs.

5. ReactJS – Render HTML, JSX, Components – function & Class

- a. Develop a program to render HTML to a web page.
- b. Develop a program for writing markup with JSX.
- c. Develop a program for creating and nesting components (function and class).

6. ReactJS – Props and States, Styles, Respond to Events

- a. Develop a program to work with props and states.
- b. Develop a program to add styles (CSS & Sass Styling) and display data.
- c. Develop a program for responding to events.

7. ReactJS – Conditional Rendering, Rendering Lists, React Forms

- a. Develop a program for conditional rendering.
- b. Develop a program for rendering lists.
- c. Develop a program for working with different form fields using react forms.

8. ReactJS – React Router, Updating the Screen

- a. Develop a program for routing to different pages using react router.
- b. Develop a program for updating the screen.

9. ReactJS – Hooks, Sharing data between Components

- a. Develop a program to understand the importance of using hooks.
- b. Develop a program for sharing data between components.

10. MongoDB – Installation, Configuration, CRUD operations

- a. Install MongoDB and configure ATLAS
- b. Develop MongoDB queries to perform CRUD operations on document using insert(), find(), update(), remove()

11. MongoDB – Databases, Collections and Records

- a. Develop MongoDB queries to Create and drop databases and collections.
- b. Develop MongoDB queries to work with records using find(), limit(), sort(), createIndex(), aggregate().

12. Augmented Programs: (Any 2 must be completed)

- a. Design a to-do list application using NodeJS and ExpressJS.
- b. Design a Quiz app using ReactJS.
- c. Complete the MongoDB certification from MongoDB University website.

Text Books:

1. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2nd edition, APress, O'Reilly.
2. Node.Js in Action, Mike Cantelon, Mark Harter, T.J. Holowaychuk, Nathan Rajlich, Manning Publications. (Chapters 1-11)
3. React Quickly, AzatMardan, Manning Publications (Chapters 1-8, 12-14)

Web Links:

1. ExpressJS - <https://www.tutorialspoint.com/expressjs>
2. ReactJS - <https://www.w3schools.com/REACT> (and) <https://react.dev/learn#>
3. MongoDB - <https://learn.mongodb.com/learning-paths/introduction-to-mongodb> .

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23CSES01
Name of the Course	Master Coding and Competitive Programming - Part-1					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Apply Mathematical reasoning and number theory to solve real world problems in linear time. (K3)
CO2: Use of modular arithmetic, to solve complex problems in linear time , logarithmic. (K3)
CO3: Use of Prime Factorization and complex solve problems. (K3)
CO4: Analyse different techniques including sieve to find prime numbers and evaluate efficiency of these methods. (K4)
CO5: Experiment with Hashing and searching techniques to solve problems on Arrays in Linear time. (K3)

List of Experiments

1. Develop Programs to solve problems based on Mathematical logic, Reasoning and number theory
2. Develop programs using different techniques to find prime number
3. Develop programs using Sieve method and optimize Complexity of finding prime number
4. Develop Programs based on series, patterns
5. Develop programs on concept of Fibonacci series
6. Develop programs on strings including palindrome and anagram concepts
7. Develop programs to search pattern in a string
8. Develop programs for String Processing.

Text Books:

1. Java The Complete Reference - Eleventh Edition, Herbert Schildt, Oracle
2. Guide to Competitive Programming by Antti Laaksonen
3. Programming challenges by Steven S Skiena

Tools:

1. practice.geeksforgeeks.com
2. leetcode.com
3. codingninjas.com
4. Hackerrank.com
5. Interviewbit.com

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1	V23CSES02
Name of the Course	User Interface Design Using Flutter					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO 1:** Apply Dart and Flutter setup procedures to build basic mobile application components. **(K3)**
CO 2: Develop responsive user interfaces using layout widgets and media queries. **(K3)**
CO 3: Apply navigation techniques and state management strategies to enhance functionality in Flutter applications. **(K3)**
CO 4: Construct functional mobile app components using API integration, animations, custom widgets, and debugging tools. **(K3)**

List of Experiments

Students need to implement the following experiments

1. a) Show the Installation process of Flutter and Dart SDK.
 b) Develop a simple Dart program to understand the language basics.
2. a) Practice various Flutter widgets (Text, Image, Container, etc.).
 b) Develop different layout structures using Row, Column, and Stack widgets.
3. a) Develop a responsive UI that adapts to different screen sizes.
 b) Develop media queries and breakpoints for responsiveness.
4. a) Illustrate the navigation Set up process between different screens using Navigator.
 b) Develop navigation with named routes.
5. a) Explain about stateful and stateless widgets.
 b) Develop state management using set State and Provider.
6. a) Develop custom widgets for specific UI elements.
 b) Apply styling using themes and custom styles.
7. a) Develop a form with various input fields.
 b) Develop form validation and error handling.
8. a) Apply animations to UI elements using Flutter's animation framework.
 b) Experiment with different types of animations (fade, slide, etc.).
9. a) Demonstrate data Fetching from a REST API.
 b) Show the fetched data in a meaningful way in the UI.
10. a) Develop unit tests for UI components.
 b) Use Flutter's debugging tools to identify and fix issues.

Text Books:

1. Marco L. Napoli, Beginning Flutter: A Hands-on Guide to App Development.
2. Rap Payne, Beginning App Development with Flutter: Create Cross-Platform Mobile Apps 1st Edition, Apress.
6. Richard Rose, Flutter & Dart Cookbook, Developing Full stack Applications for the Cloud, Oreilly.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	2	0	0	V23ENT04
Name of the Course	English for Employability					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Demonstrate proficiency in verbal ability, including word meanings, synonyms, antonyms, and contextual usage, to enhance their communication skills and excel in competitive exams. **(K3)**

CO2: Effectively apply various reading techniques and comprehension skills, including literal, inferential, critical, evaluative, and appreciative reading, to accurately understand and interpret complex texts. **(K3)**

CO3: Apply knowledge and skills to participate effectively in group discussions on various topics, showcasing critical thinking, communication, and teamwork abilities. **(K3)**

CO4: Construct a well-organized and compelling resume tailored to specific job descriptions and industries. **(K3)**

CO5: Analyse the skills required to excel in an interview and identify areas for improvement. **(K1)**

UNIT I: VERBAL ABILITY-Words; Meanings; Synonyms; Antonyms; Contextual Usage; Various MNC mock test papers practice

UNIT II: READING COMPREHENSION-Reading Techniques; Types of RC; Literal; Inferential; Critical; Evaluative; Appreciative

UNIT III: GROUP DISCUSSION-Dos & Don'ts; Types of GD; Technical Topics; Case Studies; Abstract Topics; Current Affairs

UNIT IV: RESUME BUILDING-Tailoring resume as per job description; Making Video Resume (Visume)

UNIT V: MOCK INTERVIEWS-Tailored to Specific job roles; Dos & Don'ts; Types of Mock Interviews; Traditional-Behavioural-Technical

Reference Books:

1. Lewis, N. (1978). Word Power Made Easy. Pocket Books.
2. Wren, P. C., & Martin, H. (1990). High School English Grammar & Composition. S. Chand.
3. Krannich, C. R. (2004). 101 Dynamite Answers to Interview Questions. Impact Publications.
4. Jackson, T. (2009). Resume Magic: Trade Secrets of a Professional Resume Writer. JIST Publishing.
5. Lisa McGrimmon, The Resume Writing Guide: A Step-by-Step Workbook for Creating a Winning Resume (2013). CareerChoiceGuide; 2nd edition.
6. Puri, G. (2018). Verbal Ability and Reading Comprehension for CAT and Other MBA Entrance Examinations. Upkar Prakashan.

Semester	V	L	T	P	C	COURSE CODE
Regulation	V23	0	2	0	0	
Name of the Course	Aptitude					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Analyze appropriate methods of logical thinking on Ratio and Proportion, Partnership, LCM and HCF, Number System, Areas & Volumes. (K2)

CO2: Demonstrate problem solving skills through the concepts of Percentages, Profit and loss, Simple Interest & Compound Interest and Allegation. (K3)

CO3: Calculate the end results of Cubes, Dice and Data Analysis, Time & Work, Time & Distance, Race & Games. (K4)

UNIT I: Ratio and Proportion, Partnership, LCM and HCF, Number System, Areas & Volumes.

UNIT II: Percentages, Profit and loss, Simple Interest & Compound Interest and Allegation.

UNIT III: Cubes, Dice and Data Analysis, Time & Work, Time & Distance, Race & Games.

Text Book: WORK BOOK ON APTITUDE

Reference Books:

- ❖ Training & Placement cell, 2025, Workbook -2 on Aptitude, Sri Vasavi Engineering College.
- ❖ M Tyra, 2013, Magical Book on Quicker maths, BSC Publications.

Hyperlinks

1. <https://www.indiabix.com/>
2. <https://www.campusgate.co.in/>
3. <https://www.instacks.com/>

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST12
Name of the Course	Machine Learning					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain machine learning paradigms, data preparation, and model evaluation. **(K2)**

CO2: Apply distance-based techniques and algorithms to classify and regress data using proximity measures. **(K3)**

CO3: Construct decision tree and Bayesian models to perform classification and regression. **(K3)**

CO4: Demonstrate linear discriminants, SVMs, and neural networks to solve classification and regression problems in machine learning. **.(K3)**

CO5: Apply ensemble techniques and graphical model-based approaches to develop predictive and sequential models for real-world problems. **(K3)**

UNIT I: Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by Rate, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Stages in Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Prediction, Search and Learning, Data Sets.

UNIT II: Nearest Neighbour-Based Models: Introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures ,K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, KNN Regression, Performance of Classifiers, Performance of Regression Algorithms.

UNIT III: Models Based on Decision Trees: Decision Trees for Classification, Impurity Measures, Properties, Regression Based on Decision Trees, Bias–Variance Trade-off, Random Forests for Classification and Regression. The Bayes Classifier: Introduction to the Bayes Classifier, Bayes’ Rule and Inference, The Bayes Classifier and its Optimality, Multi-Class Classification, Class Conditional Independence and Naive Bayes Classifier (NBC)

UNIT IV: Linear Discriminants for Machine Learning: Introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Non-Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Backpropagation for Training an MLP.

UNIT V: Ensemble Learning: Boosting, Bagging, Random Forest, Different Ways to Combine Classifiers. **Graphical Models:** Bayesian Networks, Markov Random Fields, Hidden Markov Models, Tracking Methods.

Textbooks:

1. “Machine Learning Theory and Practice”, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024
2. “Machine Learning: An Algorithmic Perspective”, Stephen Marsland, 2nd Ed., CRC Press.

Reference Books:

1. “Machine Learning”, Tom M. Mitchell, McGraw-Hill Publication, 2017
2. “Machine Learning in Action”, Peter Harrington, DreamTech
3. “Introduction to Data Mining”, Pang-Ning Tan, Michel Stenbach, Vipin Kumar, 7th Edition, 2019.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST13
Name of the Course	Cloud Computing					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the evolving utility computing model called cloud computing. **(K2)**
CO2: Explain the different service models of cloud computing, such as IaaS, PaaS, and SaaS. **(K2)**
CO3: Explain the core concepts of cloud technologies like distributed computing, service-oriented architecture, and virtualization. **(K2)**
CO4: Analyze the security and other challenges in cloud computing. **(K4)**
CO5: Apply advanced concepts such as containers, serverless computing, and cloud-centric Internet of Things. **(K3)**

UNIT I: Introduction to Cloud Computing Fundamentals: Cloud computing at a glance, defining a cloud, cloud computing reference model, types of services (IaaS, PaaS, SaaS), cloud deployment models (public, private, hybrid), utility computing, cloud computing characteristics and benefits, cloud service providers (Amazon Web Services, Microsoft Azure, Google AppEngine).

UNIT II: Cloud Enabling Technologies: Ubiquitous Internet, parallel and distributed computing, elements of parallel computing, hardware architectures for parallel computing (SISD, SIMD, MISD, MIMD), elements of distributed computing, Inter-process communication, technologies for distributed computing, remote procedure calls (RPC), service-oriented architecture (SOA), Web services, virtualization.

UNIT III: Virtualization and Containers: Characteristics of virtualized environments, taxonomy of virtualization techniques, virtualization and cloud Computing, pros and cons of virtualization, technology examples (XEN, VMware), building blocks of containers, container platforms (LXC, Docker), container orchestration, Docker Swarm and Kubernetes, public cloud VM (e.g. Amazon EC2) and container (e.g. Amazon Elastic Container Service) offerings.

UNIT IV: Cloud computing challenges : Economics of the cloud, cloud interoperability and standards, scalability and fault tolerance, energy efficiency in clouds, federated clouds, cloud computing security, fundamentals of computer security, cloud security architecture, cloud shared responsibility model, security in cloud deployment models.

UNIT V: Advanced concepts in Cloud Computing: Serverless computing, Function-as-a-Service, serverless computing architecture, public cloud (e.g. AWS Lambda) and open-source (e.g. OpenFaaS) serverless platforms, Internet of Things (IoT), applications, cloud-centric IoT and layers, edge and fog computing, DevOps, infrastructure-as-code, quantum cloud computing.

Textbooks

1. Mastering Cloud Computing, Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi, ShivanandaPoojara, Satish N. Srirama, 2nd Edition, Mc Graw Hill, 2024.
2. Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.

Reference Books:

1. Cloud Computing, Theory and Practice, Dan C Marinescu, 2nd edition, MK Elsevier, 2018.
2. Essentials of cloud Computing, K. Chandrasekharan, CRC press, 2014.
3. Online documentation and tutorials from cloud service providers (e.g., AWS, Azure, GCP).

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CST14
Name of the Course	Cryptography & Network Security					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain fundamentals of security concepts and mathematical support of Cryptography. **(K2)**

CO2: Discuss Symmetric cryptosystems. **(K2)**

CO3: Discuss Asymmetric cryptosystems. **(K2)**

CO4: Apply Hash functions & Digital Signatures to provide authentication and integrity. **(K3)**

CO5: Demonstrate various methods of Mutual trust and mail security Network Security Scenarios. **(K3)**

UNIT I: Basic Principles: Security Goals, Cryptographic Attacks, Services and Mechanisms, Mathematics of Cryptography- integer arithmetic, modular arithmetic, matrices, linear congruence.

UNIT II: Symmetric Encryption: Mathematics of Symmetric Key Cryptography-algebraic structures, $GF(2^n)$ Fields, Introduction to Modern Symmetric Key Ciphers-modern block ciphers, modern stream ciphers, Data Encryption Standard- DES structure, DES analysis, Security of DES, Multiple DES, Advanced Encryption Standard-transformations, key expansions, AES ciphers, Analysis of AES.

UNIT III: Asymmetric Encryption: Mathematics of Asymmetric Key Cryptography-primality testing, factorization, CRT, Asymmetric Key Cryptography- RSA crypto system Rabin cryptosystem, Elgamal Crypto system, ECC

UNIT IV: Data Integrity, Digital Signature Schemes & Key Management: Message Integrity and Message Authentication-message integrity, Random Oracle model, Message authentication, Cryptographic Hash Functions-whirlpool, SHA-512, Digital Signature- process, services, attacks, schemes, applications, Key Management-symmetric key distribution, Kerberos.

UNIT V: Network Security-I: Security at application layer: PGP and S/MIME, Security at the Transport Layer: SSL and TLS, **Network Security-II:** Security at the Network Layer: IPSec-two modes, two security protocols, security association, IKE, ISAKMP, System Security-users, trust, trusted systems, buffer overflow, malicious software, worms, viruses, IDS, Firewalls.

Textbooks:

1. Cryptography and Network Security, 3rd Edition Behrouz A Forouzan, Deb deep Mukhopadhyay, McGraw Hill,2015
2. Cryptography and Network Security,4th Edition, William Stallings, (6e) Pearson,2006
3. Everyday Cryptography, 1st Edition, Keith M.Martin, Oxford,2016

Reference Books:

1. Network Security and Cryptography, Bernard Meneges 1st Edition, Cengage Learning,2018

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE05
Name of the Course	Software Testing Methodologies(Professional Elective-II)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:**Describe Path Testing using flowgraphs. **(K2)**
- CO2:**Demonstrate Transaction Flow, Dataflow and Domain Testing Techniques. **(K3)**
- CO3:**Apply Path Expressions and Logic-Based Testing. **(K3)**
- CO4:**Demonstrate State Transition Testing Using State Graphs. **(K3)**
- CO5:**Implement Graph Matrices and Testing Tools for Test Case Generation, Automation. **(K3)**

UNIT I: Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing. **(Textbook-1)**

UNIT II: Transaction Flow Testing: transaction flows, transaction flow testing techniques.

Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing.

Domain Testing: domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability. **(Textbook-1)**

UNIT III: Paths, Path products and Regular expressions: path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection.

Logic Based Testing: overview, decision tables, path expressions, kv charts, specifications. **(Textbook-1)**

UNIT IV: State, State Graphs and Transition testing: state graphs, good & bad state graphs, state testing, Testability tips. **(Textbook-1)**

UNIT V: Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools. (Student should be given an exposure to a tool like Jmeter/selenium/soapUI/Catalon). **(Textbook-2)**

Textbooks:

1. Software Testing techniques - BarisBeizer, Dreamtech, second edition.
2. Software Testing Tools – Dr. K. V. K. K. Prasad, Dreamtech.

Reference Books:

1. The craft of software testing - Brian Marick, Pearson Education.
2. Software Testing Techniques – SPD(Oreille)
3. Software Testing in the Real World – Edward Kit, Pearson.
4. Effective methods of Software Testing, Perry, John Wiley.
5. Art of Software Testing – Meyers, John Wiley.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE06
Name of the Course	Cyber Security (Professional Elective-II)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the fundamental concepts of cybercrime, its origins, types, and the challenges posed by modern devices and technologies. **(K2)**
- CO2:** Demonstrate various tools and techniques used in committing cybercrimes, including phishing, malware, spoofing, session hijacking, and social engineering. **(K2)**
- CO3:** Demonstrate knowledge of cybercrime investigation processes including digital evidence collection, eDiscovery, email/IP tracking, and the use of encryption/decryption tools. **(K2)**
- CO4:** Apply computer forensic techniques and use current tools to investigate and analyze digital evidence on different platforms such as Windows, Linux, and mobile devices. **(K3)**
- CO5:** Demonstrate the legal framework related to cybercrime, with a focus on the Indian IT Act, its challenges, amendments, and the global legal scenario. **(K2)**

UNIT I: Introduction to Cybercrime: Introduction, Cybercrime: Definition and Origins of the Word, Cybercrime and Information Security, Cybercriminals, Classifications of Cybercrime, Cyberstalking, Cybercafe and Cybercrimes, Botnets. Attack Vector, Proliferation of Mobile and Wireless Devices, Security Challenges Posed by Mobile Devices, Attacks on Mobile/Cell Phones, Network and Computer Attacks.

UNIT II: Tools and Methods: Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan Horses and Backdoors, Steganography, Sniffers, Spoofing, Session Hijacking Buffer over flow, DoS and DDoS Attacks, Attacks on Wireless Networks, Identity Theft (ID Theft), Foot Printing and Social Engineering, Port Scanning, Enumeration.

UNIT III: Cyber Crime Investigation: Introduction, Investigation Tools, eDiscovery, Digital Evidence Collection, Evidence Preservation, E-Mail Investigation, E-Mail Tracking, IP Tracking, E-Mail Recovery, Hands on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers, Recovering Deleted Evidences, Password Cracking.

UNIT IV: Computer Forensics and Investigations: Understanding Computer Forensics, Preparing for Computer Investigations. Current Computer Forensics Tools: Evaluating Computer Forensics Tools, Computer Forensics Software Tools, Computer Forensics Hardware Tools, Validating and Testing Forensics Software, Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Graphics and Network Forensics, E-mail Investigations, Cell Phone and Mobile Device Forensics.

UNIT V: Cyber Crime Legal Perspectives: Introduction, Cybercrime and the Legal Landscape around the World, The Indian IT Act, Challenges to Indian Law and Cybercrime Scenario in India, Consequences of Not Addressing the Weakness in Information Technology Act, Digital Signatures and the Indian IT Act, Amendments to the Indian IT Act, Cybercrime and Punishment, Cyber Law, Technology and Students: Indian Scenario.

Textbooks

1. Sunit Belapure Nina Godbole "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", WILEY, 2011.
2. Nelson Phillips and Enfinger Steuart, "Computer Forensics and Investigations", Cengage Learning, New Delhi, 2009.

Reference Books:

1. Michael T. Simpson, Kent Backman and James E. Corley, "Hands on Ethical Hacking and Network Defence", Cengage, 2019.
2. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi, 2009.
3. Alfred Basta, Nadine Basta, Mary Brown and Ravinder Kumar "Cyber Security and Cyber Laws", Cengage, 2018.

E-Resources:

1. CERT-In Guidelines- <http://www.cert-in.org.in/>
2. <https://www.coursera.org/learn/introduction-cybersecurity-cyber-attacks> [Online Course]
3. <https://computersecurity.stanford.edu/free-online-videos> [Free Online Videos]
4. Nickolai Zeldovich. 6.858 Computer Systems Security. Fall 2014. Massachusetts Institute of Technology: MIT OpenCourseWare, <https://ocw.mit.edu> License:Creative CommonsBY-NC-SA.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE07
Name of the Course	DevOps(Professional Elective-II)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain the architecture, principles, and lifecycle of DevOps and how it integrates with Agile practices and automation tools.. **(K2)**

CO2: Apply GIT commands and workflows to manage source code collaboratively and analyze code quality using SonarQube. **(K3)**

CO3: Develop and manage continuous integration pipelines using Jenkins to automate build and testing processes. **(K3)**

CO4: Implement containerized applications using Docker and integrate testing tools for continuous delivery. **(K3)**

CO5: Use Ansible and OpenShift to automate application deployment and manage container orchestration in CI/CD pipelines. **(K3)**

UNIT I: Introduction to DevOps: Introduction to SDLC, Agile Model. Introduction to Devops. DevOps Features, DevOps Architecture, DevOps Lifecycle, Understanding Workflow and principles, Introduction to DevOps tools, Build Automation, Delivery Automation, Understanding Code Quality, Automation of CI/ CD. Release management, Scrum, Kanban, delivery pipeline, bottlenecks, examples.

UNIT II: Source Code Management (GIT):The need for source code control, The history of source code management, Roles and code, source code management system and migrations. What is Version Control and GIT, GIT Installation, GIT features, GIT workflow, working with remote repository, GIT commands, GIT branching, GIT staging and collaboration. **Unit Testing - Code Coverage:** Junit, NUnit & Code Coverage with Sonar Qube, SonarQube - Code Quality Analysis.

UNIT III :Build Automation - Continuous Integration (CI): Build Automation, What is CI Why CI is Required, CI tools, Introduction to Jenkins (With Architecture), jenkins workflow, jenkins master slave architecture, Jenkins Pipelines, Pipeline Basics - Jenkins Master, Node, Agent, and Executor Freestyle Projects & Pipelines, Jenkins for Continuous Integration, Create and Manage Builds, User Management in Jenkins Schedule Builds, Launch Builds on Slave Nodes.

UNIT IV: Continuous Delivery (CD): Importance of Continuous Delivery, Continuous Deployment CD Flow, Containerization with Docker: Introduction to Docker, Docker installation, Docker commands, Images & Containers, DockerFile, Running containers, Working with containers and publish to Docker Hub.**Testing Tools:** Introduction to Selenium and its features, JavaScript testing.

UNIT V: Configuration Management - ANSIBLE: Introduction to Ansible, Ansible tasks, Roles, Jinja templating, Vaults, Deployments using Ansible. **Containerization Using Kubernetes(OPENSHIFT):** Introduction to Kubernetes Namespace & Resources, CI/CD - On OCP, BC, DC & ConfigMaps, Deploying Apps on Openshift Container Pods. Introduction to Puppet master and Chef.

Textbooks:

1. Joyner, Joseph., Devops for Beginners: Devops Software Development Method Guide for Software Developers and It Professionals, 1st Edition Mihails Konoplows, 2015.
2. Alisson Machado de Menezes., Hands-on DevOps with Linux,1st Edition, BPB Publications, India, 2021.

Reference Books:

1. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10
2. Gene Kim Je Humble, Patrick Debois, John Willis. The DevOps Handbook, 1st Edition, IT Revolution Press, 2016.
3. Verona, Joakim Practical DevOps, 1st Edition, Packt Publishing, 2016.
4. Joakim Verona. Practical Devops, Ingram short title; 2nd edition (2018). ISBN10: 1788392574
5. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE08
Name of the Course	Mobile Adhoc Networks (Professional Elective-III)					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1** : Explain MANETs, their uses, challenges, and types of MAC protocols. **(K2)**
CO2 : Apply suitable routing and transport protocols in ad hoc wireless networks. **(K3)**
CO3 : Describe common security threats and basic protection methods in MANETs. **(K2)**
CO4 : Use proper methods to collect and send data in wireless sensor networks. **(K3)**
CO5 : Operate tools and software to build secure wireless sensor network systems. **(K3)**

UNIT I: Introduction to Ad Hoc Wireless Networks- Cellular and Ad Hoc Wireless Networks, Characteristics of MANETs, Applications of MANETs, Issues and Challenges of MANETs, Ad Hoc Wireless Internet, MAC protocols for Ad hoc Wireless Networks-Issues, Design Goals and Classifications of the MAC Protocols.

UNIT II: Routing Protocols for Ad Hoc Wireless Networks- Issues in Designing a Routing Protocol, Classifications of Routing Protocols, Topology-based versus Position-based Approaches, Issues and design goals of a Transport layer protocol, Classification of Transport layer solutions, TCP over Ad hoc Wireless Networks, Solutions for TCP over Ad Hoc Wireless Networks, Other Transport layer protocols.

UNIT III: Security protocols for Ad hoc Wireless Networks- Security in Ad hoc Wireless Networks, Network Security Requirements, Issues and Challenges in Security Provisioning, Network Security Attacks, Key Management, Secure Routing in Ad hoc Wireless Networks, Cooperation in MANETs, Intrusion Detection Systems.

UNIT IV: Basics of Wireless Sensors and Applications- The Mica Mote, Sensing and Communication Range, Design Issues, Energy Consumption, Clustering of Sensors, Applications, Data Retrieval in Sensor Networks- Classification of WSNs, MAC layer, Routing layer, Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs.

UNIT V: Security in WSNs- Security in WSNs, Key Management in WSNs, Secure Data Aggregation in WSNs, Sensor Network Hardware-Components of Sensor Mote, Sensor Network Operating Systems-TinyOS, LA-TinyOS, SOS, RETOS, Imperative Language-nesC, **Dataflow Style Language**-TinyGALS, Node-Level Simulators, NS-2 and its sensor network extension, TOSSIM.

Text Books:

1. Ad Hoc Wireless Networks – Architectures and Protocols, 1st edition, C. Siva Ram Murthy, B. S. Murthy, Pearson Education, 2004
2. Ad Hoc and Sensor Networks – Theory and Applications, 2nd edition *Carlos Corderio Dharma P. Aggarwal*, World Scientific Publications / Cambridge University Press, March 2006

Reference Books:

1. Wireless Sensor Networks: An Information Processing Approach, 1st edition, *Feng Zhao, Leonidas Guibas*, Elsevier Science imprint, Morgan Kauffman Publishers, 2005, rp2009
2. Wireless Ad hoc Mobile Wireless Networks – Principles, Protocols and Applications, 1st edition, Subir Kumar Sarkar, et al., Auerbach Publications, Taylor & Francis Group, 2008
3. Ad hoc Networking, 1st edition, *Charles E. Perkins*, Pearson Education, 2001
4. Wireless Ad hoc Networking, 1st edition, *Shih-Lin Wu, Yu-Chee Tseng*, Auerbach Publications, Taylor & Francis Group, 2007
5. Wireless Sensor Networks – Principles and Practice, 1st edition, Fei Hu, Xiaojun Cao, An Auerbach book, CRC Press, Taylor & Francis Group, 2010.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE09
Name of the Course	Software Project Management (Professional Elective-III)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Explain conventional and agile software methods with their effect on software economics. **(K2)**

CO2: Explain modern software practices and identify life cycle phases and artifacts. **(K2)**

CO3: Describe software architectures, workflows, and checkpoints in iterative development. **(K2)**

CO4: Illustrate planning techniques and identify project structures and automation elements. **(K3)**

CO5: Describe software metrics and capability models to improve quality. **(K2)**

UNIT I: Conventional Software Management: The waterfall model, conventional software, Agile Methodology, Agiling capabilities, Management performance.

Evolution of Software Economics: Software Economics, pragmatic software cost estimation.

Improving Software Economics: Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.

The old way and the new: The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process.

UNIT II: Life cycle phases: Engineering and production stages, inception, Elaboration, construction, transition phases.

Artifacts of the process: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

UNIT III: Model based software architectures: A Management perspective and technical perspective.

Work Flows of the process: Software process workflows, Iteration workflows.

Checkpoints of the process: Majormile stones, Minor Milestones, Periodic status assessments.

Iterative Process Planning: Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

UNIT IV: Project Organizations and Responsibilities: Line-of-Business Organizations, Project Organizations, evolution of Organizations. **Process Automation:** Automation Building blocks, The Project Environment.

UNIT V: Project Control and Process instrumentation: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation.

Process Capability Models: Capability Maturity Model, Enhancing software Quality.

Textbooks:

1. Software Project Management, Walker Royce, PEA, 2005.
2. Software Engineering: A Practitioner's Approach by Roger S. Pressman and Bruce R. Maxim

Reference Books:

1. Software Project Management, Bob Hughes,3/e, Mike Cotterell, TMH
2. Software Project Management, Joel Henry, PEA
3. Software Project Management in practice, Pankaj Jalote, PEA, 2005,
4. Effective Software Project Management, Robert K.Wysocki, Wiley,2006.
5. Project Management in IT, Kathy Schwalbe, Cengage .

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE10
Name of the Course	Natural Language Processing (Professional Elective-III)					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Explain the fundamental concepts of language modeling, English morphology, and spelling correction techniques using regular expressions and finite-state automata. (K2)
- CO2:** Apply N-gram models and part-of-speech tagging techniques using stochastic and rule-based approaches to perform word-level language analysis. (K3)
- CO3:** Construct syntactic parsers using context-free grammars and apply probabilistic parsing methods such as PCFG and CYK for syntactic analysis. (K3)
- CO4:** Describe semantic representation techniques and methods for word sense disambiguation using supervise and knowledge-based approaches. (K2)
- CO5:** Apply discourse segmentation and reference resolution techniques using lexical resources like WordNet, Penn Treebank, and Brill's Tagger for effective text analysis. (K3)

UNIT I: Introduction: Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM – Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance.

UNIT II: Word Level Analysis: Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part- of-Speech Tagging, Rule-based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.

UNIT III: Syntactic Analysis: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs – Feature structures, Unification of feature structures

UNIT IV: Semantics And Pragmatics: Requirements for representation, First-Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selectional restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.

UNIT V: Discourse Analysis And Lexical Resources: Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).

Textbooks:

- Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech, 2nd Edition, Daniel Jurafsky, James H. Martin -Pearson Publication, 2014.
- Natural Language Processing with Python, First Edition, Steven Bird, Ewan Klein and Edward Loper, O'Reilly Media, 2009.

Reference Books:

- Language Processing with Java and Ling Pipe Cookbook, 1st Edition, Breck Baldwin, Atlantic Publisher, 2015.
- Natural Language Processing with Java, 2nd Edition, Richard M Reese, O'Reilly Media, 2015.
- Handbook of Natural Language Processing, Second, Nitin Indurkha and Fred J. Damerau, Chapman and Hall/CRC Press, 2010. Edition
- Natural Language Processing and Information Retrieval, 3rd Edition, Tanveer Siddiqui, U.S. Tiwary, Oxford University Press, 2008.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE11
Name of the Course	Big Data Analytics(Professional Elective-III)					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:**Describe big data concepts, applications, and related technologies. (K2)
- CO2:**Use NoSQL models and perform basic operations in Cassandra. (K3)
- CO3:**Develop data processing tasks using Hadoop and Hive. (K3)
- CO4:**Build data analytics workflows using Apache Spark. (K3)
- CO5:**Implement streaming and tuning techniques in Spark. (K3)

UNIT I: Introduction: Big data, convergence of key trends, unstructured data; **Industry examples of big data:** web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data; **Big data technologies:** Introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.

UNIT II: Introduction to NoSQL; Aggregate Data Models: aggregates, key-value and document data models; **More Details on Data Models:** relationships, graph databases, schema less databases, materialized views; **Distribution models:** sharding, master-slave replication, peer- peer replication, combining sharding and replication; **Consistency:** relaxing consistency; **Version Stamps; Working with Cassandra:** Table creation, loading and reading data.

UNIT III: MapReduce: Data formats, analyzing data with Hadoop, scaling out, Data locality, Map Reduce Architecture, Process flow; **Hadoop distributed file system (HDFS):** Architecture, fault tolerance with data replication, High availability, Java interface, data flow; **Hadoop I/O:** data integrity, compression, serialization; **Hive:** Introduction, data types and file formats, HiveQL data definition, HiveQL data manipulation, Logical joins, Window functions, Optimization, Table partitioning, Bucketing, Indexing, Join strategies.

UNIT IV: Apache spark- Advantages over Hadoop, lazy evaluation, In memory processing, DAG, Spark context, Spark Session, RDD, Transformations- Narrow and Wide, Actions, Data frames, RDD to Data frames, Catalyst optimizer, Data Frame Transformations, Working with Dates and Timestamps, Working with Nulls in Data, Working with Complex Types, Working with JSON, Grouping, Window Functions, Joins, Data Sources, Broadcast Variables, Accumulators, Deploying Spark- On-Premises Cluster Deployments, Cluster Managers- Standalone Mode, Spark on YARN , Spark Logs, The Spark UI- Spark UI History Server, Debugging and Spark First Aid.

UNIT V: Spark-Performance Tuning, Stream Processing Fundamentals, Event-Time and State full Processing - Event Time, State full Processing, Windows on Event Time- Tumbling Windows, Handling Late Data with Watermarks, Dropping Duplicates in a Stream, Structured Streaming Basics - Core Concepts, Structured Streaming in Action, Transformations on Streams, Input and Output.

Textbooks:

1. Big Data, Big Analytics: Emerging, Michael Minnelli, Michelle Chambers, and AmbigaDhiraj, 1st edition ,2013
2. SPARK: The Definitive Guide, Bill Chambers & Matei Zaharia, O'Reilly, 2018-first Edition.
3. Business Intelligence and Analytic Trends for Today's Businesses", Wiley, First edition-2013.
4. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the Emerging World Polyglot Persistence", Addison-Wesley Professional, 2012
5. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilly, 2012

Reference Books:

1. Hadoop Operations, Eric Sammer, 1st Edition, O'Reilly
2. Programming Hive, E. Capriolo, D. Wampler, J. Rutherglen, O'Reilly
3. HBase: The Definitive Guide, Lars George, 1st Edition, O'Reilly
4. Cassandra: The Definitive Guide, Eben Hewitt, O'Reilly
5. Programming Pig, Alan Gates, 1st Edition, O'Reilly.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	3	0	0	3	V23CSTPE12
Name of the Course	Distributed Operating Systems(Professional Elective-III)					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Describe the basic concepts of distributed operating systems. **(K2)**
- CO2:** Understand the Communication using RPC. **(K2)**
- CO3:** Demonstrate the concept of shared memory. **(K3)**
- CO4:** Explain resource management techniques. **(K2)**
- CO5:** Illustrate Distributed file system and fault tolerance performance. **(K2)**

UNIT I: Fundamentals: What is Distributed Computing Systems? Evolution of Distributed Computing System; Distributed Computing System Models; What is Distributed Operating System? Issues in Designing a Distributed Operating System; Introduction to Distributed Computing Environment(DCE).

Message Passing: Introduction, Desirable features of a Good Message Passing System, Issues in PC by Message Passing, Synchronization, Buffering, Multi-datagram Messages, Encoding and Decoding of Message Data, Process Addressing, Failure Handling, Group Communication, Case Study: 4.3 BSD UNIX IPC Mechanism.

UNIT II: Remote Procedure Calls: Introduction, The RPC Model, Transparency of RPC, Implementing RPC Mechanism, Stub Generation, RPC Messages, Marshaling Arguments and Results, Server Management, Parameter-Passing Semantics, Call Semantics, Communication Protocols for RPCs, Complicated RPCs, Client-Server Binding, Exception Handling, Security, Some Special Types of RPCs, RPC in Heterogeneous Environments, Lightweight RPC, Optimization for Better Performance, Case Studies: Sun RPC

UNIT III: Distributed Shared Memory: Introduction, General Architecture of DSM systems, Design and Implementation Issues of DSM, Granularity, Structure of Shared Memory Space, Consistency Models, Replacement Strategy, Thrashing, Other approaches to DSM, Heterogeneous DSM, Advantages of DSM. Synchronization: Introduction, Clock Synchronization, Event Ordering, Mutual Exclusion, Dead Lock, Election Algorithms

UNIT IV: Resource Management: Introduction, Desirable Features of a Good Global Scheduling Algorithm, Task Assignment Approach, Load – Balancing Approach, Load – Sharing Approach Process Management: Introduction, Process Migration, Threads.

UNIT V: Distributed File Systems: Introduction, Desirable Features of a Good Distributed File System, File models, File–Accessing Models, File – Sharing Semantics, File – Caching Schemes, File Replication, Fault Tolerance, Atomic Transactions and Design Principles.

Textbooks:

1. Pradeep. K. Sinha: Distributed Operating Systems: Concepts and Design, PHI, 2007.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, Pearson Education, 2013.
2. Ajay D. Kshem kalyani and Mukesh Singhal, Distributed Computing: Principles, Algorithms and Systems, Cambridge University Press, 2008
3. Sunita Mahajan, Seema Shan, “ Distributed Computing”, Oxford University Press, 2015.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL10
Name of the Course	Cloud Computing Lab					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Demonstrate various service types, delivery models and technologies of a cloud computing environment. **(K3)**
- CO2:** Classify cloud services based on virtual machines and containers in the cloud offerings. **(K2)**
- CO3:** Identify and outline key challenges associated with a cloud-based application. **(K2)**
- CO4:** Illustrate advanced cloud concepts such as server less computing and cloud simulation. **(K3)**
- CO5:** Apply suitable programming paradigms to solve real world and scientific problems using cloud services. **(K3)**

List of Experiments

1. Develop and deploy basic web services
2. Develop IPC mechanisms and message communication using publish/subscribe model.
3. Demonstrate Configuration and Installation of VirtualBox/VMware Workstation with different flavors of Linux or windows OS on top of windows8 or above.
4. Demonstrate Installation of C compiler in the virtual machine created using VirtualBox and execute Simple Programs.
5. Demonstrate Creation of an Amazon EC2 instance and set up a webserver on the instance and associate an IP address with the instance. In the process, create a security group allowing access to port 80 on the instance.

OR

6. Demonstrate EC2 setup on OpenStack environment (*Hint: perform experiment 5 using OpenStack*)
7. Demonstrate Installation of Google App Engine. Create a Hello World app and other simple web applications using python/java.
8. Operate a Docker container and set up a webserver (e.g. apache2 or Python based Flask micro web framework) on the instance. Map the host directory as a data volume for the container.
9. Demonstrate file transfer from one virtual machine to another virtual machine. Similarly, from one container to another container.
10. Operate and verify virtual machine on Trystack. (Online Openstack Demo Version)
11. Demonstrate Installation of Hadoop single node cluster and run simple applications like word count.
12. Employ OpenFaaS – Serverless computing framework and demonstrate basic event driven function invocation.
13. Demonstrate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in Cloud Sim.

Textbooks:

1. Mastering Cloud Computing, Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi, Shivananda Poojara, Satish N. Srirama, 2nd Edition, McGraw Hill, 2024.
2. Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.

Reference Books:

1. Cloud Computing, Theory and Practice, Dan C Marinescu, 2nd edition, MK Elsevier, 2018.
2. Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg and Andrzej M. Goscinski, Wiley, 2011.
3. Online documentation and tutorials from cloud service providers (e.g. AWS, Google App Engine)
4. Docker, Reference documentation, <https://docs.docker.com/reference/>
5. OpenFaaS, Serverless Functions Made Simple, <https://docs.openfaas.com/>

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	0	0	3	1.5	V23CSL11
Name of the Course	Machine Learning Lab using Python					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After successful completion of the course, the student will be able to:

- CO1:**Identify various Python libraries used in Machine Learning. **(K2)**
CO2:Construct non-probabilistic classifiers using Python Programming. **(K3)**
CO3:Implement probabilistic classifiers using Python Programming. **(K3)**
CO4:Apply ensemble methods for classification tasks. **(K3)**
CO5:Demonstrate neural and sequential models using Python. **(K3)**

LIST OF EXPERIMENTS

1. Introduction to required python libraries such as Numpy, Pandas, Scipy, Matplotlib and Scikit-learn.
2. Import, preprocess, and split the datasets using scikit-learn.
3. Implement the KNN algorithm for classification and demonstrate the process of finding out optimal “K” value using Python Programming.
4. Demonstrate the process of the Decision Tree construction for classification problems using python programming.
5. Construct a classification model using the Bayes classifier using Python Programming.
6. Implement a Logistic Regression algorithm for binary classification using Python Programming.
7. Construct an SVM classifier using python programming.
8. Demonstrate the working of Multi-layer perceptron with MLPClassifier() using Python programming.
9. Implement an Ensemble Learner using Random Forest Algorithm using python programming.
10. Implement an Ensemble Learner using Adaboost Algorithm using Python programming.
11. Construct and Analyze a Bayesian Network using pgmpy for Probabilistic Inference.
12. Implement and Evaluate a Hidden Markov Model using hmmlearn for Sequential Pattern Recognition.

Text Books:

1. “Machine Learning Theory and Practice”, M N Murthy, V S Ananthanarayana, Universities Press (India), 2024
2. “Machine Learning: An Algorithmic Perspective, Stephen Marsland, 2nd Ed., CRC Press.
3. “Introduction to Machine Learning with Python”, Andreas C. Muller and Sarah Guido, First Edition, O’Reilly.

Reference Books:

1. “Practical Machine Learning with Python”, Dipanjan Sarkar, Raghav Bali and Tushar Sharma, First Edition, A Press.

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	0	1	2	2	V23CSSE04
Name of the Course	Master Coding and Competitive Programming - Part-2					
Branch	Common to CSE, CST, CAI, AIM and CDS					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

CO1: Apply Divide and Conquer algorithm technique to solve complex in logarithmic time. **(K3)**

CO2: Apply Greedy method to solve Optimization and decision making problems. **(K3)**

CO3: Apply Backtracking Algorithm technique to find combinatorial problems. **(K4)**

CO4: Experiment with Dynamic Programming Algorithm technique to solve Problems that uses Optimal substructures. **(K3)**

CO5: Develop programs using Linked List Graphs, DFS and BFS techniques. **(K3)**

List of Experiments

1. Develop Programs to solve problems based on Divide and Conquer Algorithm Technique.
2. Develop programs using two pointer and sliding window algorithms.
3. Illustrate Problem Solving using Greedy Algorithm technique.
4. Illustrate Problem Solving using Backtracking.
5. Develop programs using Dynamic Programming and Kadane Algorithm.
6. Develop programs using Linked List and its applications.
7. Develop programs using Graphs and Graph Searching Techniques.

Textbooks:

1. Introduction to Algorithms, Second Edition, Thomas H. Cormen Charles E. Leiserson.
2. Data Structures and Algorithms Made Easy: Narasimha Karumanchi .
3. The Algorithm Design Manual, Springer series, Steven Skiena.

Tools:

1. practice.geeksforgeeks.com
2. leetcode.com
3. codingninjas.com
4. [Hackerrank.com](https://hackerrank.com)
5. [Interviewbit.com](https://interviewbit.com)

Semester	VI	L	T	P	C	COURSE CODE
Regulation	V23	2	0	0	-	V23ENT05
Name of the Course	Technical Paper Writing and IPR					
Branch	Common to CSE and CST					

Syllabus Details

Course Outcomes: After Successful completion of the Course, the student will be able to:

- CO1:** Understand the structure of the technical paper and its components. (K2)
- CO2:** Review the literature and acquire the skills to write a technical paper for first submission. (K2)
- CO3:** Understand the process and development of IPR. (K2)
- CO4:** Create awareness about the scope of patent rights. (K6)
- CO5:** Analyze the new developments in IPR include latest software. (K4)

UNIT I: Planning and preparation : Planning and Preparation, Word Order, breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness. Clarifying Who Did What, Highlighting Your Findings, Hedging and Criticizing, Paraphrasing and Plagiarism, Sections of a Paper, Abstracts, Introduction.

UNIT II: Literature review: Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check. Key skills needed when writing a Title, Abstract, Introduction, a Review of the Literature, the Methods, the Results, the Discussion, and the Conclusions. Useful phrases, how to ensure paper is as good as it could possibly be the first- time submission.

UNIT III: Process and Development: Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, patenting under PCT.

UNIT IV: Patent Rights: Scope of Patent Rights. Licensing and transfer of technology, Patent information and databases, Geographical Indications.

UNIT V: New Developments In IPR: New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies.

Textbooks:

1. Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press.
2. Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd, 2007.

References:

1. Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)
2. Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman's book.
3. Adrian Wallwork, English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011
4. Mayall, "Industrial Design", McGraw Hill, 1992.
5. Robert P. Merges, Peter S. Menell, Mark A. Lemley, "Intellectual Property in New Technological Age" 2016.
6. T. Ramappa, "Intellectual Property Rights Under WTO", S. Chand, 2008.